

Federated Adaptive Causal Estimation with High Dimensional Covariates (FACE-HD)

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August 25, 2026

Abstract

High dimensional observational data, such as electronic health records (EHRs), are common in causal inference and comparative effectiveness research. When patient level data are distributed across clinical sites and covariate distributions differ by site, naive pooling or separate site analyses can give unstable or biased estimates for a target population. We propose a federated adaptive causal estimation framework for high dimensional covariates (FACE-HD), extending Federated Adaptive Causal Estimation (FACE) ([Han et al. 2025](#)) to target average treatment effects with many candidate confounders. FACE-HD uses one calibration model for the propensity score and density ratio, combines penalized nuisance estimation with cross-fitting to form doubly robust source assisted estimators, and aggregates sources with a standardized Wald penalty. Simulations with nontransportable sources show that FACE-HD maintains substantially more stable bias and coverage than non-adaptive aggregation while improving precision over target-only estimation. We also apply FACE-HD to the Right Heart Catheterization study.

Keywords: causal inference, transfer learning, doubly robust estimation, transportability, observational studies

1 Introduction

Real world data, including electronic health records (EHRs), administrative claims, and patient registries, are increasingly used to estimate causal effects in large populations (Hernán & Robins 2016, Sherman et al. 2016). In practice, these data are often distributed across sites. Patient level records may be held locally because of privacy, governance, or infrastructure constraints, and the number of candidate confounders can be large enough that standard low dimensional methods become unstable (Belloni et al. 2014, Chernozhukov et al. 2018, Tan 2020). Multisite observational studies also involve covariate shift across institutions. A source site may improve precision, but it can introduce bias if its population is not transportable to the target site (Cole & Stuart 2010, Pearl & Bareinboim 2014, Dahabreh et al. 2019).

Several strands of prior work address pieces of this problem. High dimensional causal inference methods combine regularized nuisance estimation with bias correction so that first stage regularization error does not dominate the final effect estimator. Examples include double selection, residual balancing, debiased estimation, and double machine learning procedures (Belloni et al. 2014, Athey et al. 2018, Chernozhukov et al. 2018), as well as augmented inverse probability weighting and doubly robust estimators based on orthogonal estimating equations (Robins et al. 1994, Bang & Robins 2005, Tsiatis 2006, Tan 2020, Smucler et al. 2019). A parallel literature on generalizability and transportability studies when evidence from one population can be used for another after adjustment for observed covariates (Cole & Stuart 2010, Pearl & Bareinboim 2014, Dahabreh et al. 2019). This concern is central in multisite studies, where borrowing from a source that is not transportable may reduce variance while introducing bias into the target site effect estimate. Federated learning and privacy preserving distributed analysis provide one approach to this setting: sites exchange model updates or low dimensional summaries rather than

individual level records (McMahan et al. 2017, Kairouz et al. 2021). Recent causal work has considered federated hospital quality measurement with target flexibility (Han et al. 2024), differentially private multisite aggregation of treatment effect estimates (Koga et al. 2024), one shot and multishot federated estimators beyond meta analysis (Khellaf et al. 2024), federated propensity score aggregation for IPW and AIPW estimation (Khellaf et al. 2025), and efficient collaborative learning of target ATEs under data sharing constraints (Li & Duan 2026). Broader federated and data sharing methods address robust site selection, prevailing model inference, and summary-statistic integration under heterogeneous sources (Cai et al. 2022, Guo et al. 2025), while semiparametric data fusion work characterizes efficiency gains from combining aligned or weakly aligned sources (Li & Luedtke 2023, Li et al. 2025). Closely related high dimensional distributed methods calibrate propensity score and outcome models under covariate shift (Wang et al. 2024, Tong et al. 2025), while transfer learning and multiply robust federated estimators use related source borrowing ideas to combine information across sites without pooling records (Tian & Feng 2023, Han et al. 2023). Existing approaches, however, do not fully address high dimensional confounding adjustment, target specific transportability, federated information constraints, and adaptive protection against negative transfer in one estimator.

In this paper, we extend Federated Adaptive Causal Estimation (FACE) (Han et al. 2025) to high dimensional settings. FACE-HD targets the target average treatment effect (TATE) and treats source sites as auxiliary information to be screened and weighted for the target population. The contribution is to combine doubly robust estimation, cross-fitting, calibrated weighting, federated summaries, and FACE-style adaptive aggregation for high dimensional, target specific TATE estimation. Specifically:

- (i) **Merged calibration for target specific source transport.** FACE-HD uses one exponential tilting calibration model for each source and treatment arm, combining

the source treatment propensity and the source-to-target density ratio.

- (ii) **Pairwise source assisted estimation under federated constraints.** FACE-HD derives pairwise source assisted TATE estimators and penalized calibration losses that require source sites to send only summary statistics and fitted nuisance parameters.
- (iii) **Adaptive aggregation for high dimensional FACE.** FACE-HD estimates the shared target covariance among target only and source assisted estimators and uses a self-normalized Wald penalty to downweight sources that disagree with the target.

The paper is organized as follows. [Section 2](#) introduces the multisite notation, target estimand, calibration weights, and pairwise source assisted estimators. [Section 3](#) gives the nuisance losses, aggregation objective, federated communication protocol, and tuning rules. [Section 4](#) states the regularity conditions and asymptotic results. [Section 5](#) describes the simulation study, [Section 6](#) applies FACE-HD to the Right Heart Catheterization data, and [Section 7](#) concludes. Proofs and additional details are in the Supplemental Material.

2 Notation and Setup

We consider a causal inference problem with one target site (t) and multiple source sites ($\mathcal{S} = \{s_1, s_2, \dots, s_K\}$). The goal is to estimate the target average treatment effect (TATE) using target data and source-site summaries, following the target-population perspective used in generalizability and FACE-style multi-site estimation ([Cole & Stuart 2010](#), [Dahabreh et al. 2019](#), [Han et al. 2025](#)). Throughout, $\mathbb{E}_r[\cdot]$ denotes expectation conditional on site membership $R = r$. We define the target-site potential outcome means $\mu^1 := \mathbb{E}_t[Y(1)]$ and $\mu^0 := \mathbb{E}_t[Y(0)]$, and the TATE $\tau := \mu^1 - \mu^0$. For simplicity, we first present the nuisance and pairwise construction for the treated potential-outcome mean μ^1 . The corresponding objects for μ^0 are obtained by replacing treatment level 1 by 0 and using the corresponding

nuisance parameters. Each pairwise TATE contrast is formed as $\hat{\tau}_{t,s_j} = \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$; the final FACE-HD estimator aggregates these TATE contrasts directly with one common source-weight vector, as defined in [Eq. \(9\)](#).

Each observation contains site membership $R \in \{t, s_1, \dots, s_K\}$, raw covariates $\mathbf{X}$, treatment $A \in \{0, 1\}$, and outcome Y . We write $\mathcal{D}_t$ and $\mathcal{D}_{s_j}$ for the target and source samples, with sizes N_t and N_{s_j} , and $N_{\text{all}} := N_t + \sum_{j=1}^K N_{s_j}$. The empirical average over site r is denoted by $\tilde{E}_r(\cdot)$. To allow flexible high-dimensional modeling, the nuisance models use a fixed working feature map $\phi(\mathbf{X})$; one may also read $\phi(\mathbf{X})$ simply as the transformed covariate vector used by the analysis. Throughout, $\phi(\cdot)$ denotes this feature map, $g(\cdot)$ the GLM mean (response) function, and $\Psi(\cdot)$ the GLM cumulant; the subscripted ψ_2 appears only in the sub-Gaussian Orlicz norm $\|\cdot\|_{\psi_2}$ used in the Supplemental Material. Each nuisance design includes an intercept. In every penalized nuisance-loss display below, the ℓ_1 term acts only on the non-intercept coordinates, exactly as in the implementation; unqualified vector norms used in assumptions and rate statements retain their usual mathematical meaning.

Assumption 1 (Identification and overlap). For a fixed constant $M > 1$ (equivalently, a positivity constant $1/M > 0$), assume there exists an informative source subset $\mathcal{S}^* \subseteq \mathcal{S}$ such that, for each treatment arm $a \in \{0, 1\}$, each site $r \in \{t\} \cup \mathcal{S}^*$, and each covariate value $\mathbf{x}$ in the target support:

- (a) (*Consistency*) $Y = Y(A) = AY(1) + (1 - A)Y(0)$.
- (b) (*Treatment positivity*) $1/M < P(A = a \mid R = r, \mathbf{X} = \mathbf{x}) < 1 - 1/M$.
- (c) (*Site positivity*) The target-to-source density ratios needed for transport are bounded on the target support. A sufficient pooled-superpopulation condition is that the relevant site probabilities satisfy $P(R = r \mid \mathbf{X} = \mathbf{x}) > 1/M$ together with fixed, nonvanishing marginal site probabilities; we do not claim equivalence without those additional

marginal-probability conditions.

- (d) (*Ignorability and conditional-mean transportability*) Treatment is ignorable within each relevant site, $Y(a) \perp\!\!\!\perp A \mid \mathbf{X}, R = r$, and every informative source shares the target conditional potential-outcome mean, $\mathbb{E}\{Y(a) \mid \mathbf{X} = \mathbf{x}, R = s\} = \mathbb{E}\{Y(a) \mid \mathbf{X} = \mathbf{x}, R = t\}$ for each $s \in \mathcal{S}^*$. We require equality of conditional means, not equality of the full conditional distributions of the potential outcomes across sites.

Sources outside $\mathcal{S}^*$ may violate the transportability part of this assumption; FACE-HD treats them as candidate auxiliary sources and relies on the adaptive aggregation step to shrink away sources that disagree with the target.

Remark 1 (Interpretation of transportability). [Assumption 1](#)(d) states that, after adjustment for observed covariates, the target site and an unknown informative subset of source sites $\mathcal{S}^* \subseteq \mathcal{S}$ share the relevant potential-outcome regression. This condition justifies source-assisted estimation for the TATE. FACE-HD does not assume that every source site is informative: sources in $\mathcal{S} \setminus \mathcal{S}^*$ may violate transportability, for example because of unmeasured effect modifiers, site-specific coding practices, or differences in clinical practice patterns. The adaptive aggregation step gives smaller weights to sources whose estimates disagree with the target.

As in transportability and generalizability analyses, [Assumption 1](#)(d) is not fully testable from the observed data. It should therefore be viewed as an identification condition for the informative sources, rather than as a guaranteed property of all candidate sources. Sensitivity analysis for violations of transportability is an important complement when unmeasured source-target differences are plausible.

Before specifying the working nuisance models, consider the source-assisted functional. For

any candidate regression function $m(\mathbf{X})$,

$$\mu^1 = \mathbb{E}_t\{Y(1)\} = \mathbb{E}_t\{m(\mathbf{X})\} + \mathbb{E}_t\{Y(1) - m(\mathbf{X})\}.$$

The first term only requires the target covariate distribution. Under [Assumption 1](#), the second term can be estimated from any informative source site $s_j \in \mathcal{S}^*$. Writing f_r for the covariate density at site r and $e_{s_j}^1(\mathbf{X}) = P(A = 1 \mid R = s_j, \mathbf{X})$, the transported residual identity is

$$\mathbb{E}_t\{Y(1) - m(\mathbf{X})\} = \mathbb{E}_{s_j} \left[I(A = 1) \frac{f_t(\mathbf{X})}{e_{s_j}^1(\mathbf{X}) f_{s_j}(\mathbf{X})} \{Y - m(\mathbf{X})\} \right].$$

This gives a source-assisted functional with a target regression average and a source weighted-residual correction. FACE-HD estimates the correction with a single calibration weight that combines the source treatment propensity and the source-to-target density ratio.

We model the candidate potential-outcome regression with a generalized linear working model. Let g be a GLM response function with a canonical link, mapping the linear predictor to the conditional mean of the outcome. For compactness, write

$$m^a(\mathbf{X}; \boldsymbol{\alpha}^a) := g\{\phi(\mathbf{X})^T \boldsymbol{\alpha}^a\}, \quad g(\phi(\mathbf{X}); \boldsymbol{\alpha}^a) \equiv g\{\phi(\mathbf{X})^T \boldsymbol{\alpha}^a\}, \quad a \in \{0, 1\}.$$

For each source site $s_j \in \mathcal{S}$ and treatment level $a \in \{0, 1\}$, we approximate the ideal transported inverse-probability weight by an *exponential tilting* (log-linear) *working model*, in line with calibrated weighting and density-ratio ideas for high-dimensional treatment-effect estimation ([Tan 2020](#), [Tong et al. 2025](#), [Wang et al. 2024](#)):

$$w(\mathbf{X}, A; \boldsymbol{\gamma}_{s_j, a}) := I(A = a) \exp\{-\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j, a}\}, \quad (1)$$

where $\boldsymbol{\gamma}_{s_j, a}$ is the site/treatment calibration parameter; when the treatment arm is clear, we suppress the treatment argument and write $w(\mathbf{X}; \boldsymbol{\gamma}_{s_j, a})$. This parametrization gives convex calibration losses and nonnegative weights with a strictly positive exponential tilt among arm- a source observations. We do not interpret the log-linear tilt as a model for

$P(R, A \mid \mathbf{X})$; the theory only requires the implied weights to satisfy the balance identity below. For the pairwise DR estimator $\hat{\mu}_{t,s_j}^1$, only $\gamma_{s_j,1}$ is needed. If the tilt is viewed as a multinomial inverse-odds model over site–treatment categories with the target site as reference, then $\exp\{-\phi(\mathbf{X})^T \gamma_{s_j,a}\}$ has the same target-density-over-source-treatment form as the ideal transported weight, up to a site-size ratio absorbed by the intercept. Any remaining multiplicative normalizing constant (equivalently, an additive constant in the log-tilt) is likewise absorbed into the intercept component of $\gamma_{s_j,1}$. Consequently, when $\gamma_{s_j,1}^*$ is the population solution of the calibration/balancing equation, the calibration weight satisfies the *balancing property*: for every integrable function h ,

$$\mathbb{E}_{s_j}[w(\mathbf{X}; \gamma_{s_j,1}^*) h(\mathbf{X})] = \mathbb{E}_t[h(\mathbf{X})], \quad (2)$$

which is the balance identity used by the pairwise doubly robust estimator below. The identity for *all* integrable h corresponds to the ideal population density-ratio relationship. In the high-dimensional analysis, balance is imposed over a finite-dimensional function class, typically the linear span of the chosen features $h(\mathbf{X}) \in \text{span}\{\phi_1(\mathbf{X}), \dots, \phi_p(\mathbf{X})\}$ (including an intercept). The corresponding “weight-model correctness” condition in the Supplemental Material is defined relative to this function class. This inverse-odds interpretation is therefore only a motivation for the merged form; the estimation procedure treats $\gamma_{s_j,a}$ as a calibration parameter defined by the regularized loss and its induced balance moments.

The doubly robust estimators combine outcome modeling and inverse probability weighting, following the classical AIPW/semiparametric estimating-equation construction (Robins et al. 1994, Bang & Robins 2005, Tsiatis 2006). The estimator for μ^1 assisted by source site s_j is:

$$\hat{\mu}_{t,s_j}^1 = \tilde{E}_t(g(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) + \tilde{E}_{s_j} \left(\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \hat{\gamma}_{s_j,1}}} (Y - g(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)) \right) \quad (3)$$

The estimator for the control group, $\hat{\mu}_{t,s_j}^0$, is defined symmetrically by using the nuisance

parameters $(\boldsymbol{\alpha}^0, \boldsymbol{\gamma}_{s_j,0})$ and replacing $I(A = 1)$ by $I(A = 0)$ in the weighted residual term:

$$\hat{\mu}_{t,s_j}^0 = \tilde{E}_t(g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s_j}^0)) + \tilde{E}_{s_j} \left(\frac{I(A = 0)}{e^{\phi(\mathbf{X})^T \hat{\boldsymbol{\gamma}}_{s_j,0}}} (Y - g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s_j}^0)) \right).$$

Finally, the (pairwise) TATE estimator is $\hat{\tau}_{t,s_j} := \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$, and the target-only TATE estimator is $\hat{\tau}_{ot} := \hat{\mu}_{ot}^1 - \hat{\mu}_{ot}^0$.

The target-only estimator $\hat{\mu}_{ot}^1$ is a standard AIPW estimator on the target data alone and requires the *target-site propensity score* $\pi_{ot}(\mathbf{X}) := P(A = 1 \mid R = t, \mathbf{X})$, which models treatment assignment *within* the target population. This is distinct from the cross-site calibration/tilting parameters $\boldsymbol{\gamma}_{s_j,a}$ defined above: π_{ot} is estimated locally at the target site (e.g., via ℓ_1 -penalized logistic regression on $(A, \mathbf{X})$ pairs from $\mathcal{D}_t$), and does not enter the source-assisted estimator $\hat{\mu}_{t,s_j}^1$ or its loss functions.

3 Method

Building on [Section 2](#), this section describes how FACE-HD estimates the nuisance functions, aggregates source assisted estimators, implements the procedure under federated information constraints, and selects tuning parameters.

3.1 Cross-Calibrated Nuisance Estimation

The source assisted estimator in [Eq. \(3\)](#) depends on two nuisance components: the transported outcome regression $\boldsymbol{\alpha}_{t,s_j}^a$ and the merged site/treatment tilting parameter $\boldsymbol{\gamma}_{s_j,a}$. The calibrated losses estimate these nuisance parameters and make the final source assisted estimator insensitive to first order errors. This follows the calibrated estimation approach of [Tan \(2020\)](#): the first order conditions are chosen to match the orthogonality moments of the target estimator.

Let $\boldsymbol{\alpha}_{t,s_j}^{1,*}$ and $\boldsymbol{\gamma}_{s_j,1}^*$ denote the population targets (probability limits / pseudo-true parameters)

associated with the nuisance estimation step; under correct specification of the corresponding nuisance model, these coincide with the true parameters. Let $\mu_{t,s_j}^{1,*}$ be the source-assisted functional obtained by evaluating the pairwise estimator at these population targets. The asymptotic error of $\hat{\mu}_{t,s_j}^1$ can be decomposed into terms related to the estimation errors of the parameters α and γ :

$$\hat{\mu}_{t,s_j}^1 - \mu_{t,s_j}^{1,*} = (\hat{\alpha}_{t,s_j}^1 - \alpha_{t,s_j}^{1,*})^T U_\alpha + (\hat{\gamma}_{s_j,1} - \gamma_{s_j,1}^*)^T U_\gamma + \text{h.o.t.} \quad (4)$$

The two vectors multiplying the nuisance errors are

$$\begin{aligned} U_\alpha &= \tilde{E}_t \left[\nabla_\alpha g(\phi(\mathbf{X}); \alpha_{t,s_j}^{1,*}) \right] - \tilde{E}_{s_j} \left[\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \gamma_{s_j,1}^*}} \nabla_\alpha g(\phi(\mathbf{X}); \alpha_{t,s_j}^{1,*}) \right], \\ U_\gamma &= \tilde{E}_{s_j} \left[-\frac{I(A=1)}{e^{\phi(\mathbf{X})^T \gamma_{s_j,1}^*}} \phi(\mathbf{X}) \{Y - g(\phi(\mathbf{X}); \alpha_{t,s_j}^{1,*})\} \right]. \end{aligned} \quad (5)$$

The first moment balances the weighted source treatment arm against the target average of the outcome regression derivative. The second makes the weighted source residuals orthogonal to the working features. When these moments are small at the fitted nuisances, the first order terms in [Eq. \(4\)](#) vanish and the remaining nuisance bias is second order.

We define the nuisance losses from these moments. The γ loss is calibrated against the outcome model, so its first order condition enforces $U_\alpha = \mathbf{0}$. The α loss is calibrated against the weighting model, so its first order condition enforces $U_\gamma = \mathbf{0}$.

Let $\hat{\alpha}_{\text{init}}$ and $\hat{\gamma}_{\text{init}}$ denote initial plug-in estimates of the outcome and site/treatment parameters, respectively. To ensure numerical stability and satisfy the positivity regularity conditions used in the theory, define the truncation function $\mathcal{T}(x) := \text{sign}(x) \min\{|x|, M_\tau\}$, where M_τ may depend on sample size and is chosen sufficiently large so truncation is asymptotically inactive at the population limit. The subscript recalls the TATE parameter τ defined above, and helps distinguish this numerical threshold from the treatment level. In the finite-sample implementation, truncation is used both when fitting the calibrated nuisances and when evaluating the final source correction and its influence-function variance.

More precisely, a weight displayed below as $I(A = a) \exp\{-\phi(\mathbf{X})^T \boldsymbol{\gamma}\}$ is implemented as $I(A = a) \exp[-\mathcal{T}_{M_{\tau, \text{inf}}} \{\phi(\mathbf{X})^T \boldsymbol{\gamma}\}]$; by default, $M_{\tau, \text{inf}} = M_{\tau} = 5$. The asymptotic results allow these radii to depend on sample size and assume that truncation is inactive with probability sufficiently close to one, so the implemented and untruncated estimating equations differ by $o_p(N^{-1/2})$. If clipping remains active with nonnegligible probability, the implementation instead corresponds to a clipped estimating equation, and sensitivity to both radii should be reported. For exponential-family GLMs with canonical link, write the negative log-likelihood as $-Y\theta + \Psi(\theta)$, where $\theta = \phi(\mathbf{X})^T \boldsymbol{\alpha}$ is the linear predictor, $g(\theta) = \Psi'(\theta)$ is the conditional mean, and $g'(\theta)$ denotes the derivative with respect to θ . Let $\lambda_{\gamma} \geq 0$ and $\lambda_{\alpha} \geq 0$ denote the ℓ_1 regularization parameters for the site/treatment and outcome calibrated losses, respectively.

The calibrated exponential-tilting loss for $\boldsymbol{\gamma}_{s_j, 1}$ is

$$\begin{aligned} \ell_{\gamma, \text{cal}}(\boldsymbol{\gamma}_{s_j, 1}; \hat{\boldsymbol{\alpha}}_{\text{init}}) &= \tilde{E}_t \left[\nabla_{\boldsymbol{\alpha}} g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{\text{init}})^T \boldsymbol{\gamma}_{s_j, 1} \right] \\ &\quad + \tilde{E}_{s_j} \left[I(A = 1) e^{-\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j, 1}} g'(\mathcal{T}(\phi(\mathbf{X})^T \hat{\boldsymbol{\alpha}}_{\text{init}})) \right] \\ &\quad + \lambda_{\gamma} \|\boldsymbol{\gamma}_{s_j, 1}\|_1. \end{aligned} \tag{6}$$

Symmetrically, the calibrated weighted GLM loss for $\boldsymbol{\alpha}_{t, s_j}^1$ is

$$\begin{aligned} \ell_{\alpha, \text{cal}}(\boldsymbol{\alpha}_{t, s_j}^1; \hat{\boldsymbol{\gamma}}_{\text{init}}) &= \tilde{E}_{s_j} \left[\frac{I(A = 1)}{e^{\mathcal{T}(\phi(\mathbf{X})^T \hat{\boldsymbol{\gamma}}_{\text{init}})}} \left\{ -Y \cdot (\phi(\mathbf{X})^T \boldsymbol{\alpha}_{t, s_j}^1) + \Psi(\phi(\mathbf{X})^T \boldsymbol{\alpha}_{t, s_j}^1) \right\} \right] \\ &\quad + \lambda_{\alpha} \|\boldsymbol{\alpha}_{t, s_j}^1\|_1. \end{aligned} \tag{7}$$

The key check is the same one used in calibrated estimation. Setting aside the ℓ_1 subdifferential, the first-order condition of [Eq. \(6\)](#) is the sample analogue of $\mathbf{U}_{\alpha} = \mathbf{0}$ because a canonical-link GLM satisfies $\nabla_{\boldsymbol{\alpha}} g(\phi(\mathbf{X}); \boldsymbol{\alpha}) = g'(\phi(\mathbf{X})^T \boldsymbol{\alpha}) \phi(\mathbf{X})$. Similarly, the first-order condition of [Eq. \(7\)](#) is the sample analogue of $\mathbf{U}_{\gamma} = \mathbf{0}$. Thus the calibrated losses make the nuisance scores in [Eq. \(4\)](#) vanish at the population level, leaving a second-order error of order $O_p(\|\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*\| \cdot \|\hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma}^*\|)$ under standard high-dimensional rates ([Bühlmann & Van De Geer 2011](#), [Negahban et al. 2012](#)); see the Supplemental Material for details.

In practice we first obtain initial estimates to supply the plug-ins. Let ℓ_{GLM} be the standard loss for the chosen GLM, for example squared error for linear outcomes or log-loss for binary outcomes. The initial outcome model and the initial site/treatment tilting model are obtained from

$$\begin{aligned}\ell_{\alpha, \text{init}}(\boldsymbol{\alpha}_{t, s_j}^1) &= \tilde{E}_{s_j} \left[I(A=1) \ell_{\text{GLM}} \{Y, g(\phi(\mathbf{X}); \boldsymbol{\alpha}_{t, s_j}^1)\} \right] + \lambda_{\alpha} \|\boldsymbol{\alpha}_{t, s_j}^1\|_1, \\ \ell_{\gamma, \text{init}}(\boldsymbol{\gamma}_{s_j, 1}) &= \{\tilde{E}_t[\phi(\mathbf{X})]\}^T \boldsymbol{\gamma}_{s_j, 1} + \tilde{E}_{s_j} \left[I(A=1) e^{-\phi(\mathbf{X})^T \boldsymbol{\gamma}_{s_j, 1}} \right] + \lambda_{\gamma} \|\boldsymbol{\gamma}_{s_j, 1}\|_1.\end{aligned}\tag{8}$$

Using these initial estimates, the source site performs calibrated updates in a two-level cross-fitted way, combining the general cross-fitting idea from double/debiased machine learning with the SMMAL-style two-level calibration logic used in semi-supervised ATE estimation (Chernozhukov et al. 2018, Hou et al. 2025). For an outer evaluation fold k_1 , an initial nuisance plug-in is trained for each secondary fold $k_2 \neq k_1$ using data excluding both k_1 and k_2 . The final calibrated nuisance estimator for fold k_1 is then obtained by minimizing the average of the fold-specific calibrated losses over all $k_2 \neq k_1$: the k_2 loss contribution is evaluated on fold k_2 and uses the initial plug-in fitted on the remaining training folds. Thus calibration uses the full outer-training sample $\mathcal{D} \setminus \mathcal{D}_{k_1}$ through $K_f - 1$ fold-specific loss contributions, while the final evaluation fold $\mathcal{D}_{k_1}$ remains untouched. The plug-in order is intentional in the two-level cross-fitting construction and does not affect first-order asymptotics under the alignment condition in the Supplemental Material.

3.2 Influence Functions and Adaptive Aggregation

Following FACE, FACE-HD aggregates the TATE contrast itself. Let $\hat{\tau}_{ot} = \hat{\mu}_{ot}^1 - \hat{\mu}_{ot}^0$ be the target-only estimator and $\hat{\tau}_{t, s_j} = \hat{\mu}_{t, s_j}^1 - \hat{\mu}_{t, s_j}^0$ the source-assisted estimator from site s_j . We use one common source-weight vector for both treatment arms:

$$\hat{\tau}_{\text{agg}}(\boldsymbol{\eta}) = \hat{\tau}_{ot} + \sum_{j=1}^K \eta_j (\hat{\tau}_{t, s_j} - \hat{\tau}_{ot}) = \left(1 - \sum_{j=1}^K \eta_j\right) \hat{\tau}_{ot} + \sum_{j=1}^K \eta_j \hat{\tau}_{t, s_j}.\tag{9}$$

Thus $\boldsymbol{\eta} = \mathbf{0}$ is exactly the target-only TATE estimator. Unlike separate arm-wise aggregation, [Eq. \(9\)](#) optimizes the variance of the scientific estimand and therefore includes the treated-control covariance.

To construct the variance objective, first form the arm-specific cross-fitted components, for $a \in \{0, 1\}$,

$$\begin{aligned}\hat{\varphi}_{ot,i}^a &= \frac{I(A_i = a)}{\hat{\pi}_{ot,a}(\mathbf{X}_i)} \{Y_i - \hat{g}_{ot}^a(\mathbf{X}_i)\} + \hat{g}_{ot}^a(\mathbf{X}_i) - \hat{\mu}_{ot}^a, \quad i \in \mathcal{D}_t, \\ \hat{\zeta}_{t,s_j,i}^a &= \hat{g}_{t,s_j}^a(\mathbf{X}_i) - \tilde{E}_t\{\hat{g}_{t,s_j}^a(\mathbf{X})\}, \quad i \in \mathcal{D}_t, \\ \hat{\zeta}_{t,s_j,i}^a &= w(\mathbf{X}_i; \hat{\gamma}_{s_j,a})\{Y_i - \hat{g}_{t,s_j}^a(\mathbf{X}_i)\} - \hat{\delta}_{t,s_j}^a, \quad i \in \mathcal{D}_{s_j},\end{aligned}$$

where $\hat{\pi}_{ot,a}(\mathbf{X}) = a\hat{\pi}_{ot}(\mathbf{X}) + (1-a)\{1 - \hat{\pi}_{ot}(\mathbf{X})\}$ and $\hat{g}_{t,s_j}^a(\mathbf{X}) = g\{\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s_j}^a\}$. Crucially, FACE-HD takes the treatment contrast *before* calculating any variance or covariance:

$$\hat{\varphi}_{ot,i}^\tau = \hat{\varphi}_{ot,i}^1 - \hat{\varphi}_{ot,i}^0, \quad \hat{\zeta}_{t,s_j,i}^\tau = \hat{\zeta}_{t,s_j,i}^1 - \hat{\zeta}_{t,s_j,i}^0, \quad \hat{\xi}_{t,s_j,i}^\tau = \hat{\xi}_{t,s_j,i}^1 - \hat{\xi}_{t,s_j,i}^0.$$

For example, $\widehat{\text{Var}}_t(\hat{\varphi}_{ot}^\tau) = \widehat{\text{Var}}_t(\hat{\varphi}_{ot}^1) + \widehat{\text{Var}}_t(\hat{\varphi}_{ot}^0) - 2\widehat{\text{Cov}}_t(\hat{\varphi}_{ot}^1, \hat{\varphi}_{ot}^0)$, so the cross-arm covariance is retained.

For any site r , write $\widehat{\text{Cov}}_r(U, V)$ for the within-site empirical covariance. Define

$$\begin{aligned}\hat{V}_{ot}^\tau &= \widehat{\text{Var}}_t(\hat{\varphi}_{ot}^\tau), \quad \hat{V}_{t,s_j}^\tau = \widehat{\text{Var}}_t(\hat{\zeta}_{t,s_j}^\tau), \quad \hat{V}_{s_j}^\tau = \widehat{\text{Var}}_{s_j}(\hat{\xi}_{t,s_j}^\tau), \\ \hat{C}_{ot,s_j}^\tau &= \widehat{\text{Cov}}_t(\hat{\varphi}_{ot}^\tau, \hat{\zeta}_{t,s_j}^\tau), \quad \hat{C}_{t,s_j,s_k}^\tau = \widehat{\text{Cov}}_t(\hat{\zeta}_{t,s_j}^\tau, \hat{\zeta}_{t,s_k}^\tau).\end{aligned}$$

The cross-source term is required because every source-assisted TATE reuses the same target observations, as in FACE ([Han et al. 2025](#)).

In finite samples, the plug-in variance/covariance components (especially the shared-target covariance terms) can yield a quadratic objective in [Eq. \(10\)](#) that is only approximately convex due to estimation noise. For numerical stability, the estimated covariance inputs may be symmetrized and minimally ridge-regularized when needed to keep the weight optimization well posed; this ridge vanishes asymptotically under the regularity conditions in the Supplemental Material.

The aggregation weights $\hat{\boldsymbol{\eta}} = (\hat{\eta}_1, \dots, \hat{\eta}_K)$ minimize an estimated asymptotic variance plus a source-selection penalty over $\mathbb{R}^K$, matching the implementation. Under the positive-definite population curvature condition below, the variance objective is coercive and has a finite unique minimizer. The minimally ridged empirical quadratic is likewise coercive; the resulting weights are $O_p(1)$, so no arbitrary finite-sample radius C_η is introduced. The objective is

$$\begin{aligned} \min_{\boldsymbol{\eta} \in \mathbb{R}^K} \quad & N_{\text{all}} \left[\frac{1}{N_t} \left(1 - \sum_{j=1}^K \eta_j \right)^2 \hat{V}_{ot}^\tau + \sum_{j=1}^K \frac{\eta_j^2}{N_t} \hat{V}_{t,s_j}^\tau + \sum_{j=1}^K \frac{\eta_j^2}{N_{s_j}} \hat{V}_{s_j}^\tau \right. \\ & + \frac{2}{N_t} \sum_{j=1}^K \eta_j \left(1 - \sum_{k=1}^K \eta_k \right) \hat{C}_{ot,s_j}^\tau + \frac{2}{N_t} \sum_{j < k} \eta_j \eta_k \hat{C}_{t,s_j,s_k}^\tau \\ & \left. + \sum_{j=1}^K (\lambda \hat{t}_j^\tau - 1)_+ |\eta_j| \right] \end{aligned} \quad (10)$$

Here $(x)_+ = \max(x, 0)$ and $\text{disc}_j^\tau := \hat{\tau}_{ot} - \hat{\tau}_{t,s_j}$ is the TATE discrepancy. Its Wald statistic is

$$\hat{t}_j^\tau = \frac{|\hat{\tau}_{ot} - \hat{\tau}_{t,s_j}|}{\widehat{\text{SE}}(\text{disc}_j^\tau)}, \quad \widehat{\text{SE}}(\text{disc}_j^\tau)^2 = \frac{\hat{V}_{ot}^\tau}{N_t} + \frac{\hat{V}_{t,s_j}^\tau}{N_t} + \frac{\hat{V}_{s_j}^\tau}{N_{s_j}} - \frac{2 \hat{C}_{ot,s_j}^\tau}{N_t} = \widehat{\text{Var}}(\hat{\tau}_{t,s_j} - \hat{\tau}_{ot}).$$

The ℓ_1 coefficient on $|\eta_j|$ is zero when $\hat{t}_j^\tau \leq 1/\lambda$ and positive otherwise, so $1/\lambda$ is a TATE-level Wald penalty-activation threshold. Crossing this threshold activates shrinkage but does not, by itself, guarantee an exactly zero finite-sample weight.

Because $\hat{t}_j^\tau$ is self-normalized, the decision to activate a source's penalty at the cutoff $1/\lambda$ is invariant to a common rescaling of the outcome. The full finite-sample minimizer need not be invariant to such a rescaling, because the quadratic variance and the dimensionless penalty then change relative scale; throughout the simulations and RHC analysis the binary outcome has its natural probability scale. A TATE-incompatible source with separation $b_j^\tau > 0$ has $\hat{t}_j^\tau = \Theta_p(\sqrt{N_{\text{all}}} b_j^\tau)$, whereas a TATE-compatible source has $\hat{t}_j^\tau = O_p(1)$.

Remark 2 (Finite-sample Wald cutoff). The asymptotic oracle-selection proof uses a growing Wald penalty-activation threshold $1/\lambda_N$. The production implementation uses $c = 1/\lambda = 1$: source s_j is unpenalized when $\hat{t}_j^\tau \leq 1$ and shrunk toward zero otherwise. This cutoff was locked by the coverage-blind pilot rule described in [Section 5](#), using simulation seeds disjoint

from the production replicates. Sensitivity analyses use the full pre-specified cutoff grid; within an analyzed data set, the cutoff remains fixed and the weights are estimated using only the outer-training sample. Accordingly, the oracle-selection and normal-limit theorem below applies to the rate-indexed sequence λ_N , not automatically to the reported fixed cutoff $c = 1$. Coverage for that finite-sample implementation is assessed by the Monte Carlo and cutoff-sensitivity analyses; extending the same oracle limit to a fixed cutoff would require an additional argument.

Remark 3 (Relation to the original FACE objective). The original FACE adaptive-aggregation objective (Han et al. 2025, Equation (11)) uses the raw squared discrepancy as the adaptive penalty factor. In the present notation, its direct FACE-HD analogue would replace the last term in Eq. (10) by

$$\lambda_{\text{sq},N} \sum_{j=1}^K |\eta_j| \left(\hat{\tau}_{ot} - \hat{\tau}_{t,s_j} \right)^2,$$

where $\lambda_{\text{sq},N}$ is a diverging tuning parameter. We instead use the self-normalized truncated-Wald version in Eq. (10), following FACE (Han et al. 2025, Remark 9), because it puts source discrepancies on a standard-error scale. The Supplemental Material gives the corresponding squared-discrepancy objective and the stronger tuning/separation conditions under which it gives the same oracle selection conclusion.

For a cross-fitted observation, let $\hat{\Phi}_i^\tau$ denote the uncentered, site-scaled pseudo-value induced by Eq. (9): target observations carry the target-only and target-side regression components, and observations at source s_j carry $\hat{\eta}_j(\hat{\xi}_{t,s_j,i}^\tau + \hat{\delta}_{t,s_j}^\tau)$, with factors N_{all}/N_t and N_{all}/N_{s_j} , respectively. The implementation computes the point estimate and variance from these same pseudo-values:

$$\hat{\tau}_{\text{agg}} = N_{\text{all}}^{-1} \sum_r \sum_{i \in \mathcal{D}_r} \hat{\Phi}_i^\tau, \quad \hat{V}_\tau = N_{\text{all}}^{-1} \sum_r \sum_{i \in \mathcal{D}_r} \left(\hat{\Phi}_i^\tau - \overline{\hat{\Phi}}_r^\tau \right)^2. \quad (11)$$

Centering is within site, as required by the fixed-site multisample asymptotic regime. Each

site need only return its sample size, pseudo-value sum, and centered sum of squares. The reported standard error is $\{\widehat{V}_\tau/N_{\text{all}}\}^{1/2}$.

3.3 Federated Algorithm

FACE-HD can be implemented as a summary-exchange algorithm. Each site keeps its individual-level records locally, while the target and source sites pass only fitted nuisance parameters, empirical moments, sample sizes, and variance summaries (McMahan et al. 2017, Kairouz et al. 2021, Han et al. 2025). This data-locality property is not, by itself, a formal privacy guarantee; differential privacy, secure aggregation, and disclosure-risk control are outside the present scope. The two-level cross-fitting separates the outer evaluation fold from the secondary calibration folds. We consider two communication patterns that differ only in how each source’s nuisances are initialized. The default one-round algorithm uses the target-only outcome model as the shared initialization, and each source returns its calibrated summaries in a single round. A two-round version instead uses a first round for source-specific initialization. In an archived continuous-outcome communication diagnostic, the preceding arm-wise TATE implementations of the two variants track one another closely when the sources are mean-compatible for the TATE, even though their conditional outcome regressions differ in shape from the target’s (figure 7). This provides descriptive support for the communication-light initialization, but the modest archived Monte Carlo size is not a formal equivalence study and the comparison is not relabeled as a common-weight direct-TATE experiment. Nuisance fitting is arm-specific, but the current estimator contrasts the two arm-specific influence components before aggregation and estimates only one TATE-level weight vector. Below we display the treated-arm nuisance step, with the control arm obtained by replacing $I(A = 1)$ by $I(A = 0)$. The expanded two-round fold bookkeeping and the linear-outcome specialization are given in Section S1 and Section S1.3.

The following paragraphs describe the one-round algorithm, first for a single training/evaluation split and then for the fold-level implementation used for inference. The expanded two-round variant is given in [Section S1.2](#).

Local inputs. At the beginning, the target site holds $\mathcal{D}_t$ and each source site s_j holds $\mathcal{D}_{s_j}$. All sites agree on the working feature map $\phi(\mathbf{X})$, the GLM response function g , the fold partition used for two-level cross-fitting, candidate grids for the nuisance penalties $(\lambda_\gamma, \lambda_\alpha)$, and the fixed TATE-level Wald threshold $1/\lambda$. The target fits the target-only nuisance models for both arms and constructs $\hat{\tau}_{ot}$ and $\hat{\varphi}_{ot}^\tau$.

One communication round: target initialization. The target first fits a target-only initial treated-outcome model on its treated observations:

$$\hat{\alpha}_{ot}^{1,(0)} \in \arg \min_{\alpha} \left\{ \tilde{E}_t[I(A=1)\ell_{\text{GLM}}\{Y, g(\phi(\mathbf{X}); \alpha)\}] + \lambda_\alpha \|\alpha\|_1 \right\}.$$

It then evaluates the target-side summaries $\tilde{E}_t[\nabla_{\alpha} g\{\phi(\mathbf{X}); \hat{\alpha}_{ot}^{1,(0)}\}]$ and $\tilde{E}_t\{\phi(\mathbf{X})\}$, and broadcasts $\hat{\alpha}_{ot}^{1,(0)}$ together with these summaries to all source sites. These quantities are low-dimensional or p -dimensional summaries determined by the chosen feature map, rather than individual-level observations.

Source calibration and return summaries. After receiving the target summaries, each source solves the two calibrated nuisance problems locally. First, it computes an initial merged site/treatment tilting fit using the initialization loss in [Eq. \(8\)](#) with the received target covariate mean. It then estimates the calibrated tilting parameter $\hat{\gamma}_{s_j,1}$ using the calibrated exponential-tilting loss in [Eq. \(6\)](#), with the target-only outcome fit $\hat{\alpha}_{ot}^{1,(0)}$ as the shared initial outcome plug-in. Second, using the initial tilting fit as the plug-in weight model, the source estimates the calibrated outcome parameter $\hat{\alpha}_{t,s_j}^1$ from [Eq. \(7\)](#). This ordering is the finite-sample realization of the cross-calibration logic: the γ update targets covariate/score balance against the target, and the α update targets a weighted residual equation at the source.

The source then forms the source-side correction and variance summaries

$$\begin{aligned}\hat{\delta}_{t,s_j}^1 &= \tilde{E}_{s_j} \left[w(\mathbf{X}; \hat{\gamma}_{s_j,1}) \{Y - g(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^1)\} \right], \\ \hat{V}_{s_j} &= N_{s_j}^{-1} \sum_{i \in \mathcal{D}_{s_j}} \left[w(\mathbf{X}_i; \hat{\gamma}_{s_j,1}) \{Y_i - g(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^1)\} - \hat{\delta}_{t,s_j}^1 \right]^2.\end{aligned}$$

The source repeats this nuisance step for $a = 0$, forms the contrasts $\hat{\delta}_{t,s_j}^\tau = \hat{\delta}_{t,s_j}^1 - \hat{\delta}_{t,s_j}^0$ and $\hat{\xi}_{t,s_j}^\tau = \hat{\xi}_{t,s_j}^1 - \hat{\xi}_{t,s_j}^0$, and sends the two outcome-parameter vectors, $\hat{\delta}_{t,s_j}^\tau$, $\hat{V}_{s_j}^\tau$, and N_{s_j} . The site/treatment parameters can remain local because the target does not need to reconstruct individual source residuals once the contrast summaries have been computed.

Target-side assembly. With the returned source summaries, the target builds $\hat{\tau}_{t,s_j} = \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$ for every source. It then contrasts the target-side regression components observation by observation and computes $\hat{V}_{t,s_j}^\tau$, $\hat{C}_{ot,s_j}^\tau$, and $\hat{C}_{t,s_j,s_k}^\tau$ locally. These quantities, together with $\hat{V}_{ot}^\tau$ and the source-side contrast variances, are exactly the inputs to [Eq. \(10\)](#). The target selects one $\hat{\eta}$, forms [Eq. \(9\)](#), and reports the variance in [Eq. \(11\)](#).

Two-level cross-fitted construction. The description above gives one training/evaluation split. The estimator analyzed in the theory uses a fold-specific, two-level cross-fitted construction following the SMMAL-style calibration construction of [Hou et al. \(2025\)](#). For an outer evaluation fold k_1 , all nuisance models and aggregation weights are learned only on $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$. Inside $\mathcal{T}_{k_1}$, each secondary fold $k_2 \neq k_1$ supplies a calibration fold. In the one-round procedure, the target trains the fold-specific initial outcome plug-in and target summaries on data excluding both k_1 and k_2 , broadcasts those plug-ins and summaries to the sources, and each source contributes the corresponding calibrated loss on its own secondary fold. The calibrated nuisance estimator for evaluating $\mathcal{D}_{k_1}$ is obtained by minimizing the average of these secondary-fold calibrated losses. The aggregation penalty parameter λ is fixed at the Wald screening threshold described in [Remark 2](#). The variance/discrepancy summaries and the final weights $\hat{\eta}^{(-k_1)}$ are computed only inside $\mathcal{T}_{k_1}$, and the weights are applied only to observations in the held-out outer fold. [Section S1](#) gives the fold-

level bookkeeping and the two-round variant, and [Section S1.3](#) gives the linear-outcome simplification.

3.4 Cross-Validation and Tuning Parameters

The framework involves two types of regularization parameters. The nuisance penalties $(\lambda_\gamma, \lambda_\alpha)$ are selected by cross-validation, while the aggregation cutoff $c = 1/\lambda$ controls the finite-sample Wald screening rule. The production simulations use the minimum-validation-loss nuisance rule (`lambda.min`). In the RHC application, a pre-specified numerical audit first evaluates that rule and switches the complete analysis to the stronger one-standard-error rule (`lambda.1se`) if any selected nuisance fit reaches the numerical coefficient boundary or otherwise fails the strict fit audit; the failed fit is retained as a named sensitivity result. In the main analysis we fix the aggregation cutoff at the value locked by the disjoint, coverage-blind simulation pilot described in [Section 5](#); the remaining candidate values are retained as a sensitivity analysis.

For each nuisance model (parameters $\boldsymbol{\gamma}$ and $\boldsymbol{\alpha}$), the regularization parameters $(\lambda_\gamma, \lambda_\alpha)$ are selected via K_f -fold cross-validation, where K_f is a fixed constant. The criterion for selection is the minimization of the corresponding calibrated loss function ([Eq. \(6\)](#) or [Eq. \(7\)](#)) on the validation folds. This is performed locally at each site during the federated step where the model is estimated.

The density-ratio path also includes a finite-sample coercivity safeguard. Let $\boldsymbol{m}$ denote the target-side linear moment in the relevant exponential-tilting loss, including its intercept coordinate m_0 . If non-intercept feature j is constant at c_j within a source treatment arm used for fitting, then the direction $(-c_j, \boldsymbol{e}_j)$ leaves every source linear predictor unchanged. Along this direction the unpenalized linear part has slope $m_j - c_j m_0$; because the intercept is unpenalized, a candidate penalty below $|m_j - c_j m_0|$ can make the empirical loss noncoercive.

We therefore lower-bound the local λ_γ path by the largest such absolute gap, plus a negligible strict numerical margin, and record every activation. This safeguard prevents a sparse fold from driving a coefficient to the solver boundary; it does not manufacture covariate support or replace the overlap assumption. A whole site/arm with a persistent support failure is flagged for exclusion from the primary analysis and retained only as a sensitivity analysis. Under the stated positivity conditions, the probability of an empirically empty fixed-probability cell vanishes, so this floor is asymptotically inactive.

The parameter λ in the aggregation objective [Eq. \(10\)](#) controls the Wald penalty-activation cutoff $c = 1/\lambda$: source s_j is unpenalized when its TATE discrepancy satisfies $\hat{t}_j^\tau \leq c$, and shrunk toward zero when $\hat{t}_j^\tau > c$. A larger cutoff leaves more sources unpenalized and can reduce variance when source-assisted TATE estimators are compatible with the target, but it can also increase bias from incompatible sources. A smaller cutoff penalizes more sources and generally moves the estimator toward target-only TATE; exact zero weights are an optimization outcome, not a finite-sample consequence of crossing the cutoff alone. The oracle-selection theorem uses an asymptotic sequence $\lambda_N \rightarrow 0$ with $\lambda_N \sqrt{N_{\text{all}}} \rightarrow \infty$ under fixed TATE separation; equivalently, the cutoff $1/\lambda_N$ grows without bound but slower than $\sqrt{N_{\text{all}}}$.

Because the aggregation penalty acts through the self-normalized statistic $\hat{t}_j^\tau$ in [Eq. \(10\)](#), the cutoff is scale-free and does not require a problem-scale-dependent grid. We use the locked default $c = 1$ described in [Remark 2](#). For a new application, selecting c from a grid using the observed application data would instead be part of weight estimation and must use only training-fold summaries.

For an outer evaluation fold k_1 , let $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$ be the data available for nuisance fitting and weight estimation. All TATE variance, covariance, and discrepancy summaries entering [Eq. \(10\)](#), and hence $\hat{t}_j^\tau$, are computed on $\mathcal{T}_{k_1}$; the sample sizes used to studentize that

discrepancy are likewise the site-specific sizes in $\mathcal{T}_{k_1}$. Under the balanced-fold convention, replacing every full-site size in $N_{\text{all}}\widehat{\text{Var}}(\boldsymbol{\eta})$ by its common outer-training fraction leaves that normalized variance objective unchanged, while it gives the correct training-sample Wald standard error. The common weights $\widehat{\boldsymbol{\eta}}^{(-k_1)}$ are then applied only to the outer evaluation fold $\mathcal{D}_{k_1}$. No information from $\mathcal{D}_{k_1}$ is used to choose the cutoff. The optional FACE-style validation criterion $Q_{\text{val},m}(\lambda)$ that would be used if one tuned λ over a grid is recorded in the Supplemental Material for completeness.

4 Theory

This section states the main theoretical guarantees in the same order as the method is built: first the conditions under which each target-source estimator is valid, then the aggregation result that protects against negative transfer, and finally the implication for the TATE. Detailed proofs, including empirical-process and selection arguments, are provided in the Supplemental Material. Throughout this section K and K_f are fixed, $N_{\text{all}} \rightarrow \infty$, the site fractions N_t/N_{all} and N_{s_j}/N_{all} stay bounded away from zero, and the covariate dimension p may grow with sample size. We first record the assumptions needed to read the theorem statements; the Supplemental Material gives sharper versions and proof-level empirical-process conditions.

Assumption 2 (Sampling, moments, and cross-fitting). For fixed constants $M > 1$ and $\delta > 0$, assume the following conditions hold for the target site and the candidate source sites:

- (a) (*Fixed site and fold regime*) The number of source sites K and the number of cross-fitting folds K_f are fixed.
- (b) (*Independent multisite sampling*) Target and source samples are mutually independent,

observations are i.i.d. within each site, and the site fractions satisfy $N_t/N_{\text{all}} \geq 1/M$ and $N_{s_j}/N_{\text{all}} \geq 1/M$ asymptotically for every fixed source s_j .

- (c) (*Working-feature tails*) The feature map $\phi(\mathbf{X})$ is fixed across sites, includes an intercept, and satisfies $\max_{1 \leq \ell \leq p} \|\phi_\ell(\mathbf{X})\|_{\psi_2} \leq M$ under each participating site distribution.
- (d) (*Moment and weight regularity*) The exponential tilting weights and the influence-function components used for pairwise estimation and aggregation have uniformly bounded moments of order $2 + \delta$ by M . Products of any two influence-function components used in the aggregation covariance estimators are uniformly integrable.
- (e) (*Fold independence*) All nuisance functions and aggregation weights applied to an evaluation fold are trained using data independent of that fold.

Assumption 3 (High dimensional nuisance estimation). For each informative source s_j and treatment arm a , the calibrated nuisance estimation step satisfies the following conditions.

- (a) (*Sparse nuisance targets*) The calibrated outcome target $\boldsymbol{\alpha}_{t,s_j,a}^*$ and the calibrated site/treatment tilting target $\boldsymbol{\gamma}_{s_j,a}^*$ are sparse. Let $s_{\alpha,j}$ and $s_{\gamma,j}$ denote their sparsities, and let $s_{\text{tot},j} = s_{\alpha,j} + s_{\gamma,j}$.
- (b) (*Calibrated target uniqueness*) The population calibrated losses have unique local targets; under correct specification of the corresponding nuisance model, these targets coincide with the true outcome regression or the true balancing weight.
- (c) (*Curvature and initialization alignment*) The population calibrated losses satisfy restricted strong convexity on the relevant sparse cones, and the two-level cross-fitting construction aligns the population limits of the initial nuisance estimators with the final calibrated targets.

(d) (*Rate of estimation*) With $N_{t,s_j} = N_t + N_{s_j}$, the calibrated ℓ_1 nuisance estimators obey

$$\|\hat{\alpha} - \alpha^*\|_2 + \|\hat{\gamma} - \gamma^*\|_2 = O_p\left(\sqrt{s_{\text{tot},j} \log(p)/N_{t,s_j}}\right),$$

with analogous ℓ_1 rates of order $s_{\text{tot},j} \sqrt{\log(p)/N_{t,s_j}}$.

(e) (*Remainder scaling*) The sparsity scaling required for the first order theory is

$$s_{\text{tot},j}^2 \log^2(p)/N_{t,s_j} \rightarrow 0 \text{ for each informative source.}$$

4.1 Pairwise Source-Assisted Estimation

For a fixed source site s_j and treatment arm $a \in \{0, 1\}$, the calibrated source-assisted estimator has the same first-order structure as an orthogonal doubly robust estimator. The calibration losses are designed so that the derivative of the estimating equation with respect to one nuisance component vanishes when the other nuisance component is evaluated at its calibrated target. As a result, the first-order nuisance error cancels and the remainder is second order, of size

$$O_p(\|\hat{\alpha} - \alpha^*\|_2 \|\hat{\gamma} - \gamma^*\|_2) = o_p(N_{t,s_j}^{-1/2})$$

under [Assumption 3\(d\)](#)–[Assumption 3\(e\)](#). In the theorem below, [Assumption 1](#) identifies μ^a for the target and informative sources; [Assumption 2\(a\)](#)–[Assumption 2\(b\)](#) gives the multisite CLT regime; [Assumption 2\(c\)](#)–[Assumption 2\(d\)](#) controls the influence function expansion and variance estimation; [Assumption 2\(e\)](#) justifies conditioning on training folds; and [Assumption 3\(a\)](#)–[Assumption 3\(e\)](#) supplies the sparse calibrated nuisance rates that make the orthogonal remainder negligible.

Theorem 1 (Pairwise doubly robust asymptotic normality). *Under [Assumptions 1 to 3](#), suppose that for source s_j at least one of the two nuisance models is correctly specified for arm a : either the calibrated outcome regression equals the target-site conditional mean, or the exponential-tilting weight satisfies the target-source balancing identity for the relevant*

function class. If $s_{\text{tot},j}^2 \log^2(p)/N_{t,s_j} \rightarrow 0$ and the asymptotic variance V_{t,s_j}^a is positive, then

$$\sqrt{N_{t,s_j}}(\hat{\mu}_{t,s_j}^a - \mu^a) \xrightarrow{d} \mathcal{N}(0, V_{t,s_j}^a).$$

The plug-in variance estimator based on the cross-fitted influence-function components is consistent for V_{t,s_j}^a , the total pairwise asymptotic variance of $\hat{\mu}_{t,s_j}^a$ (not to be confused with the target-side outcome-regression component $\hat{V}_{t,s_j}$ entering the aggregation objective in [Eq. \(10\)](#)).

The theorem gives the pairwise protection that FACE-HD needs before aggregation: a source can be useful even if one nuisance model is misspecified, provided the other nuisance model identifies the target mean and the high-dimensional nuisance rates are fast enough. Applying the same statement to $a = 1$ and $a = 0$ yields the pairwise TATE estimator $\hat{\tau}_{t,s_j} = \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$, whose variance is obtained from the difference of the two arm-specific influence functions. This arm-wise identification route is a sufficient condition for the TATE expansion below. A source whose two arm-specific probability-limit biases cancel can also be TATE-compatible, but only when the contrast itself satisfies the stated first-order linear expansion about τ ; cancellation of probability limits alone is not used as a substitute for that regularity condition.

4.2 Adaptive Aggregation and Source Selection

The aggregation step decides how much each source-assisted TATE estimator should contribute after accounting for both variance and possible bias. Define the TATE-compatible source set

$$\mathcal{J}^{\tau,*} = \{j : \text{plim } \hat{\tau}_{t,s_j} = \tau\}.$$

Let $\boldsymbol{\eta}^{\tau,*}$ minimize the population TATE variance $L_{\tau}^*(\boldsymbol{\eta})$ over weights supported on $\mathcal{J}^{\tau,*}$. This definition is intentionally at the estimand level: a source can be compatible for the TATE

even if equal arm-specific biases cancel, provided its contrast has the compatible-source linear expansion required below.

Assumption 4 (Aggregation separation and penalty). Assume:

- (a) (*Global curvature and weight stability*) The population TATE-variance objective is a coercive quadratic on $\mathbb{R}^K$ with a positive-definite Hessian whose minimum eigenvalue is bounded away from zero. Consequently, its restriction to the oracle subspace supported on $\mathcal{J}^{\tau,*}$ has a finite unique minimizer $\boldsymbol{\eta}^{\tau,*}$; coefficient consistency and the vanishing numerical ridge imply $\max_{k \leq K_f} \|\hat{\boldsymbol{\eta}}^{(-k)}\|_2 = O_p(1)$ for the fixed number of outer folds.
- (b) (*TATE separation*) There is a fixed $b_0^\tau > 0$ such that $|\text{plim } \hat{\tau}_{t,s_j} - \tau| \geq b_0^\tau$ for every $j \notin \mathcal{J}^{\tau,*}$.
- (c) (*Threshold rate*) The aggregation penalty sequence satisfies $\lambda_N \rightarrow 0$ and $\lambda_N \sqrt{N_{\text{all}}} \rightarrow \infty$, equivalently the Wald penalty-activation threshold $1/\lambda_N \rightarrow \infty$ but grows slower than $\sqrt{N_{\text{all}}}$ under the fixed-separation condition above.
- (d) (*Studentization and variance-component consistency*) The TATE variance and covariance components entering [Eq. \(10\)](#) are uniformly consistent on the candidate source set. The asymptotic variance of each studentized TATE source–target discrepancy is bounded away from zero, so $\widehat{\text{SE}}(\text{disc}_j^\tau)$ is well defined with probability tending to one.
- (e) (*Compatible-source contrast expansion*) For every $j \in \mathcal{J}^{\tau,*}$, the source-assisted contrast is asymptotically linear about τ with influence function $\varphi_{t,s_j}^\tau = \varphi_{t,s_j}^1 - \varphi_{t,s_j}^0$ and an $o_p(N_{t,s_j}^{-1/2})$ remainder. This follows from the two arm-wise pairwise theorems when both arms are identified; for a source admitted only through cancellation of arm-specific probability-limit biases, it is an additional TATE-level regularity condition.

The truncated-Wald penalty $(\lambda_N \hat{t}_j^\tau - 1)_+$ uses self-normalized TATE discrepancies. For an incompatible source, $\hat{t}_j^\tau = \Theta_p(\sqrt{N_{\text{all}}} b_j^\tau)$, so the active penalty dominates the $O(1)$ variance

gradient at zero. For a compatible source, $\hat{t}_j^\tau = O_p(1)$ and the penalty is asymptotically inactive.

Theorem 2 (Direct-TATE FACE-HD aggregation). *Suppose [Assumptions 1 to 4](#) hold and $\max_{j \in \mathcal{J}^{\tau,*}} s_{\text{tot},j}^2 \log^2(p)/N_{t,s_j} \rightarrow 0$. Then*

$$\max_{k \leq K_f} \|\hat{\boldsymbol{\eta}}^{(-k)} - \boldsymbol{\eta}^{\tau,*}\|_2 = o_p(1), \quad \hat{\eta}_j^{(-k)} = 0 \quad \text{for all } j \notin \mathcal{J}^{\tau,*} \text{ and all } k \leq K_f$$

with probability tending to one, and

$$\sqrt{N_{\text{all}}}(\hat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}\{0, L_\tau^*(\boldsymbol{\eta}^{\tau,*})\}.$$

Moreover, the site-centered estimator in [Eq. \(11\)](#) satisfies $\hat{V}_\tau \xrightarrow{p} L_\tau^*(\boldsymbol{\eta}^{\tau,*})$.

4.3 Inference for the TATE and Efficiency

The direct-TATE theorem gives the Wald interval

$$\hat{\tau}_{\text{agg}} \pm z_{1-\alpha/2} \sqrt{\hat{V}_\tau / N_{\text{all}}},$$

where $\hat{V}_\tau$ is computed from the same contrast pseudo-values as the point estimator. Because $\boldsymbol{\eta} = \mathbf{0}$ is feasible and equals target-only TATE,

$$L_\tau^*(\boldsymbol{\eta}^{\tau,*}) \leq L_\tau^*(\mathbf{0}) = V_{\tau, \text{target}}.$$

This non-inferiority statement is now a TATE-level result; it does not rely on separately minimizing the two arm-specific variances. It is an asymptotic comparison with target-only TATE, not a claim that adaptive FACE-HD has the smallest finite-sample RMSE among all non-adaptive estimators when every source is known to be compatible.

5 Simulation Study

We conduct simulations with one target site and K source sites to evaluate bias, root mean squared error (RMSE), and 95% confidence-interval coverage under controlled source

heterogeneity and nuisance-model misspecification. The design follows the FACE simulation structure. Each site contains 1,000 observations. The new direct-TATE experiment uses $p = 100$ and $K \in \{2, 4, 8\}$; only the first four covariates have nonzero coefficients. The earlier $p = 50$ arm-wise results are retained in the Supplemental Material as legacy reproducibility results but are not rerun under direct-TATE aggregation.

Covariates are generated independently by coordinate from site-specific skew-normal laws,

$$X_{r\ell} \sim \text{SN}(\kappa, 1, \nu_r), \quad \kappa = 0.125,$$

where $\nu_t = 0$ at the target and the K source values are equally spaced between $0.2/K$ and 0.2 . Treatment is generated at every site from

$$\Pr(A = 1 \mid \mathbf{X}) = \text{expit}\{\mathbf{X}^\top \boldsymbol{\alpha}_1 + (\mathbf{X}^{\circ 2})^\top \boldsymbol{\alpha}_2\},$$

where the first four entries of $\boldsymbol{\alpha}_1$ are equally spaced from 0.5 to -0.5 , the first entry of $\boldsymbol{\alpha}_2$ is -0.5 , and the remaining entries are zero. Before treatment is drawn, the resulting propensity is truncated to $[0.1, 0.9]$. This enforces finite-sample positivity but also means that the untruncated logistic working equation is not literally exact at the boundary.

For the binary outcome used in the reported experiment, define

$$q(\mathbf{X}) = (\mathbf{X} - \kappa \mathbf{1})^\top \boldsymbol{\beta} + (\mathbf{X}^{\circ 2})^\top \boldsymbol{\beta}, \quad g(\mathbf{X}) = \frac{q(\mathbf{X}) - \mathbb{E}_t\{q(\mathbf{X})\}}{\text{sd}_t\{q(\mathbf{X})\}},$$

where the first four entries of $\boldsymbol{\beta}$ are equally spaced from 0.4 to 1.2 . Outcomes satisfy

$$\Pr\{Y(a) = 1 \mid \mathbf{X}, R = r\} = \text{expit}\{g(\mathbf{X}) + \Delta_r a\}.$$

The target and informative sources use $\Delta_r = 1$. In the negative-transfer experiment below, source s_1 instead uses $\Delta_{s_1} = 1 + \rho$. Thus ρ is a source-specific log-odds shift; the reported TATE is the target risk difference, not the log-odds coefficient.

The fitted design matrices vary by configuration. C(1) uses the full quadratic bases $Z = [X, X^{\circ 2}]$ for site/treatment calibration and $W = [X - \kappa, X^{\circ 2}]$ for outcome regression.

C(2) retains the full calibration basis but fits the outcome model using $W = X$. C(3) fits the site/propensity nuisance using $Z = X$ but retains the correctly specified quadratic outcome basis. Because the source covariates are skew-normal, the quadratic exponential-tilting basis in C(1)–C(2) is a close working approximation to, rather than an exact parametric representation of, the source-to-target density ratio; the outcome regression is exactly specified in C(1) and C(3).

The direct-TATE production experiment uses 500 Monte Carlo replicates for each configuration– K – ρ combination. We compare direct-TATE FACE-HD (the one-round cross-fitted estimator) with target-only estimation, sample-size weighting (SS), inverse-variance weighting (IVW), federated doubly robust aggregation (Federated-DR), and pooled doubly robust estimation (Pooled-DR). Every method’s TATE standard error is calculated from within-site paired treated-minus-control influence values, so the cross-arm covariance is retained. FACE-HD and target-only use the corresponding plug-in influence variance. For SS, IVW, Federated-DR, and Pooled-DR, the reported standard error uses 5,000 Rademacher multiplier-bootstrap draws applied to the same paired site-level influence blocks; the analytic influence variance is retained as a diagnostic. In a fixed-block, 10-stream implementation benchmark, increasing the draw count from 2,000 to 5,000 reduced the relative Monte Carlo standard deviation of the bootstrap standard error from 1.86% to 1.16%; the 5,000-draw rule was therefore locked before the production rerun. For Federated-DR and Pooled-DR, normalized exponential-tilting density-ratio weights are clipped to $[0.1, 10]$ as a numerical safeguard; the implementation records the pre-clipping range, total clipped fraction, and largest site-specific clipped fraction in every replicate, and also emits a warning whenever clipping occurs. We also considered a tilted-AIPW baseline, but its unpenalized parametric nuisances yield unstable tilting weights in high dimension, so we omit it from the main comparison.

Before the production run, we locked the aggregation cutoff using a bounded pilot with $C(1)$, $p = 100$, $K = 2$, $\rho \in \{0, 1, 2.5\}$, candidate cutoffs $c \in \{1, 1.5, 2, 2.5, 3\}$, and ten simulation seeds 501–510. These seeds are disjoint from production seeds 1–500. The rule did not use coverage: first retain cutoffs whose $\rho = 0$ RMSE is no more than 1.05 times the best pilot $\rho = 0$ RMSE; among those candidates, minimize the worst RMSE at $\rho \in \{1, 2.5\}$, breaking ties by their mean RMSE and then the smaller cutoff. All five cutoffs passed the $\rho = 0$ tolerance, and the rule selected $c = 1$. Because this pilot is deliberately small, we report the full cutoff sensitivity and Monte Carlo uncertainty rather than interpreting its ranking as a universal optimum.

All reported simulations use five outer cross-fitting folds, the locked aggregation cutoff $c = 1$ (equivalently, $\lambda = 1$), truncation radius $M_\tau = 5$, and 100 candidate values in every initial and calibrated nuisance-penalty path. Monte Carlo replicate b uses random seed b , for $b = 1, \dots, 500$. Within a replicate, deterministic site-, arm-, and fold-specific random streams are used for nuisance cross-validation; hence changing the number of workers changes wall-clock time but not the fitted estimator. The primary nuisance paths use the minimum-validation-loss rule (`lambda.min`) for both target and source fits. The calibrated nuisance and aggregation problems are solved by the package’s C++ proximal coordinate-descent routines. Within each nuisance cross-validation path, a candidate penalty is eligible only when coordinate descent satisfies its complete-coordinate stopping criterion and yields a finite validation loss in every inner fold. Candidate fold fits and penalty values excluded by this rule are recorded separately from final-fit nonconvergence; a replicate fails explicitly if the entire path has no eligible penalty. Each attempted CV candidate has a 5,000-sweep budget, with the last permitted sweep forced to scan all coordinates. Within a fold’s descending penalty path, three consecutive nonconverged candidates trigger a fail-fast rule that marks the remaining, less-regularized tail ineligible; the numbers of skipped fold fits

and excluded candidates are recorded separately. The selected nuisance penalty is then refit with a 10,000-sweep budget. On the locked $p = 100$, $C(3)$, $K = 4$, $\rho = 2.5$ audit task, this rule skipped 17,850 fold fits and reduced elapsed time from 6,559 to 3,516 seconds (1.87-fold), while all 100 audited primary scientific/final-fit fields and all 97 analogous sensitivity fields were identical. The only changed legacy fields were the invalid-candidate counts, because an intentionally skipped fit is represented by an infinite score; every such increase was bounded by its explicit skip count. In the formal MSI arrays, one Slurm job owns one configuration- K -replicate seed and returns all six values of ρ . It fits the $\rho = 0$ reference independently; for positive ρ , a strict same-seed validator permits reuse only of the unchanged target, control-arm, and nondeviated-source components, while the deviated source’s treated-outcome nuisances and the full direct-TATE aggregation are recomputed. This reduces the 27,000 canonical setting-replication rows to 4,500 Slurm jobs without changing their task IDs or statistical content. Source-specific fits are parallelized across K workers and the two arm-specific nuisance pipelines for the reference run concurrently. For $K = 2, 4$, each source worker parallelizes the five nuisance-CV folds over five OpenMP threads, requiring 20 and 40 CPUs. For $K = 8$, two CV threads per source worker require 32 CPUs and give substantially better queue access than an 80-CPU request. BLAS threading is fixed at one thread per worker. Before production, an operational $p = 100$, $C(3)$, $K = 4$, $\rho = 2.5$ audit compared the reused fit with a fully independent fit from the same tested package. All 137 shared non-timing statistical and diagnostic output fields were identical. The grouped run returned all six values of ρ in 3,666 seconds; relative to the 2,866-second independent endpoint audit task, the estimated six-independent-run benchmark is 4.69 times slower. These settings are shared by the simulation driver and the MSI Slurm manifests. Quality control is performed separately for every configuration- K - ρ -method combination. In addition to bias, RMSE, and empirical coverage, we report the empirical Monte Carlo standard deviation, mean estimated standard error, their ratio, the Monte Carlo standard

error of coverage, and the normal-reference coverage implied by the observed bias and standard-error scale. Because all methods use the same generated replicate, we also report paired squared-error differences between direct-TATE FACE-HD and target-only plus every non-adaptive TATE benchmark, together with their Monte Carlo standard errors and the fraction of replicates favoring FACE-HD. We flag incomplete or nonfinite replicates and verify the RMSE moment identity before interpreting a coverage departure. We also record nuisance-fit nonconvergence counts, maximum final-fit iterations, and the final parameter-update-to-stopping-threshold ratio; a setting with any unresolved nuisance optimization is reviewed before its coverage is treated as a property of the estimator. For binary outcomes, the same replicate-level audit records the minimum site–treatment–arm–outcome cell count and the number of such cells below eight, so sparse-class warnings can be separated from estimator behavior. Because the aggregation problem is unconstrained, we additionally record the largest absolute source weight and flag values above 10 for review instead of silently truncating or resetting them. Thus a C(3) coverage flag can be distinguished as primarily bias-driven, standard-error-scale-driven, or simulation noise rather than being diagnosed from a pooled summary alone.

Reproducibility resources. The accompanying FACE-HD repository contains the R package source and test suite, the exact $p = 100$ Monte Carlo manifests, bounded MSI Slurm array and checkpoint drivers, replicate-level aggregation and per-setting diagnostic scripts, the scripted RHC cohort construction, and all figure generators. Each simulation output records its seed, tuning parameters, bootstrap size, runtime, Slurm job/array/partition/node provenance, separate SHA-256 content fingerprints for the tested package installation, simulation workflow, and exact manifest, nuisance-convergence diagnostics, aggregation-weight diagnostics, and inference-clipping diagnostics. Archived arm-wise results and new direct-TATE outputs use distinct filenames, so reproducing the new analysis does not

overwrite the earlier numerical results.

Negative-transfer robustness. To assess robustness to non-transportable sources, we make one source site progressively non-transportable by shifting its conditional treatment effect by ρ on the log-odds scale, with $\rho \in \{0, 0.5, 1, 1.5, 2, 2.5\}$, while the remaining $K - 1$ sources stay informative. [Figure 1](#) reports the archived treated-mean bias, RMSE, 95% coverage, and 95% confidence-interval length for C(1) at $p = 100$; the corresponding archived C(2)–C(3) and $p = 50$ results are reported in [Section S3](#) and labeled separately. At $\rho = 0$, all sites share the conditional outcome mechanism and treatment log-odds shift. Non-adaptive pooling can then have smaller finite-sample RMSE when the modest covariate-shift-induced differences in site-specific marginal effects are outweighed by its variance reduction, because it pays no adaptation cost. As ρ increases, FACE-HD preserves a substantially more stable bias–coverage tradeoff while shortening intervals relative to target-only; the non-adaptive aggregators trade their initial efficiency for increasing negative-transfer bias and coverage loss. The mild coverage dip near $\rho = 0.5$ is the weak-separation regime, where a small source deviation is difficult to distinguish from sampling noise. [Table 1](#) reports the corresponding numbers at $\rho \in \{0, 2.5\}$ for $K = 4$ under C(1)–C(3). The table is retained as a reproducibility benchmark for the preceding arm-specific treated-mean implementation and is not relabeled as direct TATE. Recall that C(1) uses quadratic working bases, but the skew-normal source-to-target density ratio is not exactly log-quadratic. The direct-TATE rerun uses the same data-generating settings, seeds, and comparison methods and is reported separately.

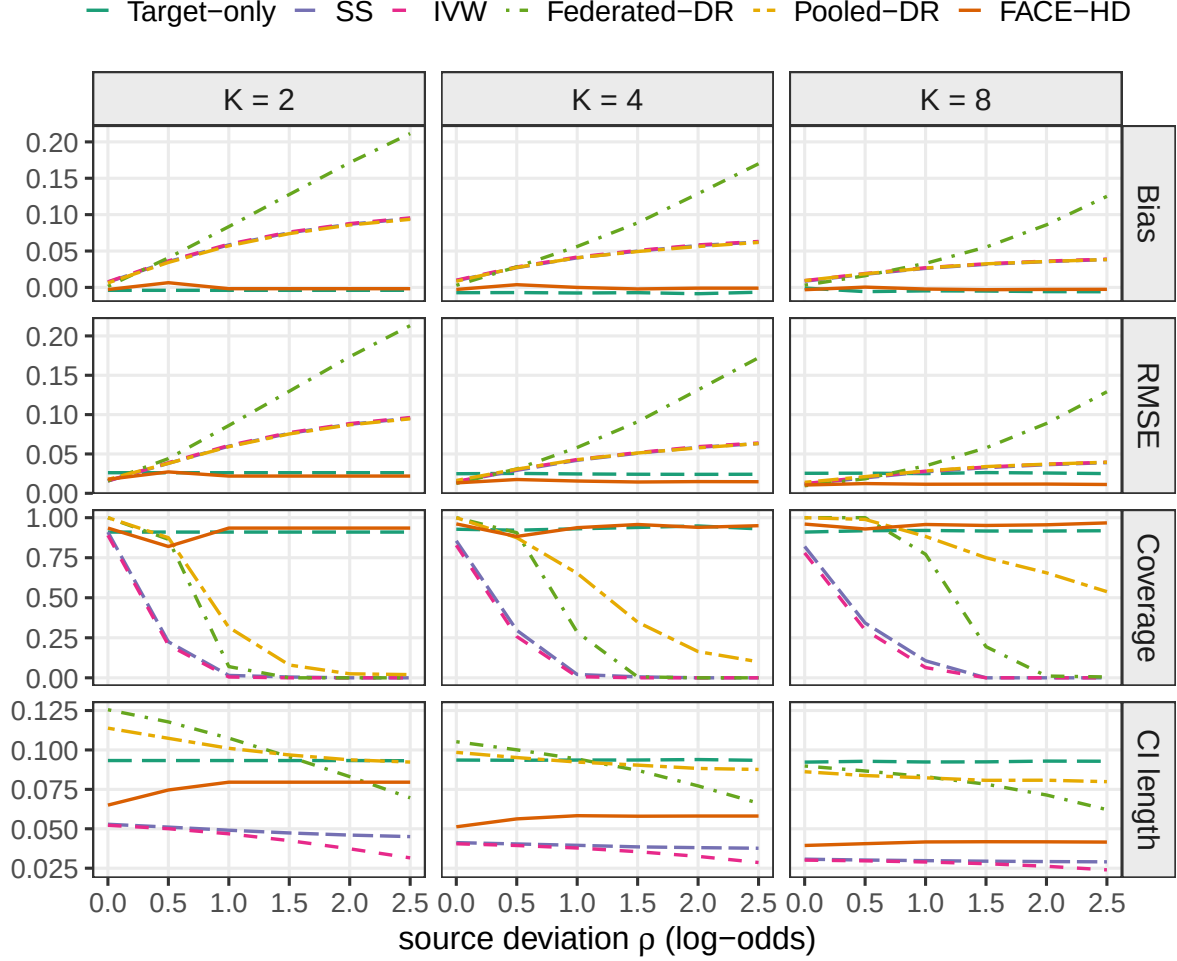


Figure 1: Legacy treated-mean negative-transfer simulation, $C(1)$, $p = 100$. Panels show bias, RMSE, 95% coverage, and CI length for μ^1 as the source deviation ρ increases, with columns for $K = 2, 4, 8$ source sites. Every curve in this archived panel uses the same treated-mean estimand; the direct-TATE rerun is reported separately.

Table 1: Legacy treated-mean negative-transfer results at $p = 100$ and $K = 4$. Entries are Monte Carlo bias, RMSE, and 95% coverage for μ^1 at $\rho \in \{0, 2.5\}$; direct-TATE results are reported separately. The archived cells use 180, 176, and 200 completed replicates for C(1), C(2), and C(3), respectively, at $\rho = 0$, and 200 at $\rho = 2.5$.

Method	$\rho = 0$ (transportable)			$\rho = 2.5$ (non-transportable)		
	Bias	RMSE	Cov ₉₅	Bias	RMSE	Cov ₉₅
<i>C(1): quadratic working bases for both nuisances</i>						
FACE-HD	−0.003	0.013	0.96	−0.001	0.015	0.95
Target-only	−0.007	0.025	0.93	−0.007	0.024	0.93
SS	+0.009	0.014	0.86	+0.062	0.063	0.00
IVW	+0.010	0.015	0.83	+0.063	0.064	0.00
Federated-DR	+0.003	0.013	1.00	+0.170	0.172	0.00
Pooled-DR	+0.008	0.016	1.00	+0.062	0.063	0.10
<i>C(2): outcome model misspecified</i>						
FACE-HD	−0.005	0.014	0.93	−0.002	0.015	0.95
Target-only	−0.008	0.024	0.94	−0.007	0.025	0.93
SS	+0.008	0.013	0.89	+0.062	0.062	0.00
IVW	+0.008	0.014	0.87	+0.062	0.063	0.00
Federated-DR	−0.002	0.012	1.00	+0.166	0.168	0.00
Pooled-DR	−0.001	0.012	1.00	+0.054	0.055	0.09
<i>C(3): site/propensity model misspecified</i>						
FACE-HD	−0.009	0.016	0.94	−0.008	0.017	0.94
Target-only	−0.011	0.025	0.93	−0.011	0.025	0.92
SS	+0.005	0.012	0.93	+0.059	0.060	0.00
IVW	+0.005	0.012	0.92	+0.060	0.060	0.00
Federated-DR	+0.000	0.013	1.00	+0.169	0.171	0.00
Pooled-DR	+0.006	0.016	0.99	+0.060	0.062	0.07

Diagnostic for the legacy C(3) TATE contrast. The earlier separately weighted TATE contrast under C(3), which is retained in the archived replicate-level files but is not the treated-mean quantity in [table 1](#), exhibits undercoverage that is primarily a finite-sample centering problem rather than standard-error underestimation. For example, at $K = 4$ and

$\rho = 0$, legacy FACE-HD has bias -0.0159 , empirical Monte Carlo standard deviation 0.0175 , mean estimated standard error 0.0192 , and coverage 0.89 . Thus its standard error is slightly conservative relative to the empirical spread, but the non-negligible bias shifts the interval away from the truth. At the other archived endpoint, $\rho = 2.5$, the corresponding legacy TATE values are bias -0.0151 , empirical standard deviation 0.0190 , mean standard error 0.0207 , and coverage 0.90 . These endpoint results are retained explicitly rather than being conflated with the treated-mean entries in [table 1](#). With 200 replicates, the binomial Monte Carlo standard errors of these two coverage estimates are approximately 0.022 and 0.021 , respectively; the simulation uncertainty does not erase the visible centering pattern. The arm decomposition makes this centering mechanism more explicit: the legacy FACE-HD treated-mean bias is -0.0091 , whereas its control-mean bias is $+0.0068$, so subtracting the two arms amplifies the TATE bias to -0.0159 . The target-only estimator in the same setting already has bias -0.0128 and coverage 0.925 , so most of the centering error is present before source aggregation. In $C(3)$, the fitted site/treatment nuisance omits the true quadratic terms. Although the outcome working basis contains the correct quadratic terms, its high-dimensional penalized estimate is not exact in finite samples; the product of the two nuisance errors therefore leaves a visible doubly robust remainder. A focused 500-replicate oracle-remainder diagnostic at $p = 100$, $K = 4$, and $\rho = 0$ confirms this mechanism in the target-only TATE anchor. With both nuisances fitted, its bias is -0.0178 , coverage is 0.918 , and the ratio of mean standard error to empirical standard deviation is 1.039 . Replacing only the outcome regression by its oracle truth reduces the bias to -0.0021 and gives coverage 0.952 ; replacing only the propensity by its oracle truth gives bias -0.0018 and coverage 0.952 ; using both oracle nuisances gives bias -0.0019 and coverage 0.950 . At the replicate level, the inclusion–exclusion contrast “fitted/fitted minus oracle–outcome minus oracle–propensity plus oracle/oracle” has mean -0.0159 (Monte Carlo SE 0.0003), directly quantifying the joint second-order nuisance remainder. Changing the target propensity and

outcome fits to the stronger one-standard-error rule does not repair this remainder. In a paired rerun using the same 500 datasets and folds, fitted/fitted bias becomes -0.0251 , RMSE becomes 0.0384 , and coverage falls to 0.872 ; the paired estimate shift relative to the minimum-loss rule is -0.00723 (Monte Carlo SE 0.00024). The corresponding interaction remainder is -0.0231 , versus -0.0159 under the minimum-loss rule, with a paired difference of -0.00721 (Monte Carlo SE 0.00017). The double-oracle estimates and standard errors are identical in every replicate across the two runs, verifying that this comparison changes only the fitted target nuisance rule. We therefore retain the minimum-loss nuisance rule rather than choosing regularization to improve the observed coverage. Consequently, the C(3) target-anchor undercoverage is primarily the finite-sample product remainder, not an underestimated standard error. The aggregation cutoff and the FACE source-nuisance path do not alter this independently fitted target anchor, so changing either one alone cannot be interpreted as removing the diagnosed remainder. The full direct-TATE rerun applies the per-setting diagnostic protocol above to determine how common-weight source aggregation changes this anchor pattern.

The same finite-sample decomposition explains why FACE-HD is not the RMSE minimum at $\rho = 0$. For the legacy treated mean at $K = 4$ under C(3), FACE-HD reduces RMSE from 0.0248 for target-only to 0.0160 , but SS and IVW attain 0.0115 and 0.0118 . At this endpoint the sites share the conditional outcome mechanism and treatment shift, and the modest covariate-shift-induced marginal mean differences are smaller than the variance reduction achieved by those non-adaptive rules. They use all sites with no source-selection or fold-specific weight-estimation cost. FACE-HD remains target-anchored and estimates adaptive weights from noisy training-fold discrepancies; together with the finite-sample nuisance bias, that protection costs RMSE in the no-negative-transfer endpoint. The TATE-level oracle inequality therefore compares asymptotic variance with target-only TATE and does

not imply minimum finite-sample RMSE among estimators that can exploit the common conditional mechanism without estimating a source-selection rule. This interpretation is also visible directly in the binary-outcome DGP. With $p = 100$, $K = 4$, and the common conditional log-odds shift equal to one, the target marginal risk difference is 0.2063, whereas the four source-law risk differences are 0.2037, 0.2014, 0.1993, and 0.1970; the largest source–target difference is 0.0093 (integration Monte Carlo SE below 9×10^{-5}). On the arm scale, the four source-law treated means exceed the target $\mu^1 = 0.6927$ by 0.0079, 0.0153, 0.0218, and 0.0292; the corresponding control-mean differences are 0.0105, 0.0202, 0.0288, and 0.0384. Their partial cancellation produces the smaller TATE differences above. Thus $\rho = 0$ removes conditional-effect heterogeneity but, because of non-collapsibility and covariate shift, does not make the raw site-specific marginal risk differences identical. The complete DGP-level calculation is reported in [table 6](#).

6 Real-Data Application

We apply FACE-HD to the Right Heart Catheterization (RHC) study from the SUPPORT investigation ([Connors et al. 1996](#)), using the covariate set of [Hirano & Imbens \(2001\)](#). The treatment $A = 1$ indicates right heart catheterization within the first 24 hours of admission, and the outcome is 30-day all-cause mortality. Insurance classes are used to emulate federated sites with different case mixes; they are not distinct data-holding hospitals. The support-screened primary analysis retains a privately insured target and three sources (Medicare, Private & Medicare, and Medicaid), and excludes the small No insurance stratum and the Medicare & Medicaid stratum implicated by the same sparse-support component. It contains $n = 5,039$ patients and $p = 61$ full-rank covariates before the intercept: 22 continuous variables, 22 binary indicators, and 17 reference-coded categorical indicators.

The working columns are centered and scaled using pooled first and second covariate

moments, which are federated-computable summaries; no outcome or individual-level record is pooled for this preprocessing. The reported analysis otherwise uses the identity feature map, so the outcome and site/treatment nuisance models have the same covariate dimension. This data-local workflow is not a formal privacy guarantee. The primary estimand is the target average treatment effect $\tau = \mathbb{E}_t\{Y(1) - Y(0)\}$, estimated with ten cross-fitting folds, 100 nuisance penalty candidates, the locked Wald cutoff $c = 1$, and truncation radius $M_\tau = 5$. The four comparison estimators use 5,000 paired multiplier-bootstrap draws. The initial minimum-loss nuisance run triggered the pre-specified numerical audit described above, so the primary RHC analysis propagates the one-standard-error rule through both target and source nuisance fits.

Figure 2 reports the primary direct-TATE results. FACE-HD estimates a 0.0315 increase in 30-day mortality (SE 0.0194; 95% CI, -0.0066 to 0.0696) under the stated observational identification assumptions. The target-only estimate is 0.0261 (SE 0.0213; 95% CI, -0.0156 to 0.0678). Thus FACE-HD’s standard error is 8.7% smaller than target-only, a modest precision gain. The non-adaptive SS and IVW estimates are 0.0604 and 0.0590, respectively. Federated-DR estimates 0.0365 (SE 0.0215), and Pooled-DR estimates 0.0628 (SE 0.0276). These differences should not be read as randomized comparisons of estimators or as evidence of a privacy guarantee.

The common FACE-HD source weights are (0.089, 0.078, 0.108), leaving target anchor weight 0.725. As shown in table 2, the Medicare source has mean/max foldwise Wald statistics 2.37/2.99 and activates the penalty in all ten outer-training folds. The corresponding activation fractions for the Private & Medicare and Medicaid sources are 0 and 0.10. The conclusion is stable over the prespecified cutoff grid: $c \in \{1, 1.5, 2, 2.5, 3\}$ gives estimates from 0.0315 to 0.0367 and standard errors from 0.01935 to 0.01943; every interval includes zero. The $c = 1$ row reproduces the independently committed primary fit exactly.

All 660 recorded primary nuisance fits converged, every recorded line-search failure count is zero, and the largest absolute density-ratio and outcome coefficients are 2.22 and 1.34, respectively, far below the hard bound of 100. A separate audit of all 540 initial density-ratio fits found no empirically unbounded zero-variance direction and no coefficient near the numerical boundary. The support-floor safeguard was activated for 59 initial and 52 calibrated density-ratio fits with fold-constant indicators. Only 1 of 3,341 source correction observations (0.03%) required inference-logit truncation, and no inference safety clipping occurred. The direct-TATE pseudo-value, within-site variance, cross-arm covariance, bootstrap, and provenance identities all pass their strict audits.

For transparency, the minimum-loss nuisance run is retained but is not used for primary inference. At $c = 1$ it gives FACE-HD 0.0489 (SE 0.0221; 95% CI, 0.0056 to 0.0922) and target-only 0.0392 (SE 0.0239). Its control-arm outer fold 4, Medicaid-source inner fold 1 initial density-ratio fit places the coefficient of `incomeUnder..11k` at the hard numerical boundary 100; the ensuing calibrated density-ratio fit records 31 recovered line-search failures. This initial nuisance also enters the inner-fold variance calculation and calibrated outcome plug-in, so the strict audit correctly rejects the run even though a finite final estimate is returned. Reporting both runs avoids selecting regularization because of the observed confidence interval while preventing a boundary-contaminated fit from being presented as the primary analysis.

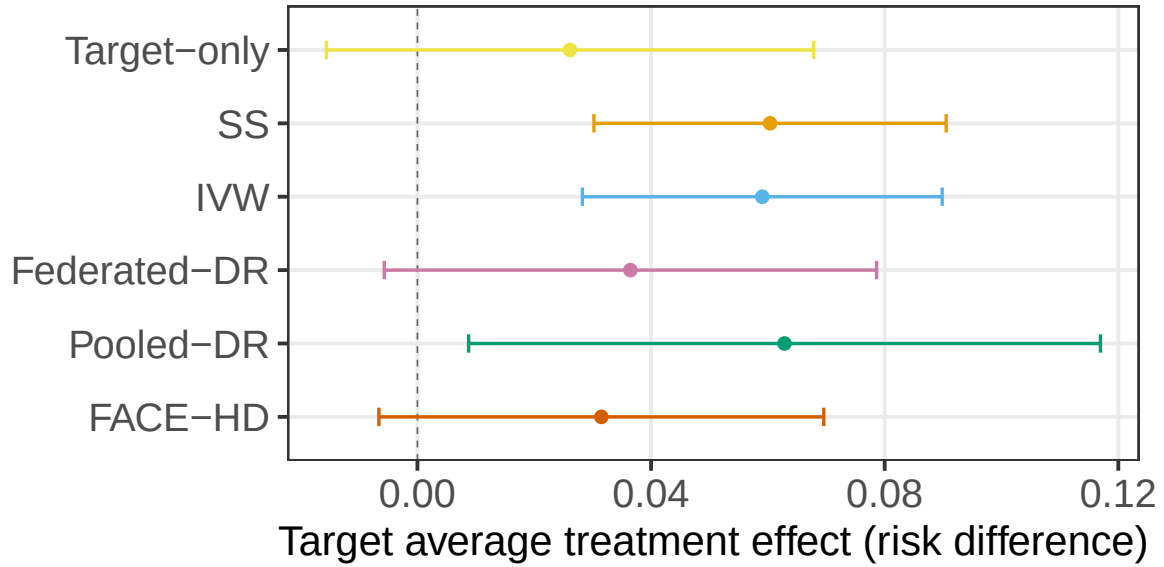


Figure 2: Primary support-screened RHC direct-TATE analysis using the one-standard-error nuisance rule and locked aggregation cutoff $c = 1$. Points and bars show estimates and 95% CIs for the target-population 30-day mortality risk difference. FACE-HD contrasts the two treatment arms before estimating one common source-weight vector; all comparison intervals use paired cross-arm influence blocks.

Table 2: Primary RHC site and direct-TATE aggregation diagnostics. The target row reports target-only TATE; source rows report source-assisted TATE, common weight, mean/max foldwise Wald statistic, and the fraction of the ten outer-training folds in which the locked primary-cutoff penalty is active.

Site	Insurance class	n	Treat.	Mort.	TATE	$\hat{\eta}_j$	Wald mean/max	Active
t	Private (target)	1698	0.431	0.309	0.026	–	–	–
s_1	Medicare	1458	0.350	0.362	0.142	0.089	2.37/2.99	1.00
s_2	Private & Medicare	1236	0.396	0.367	0.014	0.078	0.38/0.84	0.00
s_3	Medicaid	647	0.298	0.269	–0.026	0.108	0.46/1.05	0.10

Legacy arm-wise secondary analysis. To preserve the earlier results, [figure 3](#) and [table 3](#) report the archived five-stratum analysis with $n = 5,735$. It predates the final support screen and uses separate treated/control FACE-HD weights. Its arm means and their descriptive difference are not the primary direct-TATE estimator and do not receive TATE-level inference.

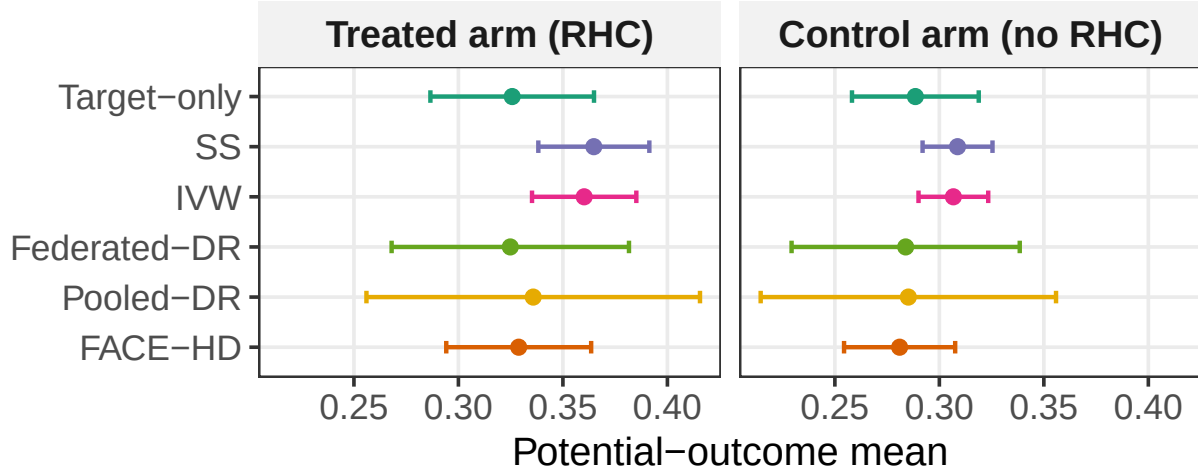


Figure 3: Legacy five-stratum RHC analysis. Point estimates and 95% CIs for treated- and control-arm potential-outcome means among privately insured target patients. The two panels use separate arm-specific FACE-HD weights and are retained as secondary reproducibility results.

Table 3: Legacy five-stratum RHC site transparency: sample size, treatment rate, mortality, separate arm-specific FACE-HD weights, and absolute source–target discrepancies. This archived construction includes the Medicare & Medicaid source, which absorbed the small No insurance stratum and is excluded from the support-screened primary analysis; its weights are not the common direct-TATE weights.

Site	Insurance class	n	Treat.	Mort.	μ^1 (treated)		μ^0 (control)	
					$\hat{\eta}_j$	$ \text{disc}_j $	$\hat{\eta}_j$	$ \text{disc}_j $
t	Private (target)	1698	0.431	0.309	–	–	–	–
s_1	Medicare	1458	0.350	0.362	0.126	0.031	0.110	0.048
s_2	Private & Medicare	1236	0.396	0.367	0.096	0.083	0.057	0.035
s_3	Medicaid	647	0.298	0.269	0.090	0.056	0.059	0.026
s_4	Medicare & Medicaid	696	0.372	0.341	0.105	0.056	0.108	0.050

7 Discussion

FACE-HD is intended for settings in which source sites may improve estimation of a target-site causal effect but cannot share individual-level records. The method combines federated calibration, doubly robust estimation, and adaptive aggregation. The calibration losses enforce orthogonality equations, the aggregation objective accounts for covariance induced by shared target data, and the penalty downweights sources whose source-assisted estimates disagree with the target-only estimate.

The simulations support these conclusions. When one source is made progressively non-transportable, FACE-HD maintains a more stable bias-coverage tradeoff across the source-deviation range and source counts $K \in \{2, 4, 8\}$ at $p = 100$. Non-adaptive pooling can have the smallest RMSE at $\rho = 0$, but its bias grows and its coverage deteriorates as the source deviates. On the Right Heart Catheterization data, FACE-HD remains close to the target-only estimate and gives a modest 6.7% reduction in standard error relative to target-only; it is not more precise than every doubly robust comparator. In the archived arm-wise communication diagnostic, the one-round estimator performs similarly to the two-round variant when sources are mean-compatible for the TATE. This is descriptive support for the one-round default, not a direct-TATE equivalence result.

The theory treats a fixed number of source sites and relies on standard high-dimensional nuisance rates, overlap/moment conditions, and a separation condition for exact source selection. Under these assumptions, FACE-HD can gain efficiency from transportable sources, retains the doubly robust property for each pairwise estimator, and moves toward the target-only estimator when source discrepancies are large. Extensions to a growing number of sources, weaker local alternatives, and more flexible nuisance learners require additional concentration and empirical-process arguments.

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Supplementary Material for Federated Adaptive Causal Estimation with High-Dimensional Covariates (FACE-HD)

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S1 Fold-Level Details for Two-Level Cross-Fitting

The main Method section describes the FACE-HD estimator with two-level cross-fitting in its one-round form. This appendix is not a separate robustness add-on; its purpose is to spell out the fold-level bookkeeping and to record the expanded two-round variant that can be used when source-specific initialization is preferred.

The two-level construction adapts the cross-fitting strategy of [Hou et al. \(2025\)](#). It uses a nested sample-splitting scheme to ensure that the data used to train an initial nuisance plug-in for an observation is disjoint from the data used for its calibrated loss contribution. As in the SMMAL calibration construction, calibration is carried out over all secondary folds rather than over one held-out fold: for each outer fold k_1 , every $k_2 \neq k_1$ supplies one calibration-fold loss, the corresponding initial fits use the out-of-two-folds data $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$, and a single calibrated nuisance estimator is obtained by minimizing the average of these fold-specific losses. This separation mitigates overfitting bias in the sequential nuisance updates and supports the Neyman-orthogonal expansion used in the theory.

S1.1 Calibrated Loss Functions

The core of the two-level cross-fitting method is the use of specially designed "calibrated" loss functions for refining the nuisance models. These losses are constructed to make the final TATE estimator insensitive to small estimation errors in the nuisance parameters. Following the principles in ([Smucler et al. 2019](#)) and ([Tan 2020](#)), we introduce a calibration step after obtaining robust initial estimates.

The truncation function, GLM notation, and calibrated losses are defined earlier in [Eqs. \(6\)](#) and [\(7\)](#). The same loss definitions are used foldwise in the two-level cross-fitting construction below. For a fixed outer fold k_1 , let $\ell_{\gamma, \text{cal}}^{(k_2)}$ and $\ell_{\alpha, \text{cal}}^{(k_2)}$ denote the fold-specific versions of [Eqs. \(6\)](#) and [\(7\)](#), with $\tilde{E}_t$ and $\tilde{E}_{s_j}$ replaced by $\tilde{E}_{t, k_2}$ and $\tilde{E}_{s_j, k_2}$ and with plug-in initial

estimators trained on $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$. The calibrated nuisance estimators for evaluating fold k_1 are

$$\hat{\gamma}_{s_j,1}^{(-k_1)} = \arg \min_{\gamma} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\gamma, \text{cal}}^{(k_2)} \left(\gamma; \hat{\alpha}_{\text{init},(k_1,k_2)}^1 \right),$$

and

$$\hat{\alpha}_{t,s_j}^{1,(-k_1)} = \arg \min_{\alpha} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\alpha, \text{cal}}^{(k_2)} \left(\alpha; \hat{\gamma}_{\text{init},(k_1,k_2)} \right).$$

Thus the secondary folds enter as loss contributions in one calibrated optimization problem, rather than as separate calibrated parameter estimates to be averaged. In the two-level algorithm, the calibrated γ -loss uses $\hat{\alpha}_{\text{init}}$ while the calibrated α -loss uses $\hat{\gamma}_{\text{init}}$. In the Supplemental Material, we assume these initial estimators converge to the same population nuisance targets as the final calibrated estimators, so this plug-in order does not change the first-order asymptotic results.

S1.2 Expanded Two-Round Variant with Two-Level Cross-Fitting

This section records the expanded fold-level version of the two-round construction. It makes explicit which observations are used for initialization, calibration, aggregation-weight estimation, and final evaluation. We describe the treated mean μ^1 ; the control mean is obtained by replacing $I(A = 1)$ with $I(A = 0)$ and using the corresponding nuisance parameters.

Outer and secondary folds. Each site $R \in \{t, s_1, \dots, s_K\}$ is partitioned into K_f folds $\{\mathcal{D}_{R,k}\}_{k=1}^{K_f}$ with aligned fold labels across sites. Fix an outer evaluation fold k_1 . The held-out data $\mathcal{D}_{R,k_1}$ are used only for evaluating the fold-specific estimating functions. All nuisance fitting, nuisance tuning, and aggregation-weight estimation are carried out inside the outer training sample $\mathcal{T}_{k_1} = \mathcal{D} \setminus \mathcal{D}_{k_1}$. Within $\mathcal{T}_{k_1}$, each $k_2 \neq k_1$ acts as a secondary calibration fold, and the corresponding initial plug-ins are trained on $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$.

Round 1: source initialization. For each source s_j and secondary fold $k_2 \neq k_1$, source s_j fits an initial treated-outcome model on $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j, k_1} \cup \mathcal{D}_{s_j, k_2})$ using the standard penalized GLM initialization loss in Eq. (8). Denote the result by $\hat{\alpha}_{\text{init}, s_j, (k_1, k_2)}^1$. The source sends the $K_f - 1$ fitted parameter vectors $\{\hat{\alpha}_{\text{init}, s_j, (k_1, k_2)}^1 : k_2 \neq k_1\}$ to the target. No source records are transmitted, and the initial site/treatment tilting model is not fit at this stage because it requires target-side summaries.

Target summaries. For each received initial outcome fit, the target computes two summaries. The first is the calibration-fold derivative summary

$$\hat{\mathbf{h}}_{t, s_j, (k_1, k_2)} = \tilde{E}_{t, k_2} \left[\nabla_{\alpha} g\{\phi(\mathbf{X}); \hat{\alpha}_{\text{init}, s_j, (k_1, k_2)}^1\} \right],$$

which enters the calibrated site/treatment loss. The second is the training-fold feature mean

$$\hat{\mathbf{b}}_{t, (k_1, k_2)} = \tilde{E}_{t, -(k_1, k_2)} \{\phi(\mathbf{X})\},$$

which enters the initial site/treatment loss. The target sends the corresponding pairs $\{(\hat{\mathbf{h}}_{t, s_j, (k_1, k_2)}, \hat{\mathbf{b}}_{t, (k_1, k_2)}) : k_2 \neq k_1\}$ back to source s_j .

Round 2: source calibration. After receiving the target summaries, source s_j first computes, for each secondary fold, an initial site/treatment parameter $\hat{\gamma}_{\text{init}, s_j, (k_1, k_2)}$ by minimizing the fold-specific site/treatment initialization loss in Eq. (8) on $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j, k_1} \cup \mathcal{D}_{s_j, k_2})$, with $\hat{\mathbf{b}}_{t, (k_1, k_2)}$ used as the target feature mean. It then solves one fold-summed calibrated problem over the secondary folds:

$$\hat{\gamma}_{s_j, 1}^{(-k_1)} = \arg \min_{\gamma} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\gamma, \text{cal}}^{(k_2)} \left(\gamma; \hat{\alpha}_{\text{init}, s_j, (k_1, k_2)}^1 \right),$$

and

$$\hat{\alpha}_{t, s_j}^{1, (-k_1)} = \arg \min_{\alpha} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\alpha, \text{cal}}^{(k_2)} \left(\alpha; \hat{\gamma}_{\text{init}, s_j, (k_1, k_2)} \right).$$

The secondary folds therefore contribute losses to a single calibrated nuisance update, rather than producing separate nuisance estimates that are averaged after fitting.

Source evaluation summaries. Using the cross-fitted nuisance pair $(\hat{\alpha}_{t,s_j}^{1,(-k_1)}, \hat{\gamma}_{s_j,1}^{(-k_1)})$, source s_j evaluates its outer fold $\mathcal{D}_{s_j,k_1}$ and forms

$$\hat{\delta}_{t,s_j,k_1}^1 = \tilde{E}_{s_j,k_1} \left[w(\mathbf{X}; \hat{\gamma}_{s_j,1}^{(-k_1)}) \{Y - g(\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^{1,(-k_1)})\} \right],$$

together with the corresponding source-side variance component computed from the centered weighted residuals on $\mathcal{D}_{s_j,k_1}$. The source sends $\hat{\alpha}_{t,s_j}^{1,(-k_1)}, \hat{\delta}_{t,s_j,k_1}^1$, the source variance component, and the relevant source fold size to the target. The tilting parameter can remain local because the target only needs the correction and variance summaries from the source.

Direct-TATE aggregation inside the outer training sample. The displayed source calibration is performed for both $a = 1$ and $a = 0$. For each arm, the target forms the fold-specific source-assisted estimate

$$\hat{\mu}_{t,s_j,k_1}^a = \tilde{E}_{t,k_1} \left[g\{\phi(\mathbf{X}); \hat{\alpha}_{t,s_j}^{a,(-k_1)}\} \right] + \hat{\delta}_{t,s_j,k_1}^a,$$

and then contrasts the arms: $\hat{\tau}_{t,s_j,k_1} = \hat{\mu}_{t,s_j,k_1}^1 - \hat{\mu}_{t,s_j,k_1}^0$ and $\hat{\tau}_{ot,k_1} = \hat{\mu}_{ot,k_1}^1 - \hat{\mu}_{ot,k_1}^0$. The aggregation weights are estimated without using $\mathcal{D}_{k_1}$. Specifically, the target recomputes the target-only component, the target-side source components, and their covariance terms inside $\mathcal{T}_{k_1}$ using the same secondary-fold cross-fitting scheme. Together with the source variance summaries, these quantities enter [Eq. \(10\)](#). The default implementation fixes the Wald threshold $1/\lambda$ in advance as described in [Remark 2](#), and solves for $\hat{\eta}^{(-k_1)}$ inside $\mathcal{T}_{k_1}$. If one instead tunes λ over a grid, the secondary folds provide the validation summaries: inner-training folds produce penalized candidate weights and the held-out secondary fold evaluates their pure variance criterion, without reapplying the training penalty. No individual-level validation records are shared.

Final fold estimate and variance. The one common TATE weight vector $\hat{\boldsymbol{\eta}}^{(-k_1)}$ is applied only to the held-out outer fold, yielding

$$\hat{\tau}_{\text{agg},k_1} = \hat{\tau}_{ot,k_1} + \sum_{j=1}^K \hat{\eta}_j^{(-k_1)} (\hat{\tau}_{t,s_j,k_1} - \hat{\tau}_{ot,k_1}).$$

Repeating the construction over $k_1 = 1, \dots, K_f$ and pooling the observation-level pseudo-values gives $\hat{\tau}_{\text{agg}}$. The point estimate and variance are computed from the same uncentered influence-function values. For an observation i in evaluation fold $k(i)$, define

$$\hat{\Phi}_i^a := \begin{cases} \frac{N_{\text{all}}}{N_t} \left[(1 - \sum_j \hat{\eta}_j^{(-k(i))}) \tilde{Y}_{ot,i} + \sum_j \hat{\eta}_j^{(-k(i))} g\{\phi(\mathbf{X}_i); \hat{\boldsymbol{\alpha}}_{t,s_j}^{a,(-k(i))}\} \right], & i \in \mathcal{D}_t, \\ \frac{N_{\text{all}}}{N_{s_j}} \hat{\eta}_j^{(-k(i))} w(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}_{s_j,a}^{(-k(i))}) \{Y_i - g(\phi(\mathbf{X}_i); \hat{\boldsymbol{\alpha}}_{t,s_j}^{a,(-k(i))})\}, & i \in \mathcal{D}_{s_j}. \end{cases}$$

Here $\tilde{Y}_{ot,i}$ denotes the arm- a uncentered target-only AIPW pseudo-outcome. The TATE pseudo-value is formed *before* computing a variance, $\hat{\Phi}_i^\tau = \hat{\Phi}_i^1 - \hat{\Phi}_i^0$. Under independent site-stratified sampling,

$$\hat{\tau}_{\text{agg}} = N_{\text{all}}^{-1} \sum_i \hat{\Phi}_i^\tau, \quad \widehat{\text{Var}}(\hat{\tau}_{\text{agg}}) = \frac{1}{N_{\text{all}}^2} \sum_{r \in \{t, s_1, \dots, s_K\}} \sum_{i \in \mathcal{D}_r} (\hat{\Phi}_i^\tau - \bar{\Phi}_r^\tau)^2, \quad \bar{\Phi}_r^\tau = \frac{1}{N_r} \sum_{i \in \mathcal{D}_r} \hat{\Phi}_i^\tau.$$

Thus the treatment-control covariance is included automatically, and the same fold-specific weight vector enters both arm components.

S1.3 Linear Outcome One-Round Specialization with Two-Level Cross-Fitting

When the outcome model is linear, $g\{\phi(\mathbf{X}); \boldsymbol{\alpha}\} = \boldsymbol{\alpha}^T \phi(\mathbf{X})$, the communication-saving version simplifies because

$$\nabla_{\boldsymbol{\alpha}} g\{\phi(\mathbf{X}); \boldsymbol{\alpha}\} = \phi(\mathbf{X})$$

does not depend on an initial outcome parameter. Thus the target does not need to send fold-specific initial outcome models to the sources. It only sends fold-specific covariate means needed for calibration, while the source sites compute the linear calibration and

weighted least-squares updates locally. The outer fold k_1 and secondary fold k_2 are still kept separate, so this is the cross-fitted linear specialization of the one-round estimator, not the non-cross-fitted sufficient-statistics shortcut.

For a fixed target-source pair and the treated arm, define the secondary-fold linear calibration loss

$$\ell_{\gamma, \text{lin}}^{(k_1, k_2)}(\boldsymbol{\gamma}) = \tilde{E}_{t, k_2} \{ \phi(\mathbf{X}) \}^T \boldsymbol{\gamma} + \tilde{E}_{s_j, k_2} [I(A = 1) \exp\{-\phi(\mathbf{X})^T \boldsymbol{\gamma}\}] + \lambda_\gamma \|\boldsymbol{\gamma}\|_1. \quad (12)$$

This is exactly the GLM calibrated γ -loss with $g'(\cdot) = 1$. For the outcome update, each secondary fold uses an initial tilting fit $\hat{\boldsymbol{\gamma}}_{\text{init}, (k_1, k_2)}$ trained on $\mathcal{D}_{s_j} \setminus (\mathcal{D}_{s_j, k_1} \cup \mathcal{D}_{s_j, k_2})$ with the corresponding target training-fold mean $\tilde{E}_{t, -(k_1, k_2)} \{ \phi(\mathbf{X}) \}$. The fold-specific weighted least-squares loss is

$$\ell_{\alpha, \text{lin}}^{(k_1, k_2)}(\boldsymbol{\alpha}) = \tilde{E}_{s_j, k_2} [I(A = 1) \exp\{-\phi(\mathbf{X})^T \hat{\boldsymbol{\gamma}}_{\text{init}, (k_1, k_2)}\} \{Y - \boldsymbol{\alpha}^T \phi(\mathbf{X})\}^2] + \lambda_\alpha \|\boldsymbol{\alpha}\|_1. \quad (13)$$

The nuisance estimators used on the held-out outer fold are obtained by minimizing the fold-summed versions of these two losses:

$$\hat{\boldsymbol{\gamma}}_{s_j, 1}^{(-k_1)} \in \arg \min_{\boldsymbol{\gamma}} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\gamma, \text{lin}}^{(k_1, k_2)}(\boldsymbol{\gamma}), \quad \hat{\boldsymbol{\alpha}}_{t, s_j}^{1, (-k_1)} \in \arg \min_{\boldsymbol{\alpha}} \frac{1}{K_f - 1} \sum_{k_2 \neq k_1} \ell_{\alpha, \text{lin}}^{(k_1, k_2)}(\boldsymbol{\alpha}).$$

We use the same mathematical notation as the main manuscript throughout this Supplemental Material. The theory is organized in a theorem-first format: setup and notation, assumptions, main limit statements, variance estimation, and then the supporting lemmas. Aggregation is treated in its own section, where the covariance structure, penalized weight estimation, source selection, and final oracle-limit theorem are stated before the proof details.

For readability, the proofs are grouped by the statistical conclusion they support. The pairwise theory proves the source-assisted cross-fitted expansion first: identification, leading influence-function CLT, high-dimensional nuisance rates, orthogonality, second-order remainders, and variance consistency. The aggregation theory then treats FACE-HD as an oracle-selection problem: shared-target covariance decomposition, uniform convergence of the quadratic variance objective, Wald-penalized source selection, asymptotic equivalence to oracle weights, and the final CLT/Slutsky argument.

S2 Asymptotic Normality of the Cross-Fitted Estimator

S2.1 Setup and Notation

This section is organized in the standard theorem-driven order used in high-dimensional inference. We first state the setup and assumptions, then give the main pairwise asymptotic normality result and variance estimator, then collect the supporting technical lemmas in the order they are used. The control-arm and TATE extension appears after the pairwise technical lemmas, because it is a direct reuse of the treated-arm argument.

We analyze the asymptotic properties of the K_f -fold cross-fitted estimator for a single target-source pair (t, s) , derived using the two-level cross-fitting strategy described in [Section S1](#).

Let K_f be a fixed integer (e.g., 5 or 10). The data $\mathcal{D} = \mathcal{D}_t \cup \mathcal{D}_s$ is partitioned into K_f folds, $\mathcal{D} = \cup_{k=1}^{K_f} \mathcal{D}_k$, where $\mathcal{D}_k = \mathcal{D}_{t,k} \cup \mathcal{D}_{s,k}$. Let $N_k = |\mathcal{D}_k| \approx N/K_f$. For notational simplicity, we assume the folds are balanced within each site (e.g., N_t and N_s are divisible by K_f) so that $|\mathcal{D}_{t,k}| = N_t/K_f$ and $|\mathcal{D}_{s,k}| = N_s/K_f$ for all k ; the minor rounding effects in practice are $o_p(N^{-1/2})$ and do not affect the first-order results. For $r \in \{t, s\}$, $\mathbb{E}_r[\cdot]$ denotes expectation conditional on site membership $R = r$. Define the target-site potential outcome means $\mu^1 := \mathbb{E}_t[Y(1)]$ and $\mu^0 := \mathbb{E}_t[Y(0)]$, and the target average treatment effect (TATE) $\tau := \mu^1 - \mu^0$. For any function $f(D)$ of a single observation $D = (R, \mathbf{X}, A, Y)$, we write $\tilde{E}_k[f(D)] := \frac{1}{N_k} \sum_{i \in \mathcal{D}_k} f(D_i)$ for the empirical average over the combined k -th fold. Let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data used for training nuisance models evaluated on fold k . The crucial aspect of cross-fitting is that the data used for training, $\mathcal{D}^{(-k)}$, is independent of the data used for evaluation, $\mathcal{D}_k$.

Let $\vartheta = (\boldsymbol{\alpha}^1, \boldsymbol{\gamma}_1)$ be the vector of nuisance parameters. For this fixed pair (t, s) we write $\boldsymbol{\alpha}^1 \equiv \boldsymbol{\alpha}_{t,s}^1$ and $\boldsymbol{\gamma}_1 \equiv \boldsymbol{\gamma}_{s,1}$. Let $\vartheta^* = (\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ denote the population targets of the calibrated estimation step (Eq. (6) and Eq. (7) in the main manuscript), and let $\hat{\vartheta}^{(-k)} = (\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)}, \hat{\boldsymbol{\gamma}}_{s,1}^{(-k)})$ be the nuisance estimators trained on data $\mathcal{D}^{(-k)}$ using the two-level cross-fitting procedure. Here the superscript $(-k)$ denotes the *final* two-level cross-fitted nuisance estimators used for evaluating fold k (including the fold-summed calibration over secondary folds), and hence $\hat{\vartheta}^{(-k)}$ is independent of $\mathcal{D}_k$. The inner construction follows the SMMAL two-level calibration device: for each outer fold k_1 , every secondary fold $k_2 \neq k_1$ supplies one calibrated-loss contribution, and the initial nuisance fits entering that contribution are trained on $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$. The final $\hat{\vartheta}^{(-k_1)}$ is obtained by minimizing the average of these fold-specific calibrated losses, so it uses all outer-training folds while keeping the initial plug-in for each contribution disjoint from that contribution's calibration fold.

To ensure numerical stability and theoretical regularity (bounded weights), we define a

truncation function $\mathcal{T}(x) = \text{sign}(x) \min(|x|, M_\tau)$, where $M_\tau = M_{\tau,N}$ is allowed to depend on N (notation suppressed). (We avoid the symbol $\tau(\cdot)$ to prevent collision with the TATE τ in the main manuscript.) Equivalently, one may define the working covariate vector $Z = \phi(\mathbf{X})$ and write all nuisance models directly in terms of Z . We keep the notation $\phi(\mathbf{X})$ to match the FACE-style basis notation and to distinguish raw covariates from the fixed feature/basis expansion. All tail, sparsity, and restricted-curvature conditions below are imposed on this working covariate vector. The truncated weighting function is $w_\tau(\mathbf{X}; \boldsymbol{\gamma}) := I(A = 1)/e^{\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\gamma})}$. The corresponding untruncated weight is $w(\mathbf{X}; \boldsymbol{\gamma}) := I(A = 1)/e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}}$. The implementation permits separate fitting and inference radii, $M_{\tau,\text{fit}}$ and $M_{\tau,\text{inf}}$, and uses finite truncation both in the calibrated losses and in the final source correction, influence function, and variance estimator. Unless stated otherwise, both radii equal 5. Thus, in finite-sample estimator displays, an exponential source weight written as w is evaluated as w_τ with $M_\tau = M_{\tau,\text{inf}}$; retaining the shorter notation keeps the aggregation formulas readable.

The proofs use $w(\mathbf{X}; \boldsymbol{\gamma})$ as the untruncated limiting object. Under [Assumption 6\(c\)](#), the radii may increase with N so that truncation is asymptotically inactive at the population targets: in particular, the probability of clipping is $o(N^{-1})$, $w_\tau(\mathbf{X}; \boldsymbol{\gamma}_1^*) = w(\mathbf{X}; \boldsymbol{\gamma}_1^*)$ with sufficiently high probability, and $g'(\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})) = g'(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})$ likewise. The clipped and untruncated estimating equations then differ by $o_p(N^{-1/2})$, so their first-order limit theory agrees. If a fixed inference radius remains active with nonnegligible probability, the implementation instead defines a clipped estimating equation whose clipping-induced bias and variance must be assessed, for example through the stated M_τ sensitivity analysis (see also [Tan \(2020\)](#)).

The proofs below therefore use three conventions: foldwise IF components are centered when forming variance and covariance estimators; finite-sample clipping is represented by w_τ but

is asymptotically inactive under [Assumption 6\(c\)](#); and any finite-sample symmetrization or ridge stabilization used in the aggregation quadratic program is treated as a numerical device that vanishes asymptotically. Rather than relying on a standalone table, the Supplemental Material now states the role of each block in prose. [Lemma 1](#) identifies the target mean under either nuisance-model correctness; [Assumptions 5 to 7](#) collect the sampling, overlap/moment, and nuisance-rate requirements; [Theorem 3](#) gives the pairwise cross-fitted CLT; [Lemma 2](#) proves variance consistency; and [Section S4](#) extends these ingredients to adaptive aggregation and source selection.

We assume $\phi(\mathbf{X})$ includes an intercept term. Consistent with the implementation and the convention in the main manuscript, every ℓ_1 penalty in a nuisance-fitting loss excludes that intercept; ordinary norms in assumptions, bounds, and rate statements include all coordinates unless stated otherwise. For completeness, the implementation guards against an exactly noncoercive empirical density-ratio loss in a sparse source arm. Write $\mathbf{m}$ for the target-side linear moment appearing in that loss. If feature j equals a constant c_j for every observation in the fitted source arm, then $\mathbf{d}_j = (-c_j, \mathbf{e}_j)$ lies in the null space of the source design with intercept, while $\mathbf{m}^T \mathbf{d}_j = m_j - c_j m_0$. Along $t\mathbf{d}_j$, the exponential term is unchanged and the non-intercept ℓ_1 penalty grows at slope λ_γ ; hence $\lambda_\gamma > |m_j - c_j m_0|$ is sufficient for coercivity in this coordinate-null direction. Before cross-validation, the implementation replaces every candidate by at least

$$\lambda_{\gamma, \text{support}} = \max_{j: \phi_j \text{ constant in the source arm}} |m_j - c_j m_0| + \epsilon_{\text{num}},$$

with a scale-adjusted strict numerical margin ϵ_{num} , and reports the selected penalty, floor, number of constant features, and whether the candidate path was constrained. This is a finite-sample optimization diagnostic, not evidence of identification: a persistent whole-site support failure violates the intended overlap regime and is screened rather than repaired by penalization. Under fixed-dimensional cell positivity, the activation probability vanishes

with the arm-specific training size, so the safeguard is asymptotically inactive and leaves the nuisance-rate assumptions below unchanged. For every nuisance cross-validation path, the implementation additionally requires a candidate penalty to satisfy the full-coordinate stopping rule and have finite validation loss in every inner fold. Partially valid candidates are not averaged over a smaller fold set: they receive infinite validation score and are excluded. The numbers of excluded fold fits and excluded penalties are retained as numerical diagnostics, and estimation stops if no candidate is valid across all folds. An optimizer-failed candidate is rolled back before the next pathwise warm start, so an unconverged coefficient vector cannot contaminate a later candidate. To avoid repeatedly exhausting the bounded iteration budget along a terminal low-penalty tail, the implementation stops a fold-specific descending path after three consecutive nonconverged candidates and leaves the remaining candidates at their preallocated infinite score. Isolated failures therefore retain two opportunities to recover. The skipped-fold count is reported separately, and the locked $p = 100$ audit described in the main manuscript requires exact agreement of every scientific and final-fit field, including the full cutoff and inference-radius sensitivity sidecar; only the mechanically affected invalid-candidate counts may differ, and each difference must be reconciled to the reported skip count. Let $p := \dim(\phi(\mathbf{X}))$. When the feature expansion is fixed in advance, one may equivalently regard $\phi(\mathbf{X})$ as the working covariate vector. For a real-valued random variable Z , $\|Z\|_{\psi_2}$ denotes the sub-Gaussian Orlicz norm. For a vector $v \in \mathbb{R}^p$, $\|v\|_0$ denotes its sparsity level. For canonical exponential-family GLMs, we use the standard notation that the negative log-likelihood takes the form $-Y\theta + \Psi(\theta)$ with linear predictor $\theta = \phi(\mathbf{X})^T \boldsymbol{\alpha}$, and that $g(\theta) = \Psi'(\theta)$; $g'(\theta)$ denotes the derivative with respect to θ . We also use the shorthand $g(\phi(\mathbf{X}); \boldsymbol{\alpha}) \equiv g(\phi(\mathbf{X})^T \boldsymbol{\alpha})$. To avoid confusion, $\phi(\mathbf{X})$ always denotes the feature/basis expansion, $g(\cdot)$ denotes the GLM response function, and the Orlicz norm $\|\cdot\|_{\psi_2}$ is unrelated to the feature map.

As in the main manuscript, the target estimand is the target-site treated potential outcome mean $\mu^1 \equiv \mathbb{E}_t[Y(1)]$. Define the target-site treated regression function by $m_{t,1}(\mathbf{X}) := \mathbb{E}_t[Y \mid A = 1, \mathbf{X}]$. The K_f -fold cross-fitted estimator for μ^1 is:

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k} \left[g(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{1,(-k)}) \right] + \tilde{E}_{s,k} \left[w(\mathbf{X}; \hat{\gamma}_{s,1}^{(-k)}) \left(Y - g(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{1,(-k)}) \right) \right] \right\}.$$

Here,

$$\tilde{E}_{t,k}[q(D)] := \frac{1}{N_{t,k}} \sum_{i \in \mathcal{D}_{t,k}} q(D_i), \quad \tilde{E}_{s,k}[q(D)] := \frac{1}{N_{s,k}} \sum_{i \in \mathcal{D}_{s,k}} q(D_i),$$

denote sample averages over the k -th fold of the target and source data, with $N_{t,k} = |\mathcal{D}_{t,k}|$, $N_{s,k} = |\mathcal{D}_{s,k}|$, and $N_k = N_{t,k} + N_{s,k}$. Let $N = N_t + N_s$ be the total sample size, and let $\rho_t = N_t/N$ and $\rho_s = N_s/N$. We write $\mathbb{E}[\cdot]$ for expectation under the pooled distribution, i.e., $\mathbb{E}[q(D)] = \rho_t \mathbb{E}_t[q(D)] + \rho_s \mathbb{E}_s[q(D)]$. Throughout the asymptotics we consider the standard two-sample (multi-sample) regime with sampling fractions bounded away from 0 and 1, i.e., $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s \in (0, 1)$. Under overlap/positivity for the relevant treatment arm, the effective sample sizes within each arm also scale linearly with N (e.g., $\sum_{i \in \mathcal{D}_s} I(A_i = 1) = \Theta_p(N_s)$), so writing high-dimensional rates in terms of N is equivalent (up to constants) to writing them in terms of the arm-specific sample sizes. Under the balanced-fold assumption above, the estimator can equivalently be written using the combined-fold empirical average as

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k \left[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) \right],$$

which matches the $\tilde{E}_k$ notation used in the remainder decomposition below.

Define the (uncentered) score contribution at a generic nuisance parameter $\vartheta = (\alpha^1, \gamma_1)$:

$$\tilde{\varphi}_{t,s}(D_i; \vartheta) := \frac{I(R_i = t)}{\rho_t} g(\phi(\mathbf{X}_i); \alpha^1) + \frac{I(R_i = s)}{\rho_s} w(\mathbf{X}_i; \gamma_1) \{Y_i - g(\phi(\mathbf{X}_i); \alpha^1)\}.$$

Define the corresponding population estimand at ϑ as

$$\mu^1(\vartheta) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta)] = \mathbb{E}_t[g(\phi(\mathbf{X}); \alpha^1)] + \mathbb{E}_s[w(\mathbf{X}; \gamma_1) \{Y - g(\phi(\mathbf{X}); \alpha^1)\}].$$

Lemma 1 (Identification / double robustness for $\mu^1(\vartheta^*)$). *Assume the target causal parameter is identified as $\mu^1 = \mathbb{E}_t[Y(1)]$ (e.g., by consistency and ignorability/overlap on the target site), so that $\mu^1 = \mathbb{E}_t[m_{t,1}(\mathbf{X})]$. Recall $\phi(\mathbf{X})$ includes an intercept. Let $\vartheta^* = (\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ denote the population targets of the calibrated losses. Then $\mu^1(\vartheta^*) = \mu^1$ holds under either of the following conditions.*

1. **Outcome-model correctness.** *If $g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) = m_{t,1}(\mathbf{X})$ almost surely, then the calibrated outcome loss implies the population moment condition $\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X}) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] = \mathbf{0}$ (see [Lemma 5\(ii\)](#) below). In particular, the intercept component yields $\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] = 0$, and therefore $\mu^1(\vartheta^*) = \mathbb{E}_t[g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})] = \mathbb{E}_t[m_{t,1}(\mathbf{X})] = \mu^1$.*
2. **Weight-model correctness.** *Suppose the weighting model is correct in the sense that for any square-integrable measurable function $h(\mathbf{X})$,*

$$\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) h(\mathbf{X})] = \mathbb{E}_t[h(\mathbf{X})],$$

and, for the treated potential outcome, the usual covariate-shift/transportability condition holds for this pair (t, s) : $\mathbb{E}_s[Y(1) \mid \mathbf{X}] = \mathbb{E}_t[Y(1) \mid \mathbf{X}]$ almost surely. Then for any outcome model $m(\mathbf{X})$,

$$\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y(1) - m(\mathbf{X})\}] = \mathbb{E}_t[Y(1) - m(\mathbf{X})],$$

and hence $\mu^1(\boldsymbol{\alpha}, \boldsymbol{\gamma}_1^) = \mu^1$ for any $\boldsymbol{\alpha}$, in particular $\mu^1(\vartheta^*) = \mu^1$ even if the outcome model is misspecified.*

Define the centered influence function at the population parameters ϑ^* by centering each site's contribution at its own site mean:

$$\begin{aligned} \varphi_{t,s}(D_i; \vartheta^*) &= \frac{I(R_i = t)}{\rho_t} \left(g(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*}) - \mu_t^1(\boldsymbol{\alpha}^{1,*}) \right) \\ &\quad + \frac{I(R_i = s)}{\rho_s} \left(w(\mathbf{X}_i; \boldsymbol{\gamma}_1^*) \{Y_i - g(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*})\} - \delta_s^1(\vartheta^*) \right), \end{aligned}$$

where $\mu_t^1(\boldsymbol{\alpha}^{1,*}) = \mathbb{E}_t[g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})]$ and $\delta_s^1(\vartheta^*) = \mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}]$, so that $\mathbb{E}_t[\varphi_{t,s}] = \mathbb{E}_s[\varphi_{t,s}] = 0$. Its full-sample average reproduces the (uncentered) estimation error, $\frac{1}{N} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) = \tilde{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)] - \mu^1(\vartheta^*)$, because the site-specific centering constants cancel in the pooled sum. However, the within-site-centered form $\varphi_{t,s}$ must be distinguished from the pooled deviation $\tilde{\varphi}_{t,s}(D; \vartheta^*) - \mu^1(\vartheta^*)$: the two share the same mean but have different per-observation variances, because the ρ_t^{-1}, ρ_s^{-1} scaling makes the site-conditional means of $\tilde{\varphi}_{t,s}$ differ ($\mathbb{E}_t[\tilde{\varphi}_{t,s}] = \mu_t^1/\rho_t$, $\mathbb{E}_s[\tilde{\varphi}_{t,s}] = \delta_s^1/\rho_s$). More generally, for any ϑ we write $\varphi_{t,s}(D; \vartheta)$ for the same within-site-centered contribution with (μ_t^1, δ_s^1) evaluated at ϑ . The asymptotic variance is governed by the within-site-centered form:

$$\begin{aligned} V_{t,s}^* := \text{Var}(\varphi_{t,s}(D; \vartheta^*)) &= \frac{1}{\rho_t} \text{Var}_t(g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})) \\ &\quad + \frac{1}{\rho_s} \text{Var}_s(w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}). \end{aligned}$$

which differs from the pooled variance $\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta^*)) = V_{t,s}^* + \{\rho_t^{-1}(\mu_t^1)^2 + \rho_s^{-1}(\delta_s^1)^2 - (\mu^1)^2\}$ by a non-vanishing between-site term.

Definition 1 (Sparsity bookkeeping). Let

$$s_\alpha := \|\boldsymbol{\alpha}^{1,*}\|_0, \quad s_\gamma := \|\boldsymbol{\gamma}_1^*\|_0.$$

When the initial nuisance estimators have distinct population targets, define $s_{\alpha,\text{init}} := \|\boldsymbol{\alpha}_{\text{init}}^{1,*}\|_0$ and $s_{\gamma,\text{init}} := \|\boldsymbol{\gamma}_{1,\text{init}}^*\|_0$. These are bookkeeping quantities for the high-dimensional rates rather than separate modeling assumptions. The growth restrictions are imposed explicitly in the theorems and, when initial nuisance rates are needed, we assume $s_{\alpha,\text{init}} + s_{\gamma,\text{init}} = O(s_\alpha + s_\gamma)$.

S2.2 Assumptions

The assumptions below follow the compact itemized style used in FACE and SMMAL: each assumption starts with fixed constants and then lists the required pieces as labeled

subconditions. [Assumption 5](#) collects working-feature and sampling regularity; [Assumption 6](#) collects positivity, moments, smoothness, and truncation regularity; and [Assumption 7](#) collects the nuisance-estimation conditions. Sparsity levels are defined in [Definition 1](#), and their allowed growth is stated directly in the theorems.

Assumption 5 (Working-Covariate and Sampling Regularity). For a fixed constant $M < \infty$, assume:

- (a) (*Working-feature tails*) The working covariates $\phi(\mathbf{X})$ include an intercept and satisfy

$$\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq M$$

under both $R = t$ and $R = s$.

- (b) (*Independent two-site sampling*) The target and source samples are independent, observations are i.i.d. within each site, and the site fractions satisfy $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s \in (0, 1)$.

Assumption 6 (Overlap, Smoothness, and Moments). For fixed constants $M < \infty$ and $\delta > 0$, assume:

- (a) (*Overlap and bounded local weights*) The ideal density-ratio weight is well defined on the relevant target support: $P_s\{A = 1 \mid \mathbf{X}\} > 0$ almost surely on that support. In a neighborhood $\mathcal{N}(\gamma_1^*)$ of the population tilting target,

$$\sup_{\gamma \in \mathcal{N}(\gamma_1^*)} \mathbb{E}_s[w(\mathbf{X}; \gamma)^{2+\delta}] \leq M,$$

and, coordinatewise,

$$\max_{1 \leq j \leq p} \sup_{\gamma \in \mathcal{N}(\gamma_1^*)} \mathbb{E}_s[w(\mathbf{X}; \gamma)^{2+\delta} |\phi_j(\mathbf{X})|^{2+\delta}] \leq M.$$

A simple sufficient condition is an upper-overlap bound $w(\mathbf{X}; \gamma) \leq M$ almost surely on $\mathcal{N}(\gamma_1^*)$, together with coordinatewise $(2 + \delta)$ moments of $\phi(\mathbf{X})$. No uniform bound on $\|\phi(\mathbf{X})\|_2$ is imposed; in high dimensions this norm may scale as $\sqrt{p}$.

- (b) (*Smoothness and bounded moments*) The GLM mean function g is continuously differentiable with

$$\sup_{u \in \mathbb{R}} |g'(u)| \leq M, \quad |g'(u_1) - g'(u_2)| \leq M|u_1 - u_2|.$$

Moreover, under both $R = t$ and $R = s$,

$$\mathbb{E}[|Y|^{2+\delta} \mid R = r] \leq M, \quad \sup_{\vartheta \in \mathcal{N}(\vartheta^*)} \mathbb{E}[|\tilde{\varphi}_{t,s}(D; \vartheta)|^{2+\delta}] \leq M.$$

Products of any two influence-function components used in the aggregation covariance estimators are uniformly integrable. The last display is the uniform integrability condition used for the influence-function expansion and variance consistency. It follows, for example, from the weight envelope in part (a), sub-Gaussian working features, and sub-Gaussian or bounded treated outcomes.

- (c) (*Asymptotically inactive truncation*) With $M_{\tau,N}$ as in the definition of $\mathcal{T}(\cdot)$,

$$\begin{aligned} P\left(|\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}| > M_{\tau,N} \mid R = r\right) &= o(N^{-1}) \quad \text{for } r \in \{t, s\}, \\ P\left(|\phi(\mathbf{X})^T \boldsymbol{\gamma}_1^*| > M_{\tau,N} \mid R = s\right) &= o(N^{-1}). \end{aligned}$$

This condition guarantees that finite-sample clipping in the calibrated losses and final source correction has no first-order effect on the limiting estimating equation.

Assumption 7 (Nuisance Model Properties). For fixed constants $M < \infty$ and $\kappa > 0$, assume:

- (a) (*Restricted curvature*) The population versions of the calibrated losses in [Eq. \(6\)](#) and [Eq. \(7\)](#) are convex and satisfy restricted strong convexity with curvature at least κ in neighborhoods of ϑ^* compatible with the relevant sparse cones.
- (b) (*Cross-fitting and bounded fitted scores*) For each fold k , the nuisance estimator $\hat{\vartheta}^{(-k)}$ is trained on $\mathcal{D}^{(-k)}$ only and is independent of the evaluation fold $\mathcal{D}_k$. With probability tending to one, the fitted nuisance functions used on the evaluation folds lie in the local neighborhoods where the bounds in [Assumption 6\(a\)–\(b\)](#) hold.

- (c) (*Two-level alignment*) Let $(\boldsymbol{\alpha}_{\text{init}}^{1,*}, \boldsymbol{\gamma}_{1,\text{init}}^*)$ denote the population limits of the initial nuisance estimators used inside the calibrated losses. These limits coincide with the calibrated targets:

$$\boldsymbol{\alpha}_{\text{init}}^{1,*} = \boldsymbol{\alpha}^{1,*}, \quad \boldsymbol{\gamma}_{1,\text{init}}^* = \boldsymbol{\gamma}_1^*.$$

- (d) (*Rate of estimation*) Uniformly over the fixed number of outer folds,

$$\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2 = O_p\left(\sqrt{\frac{(s_\alpha + s_\gamma) \log(p)}{N}}\right),$$

with analogous ℓ_1 rates of order $O_p\{(s_\alpha + s_\gamma)\sqrt{\log(p)/N}\}$. Here $\|\cdot\|_2$ and $\|\cdot\|_1$ denote coefficient-vector norms. These rates are the high-dimensional nuisance input to the asymptotic expansion. They follow from [Lemma 4](#) under the stated restricted-curvature and score-stability conditions; the sharpened foldwise score-centering condition used in [Hou et al. \(2025\)](#) is one sufficient route, while the general version is also admissible provided the propagated plug-in error is of the same order.

Remark 4 (On the alignment condition). [Assumption 7\(c\)](#) is a technical device to avoid tracking multiple population limits induced by the nested/coupled optimization. It is satisfied, for example, when (i) the population calibrated losses have unique minimizers $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ (by strict convexity in the relevant restricted set), and (ii) the mapping from an initial plug-in value to the population minimizer of the next calibrated update is continuous at $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$, so that any consistent initialization converges to the same fixed point. In practice, the same calibrated estimating equations are solved throughout the nuisance-estimation pipeline, making the alignment natural under uniqueness.

S2.3 Main Pairwise Result and Variance Estimation

Theorem 3 (Asymptotic Normality of $\hat{\mu}_{t,s}^{1,\text{CF}}$). *Suppose [Assumptions 5 to 7](#) hold. Assume ϑ^* is the unique population target pair induced by the calibrated losses under the alignment*

condition in [Assumption 7\(c\)](#), implying [Lemma 5](#) holds. If further, at least one nuisance model (outcome or weighting) is correctly specified in the sense of [Lemma 1](#) (so $\mu^1(\vartheta^*) = \mu^1$), the calibrated nuisance sparsities satisfy

$$\frac{(s_\alpha + s_\gamma)^2 \log^2(p)}{N} \longrightarrow 0,$$

and $V_{t,s}^* > 0$, then

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

Here s_α and s_γ are the calibrated population sparsities from [Definition 1](#). If the initial nuisance estimators are estimated by separate penalized fits, their population sparsities are $s_{\alpha,\text{init}}$ and $s_{\gamma,\text{init}}$ and are assumed to be of the same order. Their stochastic effect enters only through the alignment and rate conditions in [Assumption 7\(c\)–\(d\)](#). The finite-sample KKT perturbation from the ℓ_1 penalties is likewise absorbed in these nuisance-rate conditions and is $o_p(N^{-1/2})$ under the stated sparsity scaling.

Proof. By [Lemma 1](#), the stated nuisance-correctness condition implies $\mu^1(\vartheta^*) = \mu^1$, so all expansions below are centered at the target estimand.

$$S_N := \frac{1}{N} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*), \quad \text{Rem}_N := \hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 - S_N.$$

Step 1 (Limit of the leading term). The summands in S_N are independent across the target and source samples, have mean zero, and have uniformly bounded moments of order $2 + \delta$ by [Assumptions 5](#) and [6](#). Hence [Lemma 3](#) gives

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*),$$

so $\sqrt{N}S_N$ has the desired normal limit.

Step 2 (Foldwise decomposition of the remainder). Using the foldwise representation in [Section S2.1](#),

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k \left[\tilde{\varphi}_{t,s} \{D; \hat{\vartheta}^{(-k)}\} \right].$$

Because the folds are balanced up to negligible rounding error,

$$S_N = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k \left[\tilde{\varphi}_{t,s}(D; \vartheta^*) - \mu^1 \right].$$

Since $\mu^1 = \mathbb{E}\{\tilde{\varphi}_{t,s}(D; \vartheta^*)\}$ by the identification step, we may add and subtract both $\tilde{\varphi}_{t,s}(D; \vartheta^*)$ and the conditional mean of the fold-specific plug-in difference:

$$\text{Rem}_N = \frac{1}{K_f} \sum_{k=1}^{K_f} (R_{1,k} + R_{2,k}),$$

where

$$R_{1,k} = \left(\tilde{E}_k - \mathbb{E}[\cdot \mid \mathcal{D}^{(-k)}] \right) \left[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right]$$

and

$$R_{2,k} = \mathbb{E}[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \mid \mathcal{D}^{(-k)}].$$

Step 3 (Conditional empirical-process term). Let $\Delta_k(D) = \tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*)$.

Conditional on $\mathcal{D}^{(-k)}$, cross-fitting makes the evaluation-fold summands independent and mean zero after subtracting the conditional expectation. Therefore

$$\text{Var}(R_{1,k} \mid \mathcal{D}^{(-k)}) \leq \frac{1}{N_k} \mathbb{E}[\Delta_k(D)^2 \mid \mathcal{D}^{(-k)}].$$

By the local L_2 stability in [Lemma 6](#) (Part 2),

$$\mathbb{E} \left[\left(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \mid \mathcal{D}^{(-k)} \right] \leq C \left(\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2^2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2^2 \right),$$

and [Lemma 4](#) yields, uniformly over the fixed number of folds,

$$|R_{1,k}| = O_p \left(\frac{\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2}{\sqrt{N}} \right) = O_p \left(\frac{\sqrt{(s_\alpha + s_\gamma) \log(p)}}{N} \right) = o_p(N^{-1/2}).$$

Step 4 (Orthogonal bias term). By [Lemma 6](#) (Part 1), whose proof uses the calibrated moment equations in [Lemma 5](#), the conditional bias remainder is second order:

$$|R_{2,k}| \leq C \cdot \left(\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2^2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2^2 + \|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2 \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2 \right)$$

and hence, by [Lemma 4](#),

$$|R_{2,k}| = O_p\left(\frac{s_\alpha \log(p)}{N}\right) + O_p\left(\frac{s_\gamma \log(p)}{N}\right) + O_p\left(\frac{\sqrt{s_\alpha s_\gamma} \log(p)}{N}\right) = O_p\left(\frac{(s_\alpha + s_\gamma) \log(p)}{N}\right).$$

Because K_f is fixed,

$$|\text{Rem}_N| = O_p\left(\frac{\sqrt{(s_\alpha + s_\gamma) \log(p)}}{N}\right) + O_p\left(\frac{(s_\alpha + s_\gamma) \log(p)}{N}\right).$$

The sparsity condition $(s_\alpha + s_\gamma)^2 \log^2(p)/N \rightarrow 0$ implies $\sqrt{N} \text{Rem}_N = o_p(1)$. Therefore

$$\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) + o_p(1).$$

Together with [Lemma 3](#), this yields $\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$ by Slutsky's theorem. $\square$

Lemma 2 (Variance Estimation). *The asymptotic variance $V_{t,s}^*$ is consistently estimated by the within-site-centered plug-in built from the cross-fitted, out-of-fold nuisance estimates $\hat{g}^{(-k)}$:*

$$\hat{V}_{t,s} = \frac{1}{\rho_t} \widehat{\text{Var}}_t(g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}^{(-k(i))})) + \frac{1}{\rho_s} \widehat{\text{Var}}_s(w(\mathbf{X}; \hat{\boldsymbol{\gamma}}^{(-k(i))})\{Y - g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}^{(-k(i))})\}) \xrightarrow{p} V_{t,s}^*,$$

where $\widehat{\text{Var}}_r(U) = \tilde{E}_r[U^2] - (\tilde{E}_r[U])^2$ is the sample variance over site $r \in \{t, s\}$, each observation evaluated at its out-of-fold nuisance estimate $\hat{\vartheta}^{(-k(i))}$. Centering is performed within each site (at $\tilde{E}_t$ and $\tilde{E}_s$ separately), not at the pooled mean $\hat{\mu}_{t,s}^{1,\text{CF}}$; the reason is given at the end of the proof. For consistency of the variance estimator itself no lower bound is needed beyond finiteness. For Wald inference and studentization we assume, as in [Theorem 3](#), that $V_{t,s}^*$ is bounded away from zero.

Proof. Write $\hat{V}_{t,s} = \rho_t^{-1} \widehat{\text{Var}}_t(\hat{A}) + \rho_s^{-1} \widehat{\text{Var}}_s(\hat{B})$ with the cross-fitted site integrands

$$\hat{A}_i = g(\phi(\mathbf{X}_i); \hat{\boldsymbol{\alpha}}^{(-k(i))}), \quad \hat{B}_i = w(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}^{(-k(i))})\{Y_i - g(\phi(\mathbf{X}_i); \hat{\boldsymbol{\alpha}}^{(-k(i))})\}.$$

The consistency of each within-site sample variance relies on three ingredients:

1. *Nuisance Consistency:* By [Lemma 4](#), $\|\hat{\vartheta}^{(-k)} - \vartheta^*\|_2 = o_p(1)$.

2. *L_2 -Lipschitz Continuity*: Under [Assumption 6\(a\)–\(b\)](#), the site integrands are locally Lipschitz in ϑ in mean-square: there exists $C < \infty$ such that for $U \in \{A, B\}$, $r \in \{t, s\}$, and ϑ_1, ϑ_2 in a neighborhood of ϑ^* ,

$$\mathbb{E}_r \left[(U(D; \vartheta_1) - U(D; \vartheta_2))^2 \right] \leq C \|\vartheta_1 - \vartheta_2\|_2^2,$$

by the bounded-slope/Lipschitz property of g and the weight-moment conditions, exactly as in [Lemma 6\(2\)](#).

3. *Uniform Integrability*: By the $(2 + \delta)$ moment condition in [Assumption 6\(b\)](#), $\{\hat{A}^2\}_{k,N}$ and $\{\hat{B}^2\}_{k,N}$ are uniformly integrable.

Step 1 (Conditional LLN within a site). Fix a site $r \in \{t, s\}$, a fold k , and let $U \in \{A, B\}$ be the corresponding integrand. Conditional on the training sample $\mathcal{D}^{(-k)}$, the nuisance estimate $\hat{\vartheta}^{(-k)}$ is fixed and independent of $\mathcal{D}_{r,k}$, so the summands $\{U(D_i; \hat{\vartheta}^{(-k)}) : i \in \mathcal{D}_{r,k}\}$ are i.i.d. with finite $(2 + \delta)$ moments uniformly in N by Ingredient 3. By the conditional weak law of large numbers,

$$\begin{aligned} \tilde{E}_{r,k} [U(\hat{\vartheta}^{(-k)})] - \mathbb{E}_r [U(\hat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] &= o_p(1), \\ \tilde{E}_{r,k} [U(\hat{\vartheta}^{(-k)})^2] - \mathbb{E}_r [U(\hat{\vartheta}^{(-k)})^2 \mid \mathcal{D}^{(-k)}] &= o_p(1). \end{aligned}$$

Since K_f is fixed, averaging over the evaluation folds of site r gives the same statements with $\tilde{E}_{r,k}$ replaced by the pooled within-site average $\tilde{E}_r$.

Step 2 (Plug-in stability). By nuisance consistency (Ingredient 1) and the L_2 -Lipschitz property (Ingredient 2), conditional Jensen gives

$$\left| \mathbb{E}_r [U(\hat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] - \mathbb{E}_r [U(\vartheta^*)] \right| \leq \left(\mathbb{E}_r [(U(\hat{\vartheta}^{(-k)}) - U(\vartheta^*))^2 \mid \mathcal{D}^{(-k)}] \right)^{1/2} \rightarrow_p 0,$$

and, using $(a^2 - b^2) = (a - b)(a + b)$ with conditional Cauchy–Schwarz and the $(2 + \delta)$ bound (Ingredient 3),

$$\mathbb{E}_r [U(\hat{\vartheta}^{(-k)})^2 \mid \mathcal{D}^{(-k)}] \rightarrow_p \mathbb{E}_r [U(\vartheta^*)^2].$$

Combining with Step 1,

$$\widehat{\text{Var}}_r(\widehat{U}) = \widetilde{E}_r[\widehat{U}^2] - (\widetilde{E}_r[\widehat{U}])^2 \rightarrow_p \mathbb{E}_r[U(\vartheta^*)^2] - (\mathbb{E}_r[U(\vartheta^*)])^2 = \text{Var}_r(U(\vartheta^*)).$$

Step 3 (Combine sites). Applying Step 2 with $U = A$ on $r = t$ and $U = B$ on $r = s$, and using that the site fractions ρ_t, ρ_s are fixed and bounded away from 0,

$$\begin{aligned} \widehat{V}_{t,s} &= \rho_t^{-1} \widehat{\text{Var}}_t(\widehat{A}) + \rho_s^{-1} \widehat{\text{Var}}_s(\widehat{B}) \\ &\rightarrow_p \frac{1}{\rho_t} \text{Var}_t(g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})) \\ &\quad + \frac{1}{\rho_s} \text{Var}_s(w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}) = V_{t,s}^*. \end{aligned}$$

the last equality by the variance decomposition recorded with the definition of $V_{t,s}^*$.

Why within-site centering is necessary. Centering instead at the pooled mean, i.e. using $\frac{1}{K_f} \sum_k \widetilde{E}_k[(\widetilde{\varphi}_{t,s}(D; \widehat{\vartheta}^{(-k)}) - \widehat{\mu}_{t,s}^{1,\text{CF}})^2]$, would converge to the pooled variance

$$\text{Var}(\widetilde{\varphi}_{t,s}(D; \vartheta^*)) = V_{t,s}^* + \left\{ \rho_t^{-1} (\mu_t^1)^2 + \rho_s^{-1} (\delta_s^1)^2 - (\mu^1)^2 \right\},$$

overestimating $V_{t,s}^*$ by the non-vanishing between-site term. This term arises precisely because the ρ_r^{-1} -scaled contribution $\widetilde{\varphi}_{t,s}$ has unequal site-conditional means ($\mathbb{E}_t[\widetilde{\varphi}_{t,s}] = \mu_t^1/\rho_t$ versus $\mathbb{E}_s[\widetilde{\varphi}_{t,s}] = \delta_s^1/\rho_s$); subtracting each site's own mean removes it. This is the pairwise analogue of the within-group (WSS-only) centering used for the aggregated estimator in [Remark 5](#). □

Remark 5 (Unified cross-fitted variance estimator for aggregation). This remark records the foldwise direct-TATE implementation. For arm $a \in \{0, 1\}$, let $\widehat{\Phi}_i^a(\eta)$ be the uncentered site-scaled pseudo-value obtained from the display below with $I(A = a)$, the corresponding arm nuisance estimates, and the *same* weight vector η .

Unified estimating equation. For each observation i evaluated on fold $\mathcal{D}_{k(i)}$, define the

uncentered IF

$$\hat{\Phi}_i^a(\eta) := \begin{cases} \frac{N_{\text{all}}}{N_t} \left[(1 - \sum_j \eta_j) \tilde{Y}_{ot,i}^a + \sum_j \eta_j g(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^a) \right], & i \in \mathcal{D}_t, \\ \frac{N_{\text{all}}}{N_{s_j}} \eta_j w(\mathbf{X}_i; \hat{\gamma}_{s_j,a}) \{Y_i - g(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^a)\}, & i \in \mathcal{D}_{s_j}, \end{cases}$$

where all nuisance functions are evaluated out of fold and

$$\tilde{Y}_{ot,i}^a = \frac{I(A_i = a)}{\hat{\pi}_{ot,a}(\mathbf{X}_i)} \{Y_i - \hat{g}_{ot}^a(\mathbf{X}_i)\} + \hat{g}_{ot}^a(\mathbf{X}_i).$$

The direct-TATE pseudo-value is formed before variance calculation: $\hat{\Phi}_i^\tau = \hat{\Phi}_i^1(\hat{\eta}^{(-k(i))}) - \hat{\Phi}_i^0(\hat{\eta}^{(-k(i))})$. The point estimate and variance use this same set:

$$\begin{aligned} \hat{\tau}_{\text{agg}} &= \bar{\Phi}^\tau = N_{\text{all}}^{-1} \sum_i \hat{\Phi}_i^\tau, \\ \widehat{\text{Var}}(\hat{\tau}_{\text{agg}}) &= N_{\text{all}}^{-2} \sum_r \sum_{i \in \mathcal{D}_r} (\hat{\Phi}_i^\tau - \bar{\Phi}_r^\tau)^2, \end{aligned}$$

where $\bar{\Phi}_r^\tau$ is the within-site mean. Equivalently, $\hat{V}_\tau = N_{\text{all}} \widehat{\text{Var}}(\hat{\tau}_{\text{agg}})$.

Why within-group centering is necessary. Because $\hat{\Phi}_i^\tau$ is scaled by N_{all}/N_r , its conditional mean can differ across sites. Global centering would therefore add a non-sampling between-site term. The inferential target here is

$$\frac{1}{N_{\text{all}}^2} \sum_{i=1}^{N_{\text{all}}} \text{Var}(\Phi_i^\tau),$$

rather than $\text{Var}(\Phi^\tau)$ under a pooled-mixture distribution. Global centering at $\bar{\Phi}$ estimates the latter object and introduces an additional between-group term

$$\sum_g \rho_g (\mu_g - \mu)^2,$$

equivalently the ANOVA component $\text{BSS}/N_{\text{all}}^2$, which is of the same order as $L_\tau^*(\eta_\tau^*)/N_{\text{all}}$ and therefore does not vanish. For a common fixed weight vector η (or conditionally within one outer fold), $\text{WSS}/N_{\text{all}}^2$ has the familiar decomposition

$$\widehat{\text{Var}}(\hat{\tau}_{\text{agg}}) = \frac{1}{N_t} \hat{\sigma}_{T,\tau}^2 + \sum_{j=1}^K \frac{1}{N_{s_j}} \hat{\eta}_j^2 \hat{\sigma}_{s_j}^2,$$

where

$$\hat{\sigma}_{T,\tau}^2 = \widehat{\text{Var}}_t \left[(1 - \sum_j \hat{\eta}_j) \hat{\varphi}_{ot,i}^\tau + \sum_j \hat{\eta}_j \hat{\zeta}_{t,s_j,i}^\tau \right],$$

and $\hat{\sigma}_{S_j}^2$ is the within-source variance of $\hat{\xi}_{t,s_j,i}^\tau$. Hence $\hat{\sigma}_{T,\tau}^2$ is not a target-only variance term; it already contains the shared-target cross-covariances, including terms corresponding to $\text{Cov}(\hat{\varphi}_{ot}^\tau, \hat{\zeta}_{t,s_j}^\tau)$ and $\text{Cov}(\hat{\zeta}_{t,s_j}^\tau, \hat{\zeta}_{t,s_k}^\tau)$ for $j \neq k$, fully consistent with the $\hat{C}_{ot,s_j}^\tau$ and $\hat{C}_{t,s_j,s_k}^\tau$ terms in Eq. (10).

In the implemented cross-fitted estimator, $\hat{\eta}^{(-k(i))}$ varies across outer evaluation folds, so the final finite-sample WSS is computed directly from the displayed observation-level pseudo-values and is not factored through one global weight vector. Because K_f is fixed and $\max_{k \leq K_f} \|\hat{\eta}^{(-k)} - \eta_\tau^*\|_2 = o_p(1)$, this fold-specific WSS has the same probability limit $L_\tau^*(\eta_\tau^*)$.

With balanced folds ($|\mathcal{D}_{t,k}| = N_t/K_f$, $|\mathcal{D}_{s_j,k}| = N_{s_j}/K_f$), $\bar{\Phi}^\tau$ coincides with the fold-average $(1/K_f) \sum_k \hat{\tau}_{\text{agg},k}$; with minor fold imbalance the difference is $O(N_{\text{all}}^{-1})$.

This unified form ensures $\hat{\tau}_{\text{agg}} - \tau = N_{\text{all}}^{-1} \sum_i (\hat{\Phi}_i^\tau - \tau)$, so the DML representation of the estimation error as an IF average is preserved. By the same arguments as above (nuisance consistency, L_2 -Lipschitz continuity, and $\hat{\eta}^{(-k)} \rightarrow_p \eta_\tau^*$), $\hat{V}_\tau \rightarrow_p L_\tau^*(\eta_\tau^*)$.

S2.4 Pairwise Technical Lemmas

The lemmas in this subsection are ordered by their role in the proof of Theorem 3. Lemma 3 handles the leading influence-function average. Lemma 4 supplies the fold-uniform high-dimensional rates for the calibrated nuisance estimators. Lemma 5 records the population moment equations induced by the calibrated losses, and Lemmas 6 and 7 convert those equations into the second-order bias and plug-in stability bounds used in the main expansion.

Lemma 3 (CLT for IF Average). *Under Assumption 5 and Assumption 6, the influence*

function average is asymptotically normal:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

Proof. The observations consist of two independent samples: $\{D_i : i \in \mathcal{D}_t\}$ drawn i.i.d. from the target distribution and $\{D_i : i \in \mathcal{D}_s\}$ drawn i.i.d. from the source distribution, with $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s$. Thus the summands $\{\varphi_{t,s}(D_i; \vartheta^*)\}_{i=1}^N$ are independent (but not identically distributed), have mean zero, and have finite $(2 + \delta)$ moments by [Assumption 6\(b\)](#). In particular, the Lindeberg condition holds. We verify a Lyapunov condition, which implies the Lindeberg condition for this independent triangular array. Let $X_{i,N} := \varphi_{t,s}(D_i; \vartheta^*)$ so that $\mathbb{E}[X_{i,N}] = 0$ and $\sum_{i=1}^N \text{Var}(X_{i,N}) = NV_{t,s}^* + o(N)$. By [Assumption 6\(b\)](#), $\sup_{i,N} \mathbb{E}[|X_{i,N}|^{2+\delta}] < \infty$ for some $\delta > 0$. Therefore,

$$\frac{1}{(NV_{t,s}^*)^{1+\delta/2}} \sum_{i=1}^N \mathbb{E}[|X_{i,N}|^{2+\delta}] \leq \frac{CN}{N^{1+\delta/2}} = CN^{-\delta/2} \rightarrow 0,$$

which is the Lyapunov condition. Hence the Lindeberg–Feller CLT yields $\frac{1}{\sqrt{N}} \sum_{i=1}^N X_{i,N} \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$, proving the lemma. $\square$

Lemma 4 (Cross-fitted ℓ_1 -penalized nuisance rates (adapted from [Hou et al. \(2025\)](#), Lemma A21)). *Fix a constant number of folds K_f . Let the full sample $\mathcal{D} = \{D_1, \dots, D_N\}$ be partitioned into folds $\mathcal{D} = \bigcup_{k=1}^{K_f} \mathcal{D}_k$, and let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data. For each fold k , let $\hat{\gamma}^{(-k)}$ be an estimator trained on $\mathcal{D}^{(-k)}$ (hence independent of $\mathcal{D}_k$). For an observation index $i \in \{1, \dots, N\}$, let $\text{fold}(i) \in \{1, \dots, K_f\}$ denote the fold index such that $D_i \in \mathcal{D}_{\text{fold}(i)}$; we use $\hat{\gamma}^{(-\text{fold}(i))}$ to denote the out-of-fold nuisance estimate used when evaluating the loss contribution from D_i . Consider the cross-fitted ℓ_1 -penalized estimator*

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}) + \lambda \|\beta\|_1 \right\}, \quad \ell_k(\beta; \gamma) := \tilde{E}_k[\ell(D; \beta, \gamma)],$$

where $\ell(D; \beta, \gamma)$ is convex in β . Define also the fold-averaged empirical risk

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}).$$

Let $\bar{\beta}$ be the population minimizer at a population nuisance target $\bar{\gamma}$,

$$\bar{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \mathbb{E}[\ell(D; \beta, \bar{\gamma})], \quad s_\beta := \|\bar{\beta}\|_0.$$

Assume the following high-dimensional regularity conditions (cf. Assumption 6 of [Hou et al. \(2025\)](#) and the RSC framework of [Negahban et al. \(2012\)](#)):

- (a) (Sub-Gaussian covariates) Each coordinate of $\phi(\mathbf{X})$ is sub-Gaussian with $\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq K_\psi$ (allowing $\|\phi(\mathbf{X})\|_\infty$ to grow at the usual $\sqrt{\log p}$ rate).
- (b) (Response/link tails and smoothness) Y is uniformly sub-Gaussian (or bounded), and the GLM mean function and its derivative are uniformly bounded on a neighborhood of $\bar{\beta}$.
- (c) (Identifiability) The population Gram matrix is non-degenerate:

$$\lambda_{\min}(\mathbb{E}[\phi(\mathbf{X})\phi(\mathbf{X})^T]) \geq c > 0.$$

- (d) (Weight regularity) Any weights induced by γ (e.g., density-ratio type weights) have finite moments and are locally Lipschitz in γ on a neighborhood of $\bar{\gamma}$ (sufficient for the score stability bound used below).
- (e) (Restricted strong convexity) With probability tending to 1, $L_N(\cdot)$ satisfies an RSC inequality on the standard cone associated with $\text{supp}(\bar{\beta})$.
- (f) (Plug-in score domination) For the penalty level used in the generic update, the cross-fitted plug-in score term B defined in the proof satisfies $\Pr\{\|B\|_\infty \leq \lambda/4\} \rightarrow 1$. In the foldwise score-centered special case below, this condition holds with $B = \mathbf{0}$.

In our FACE-HD setting, we impose these regularity conditions in addition to [Assumption 5](#), [Assumption 6](#), and [Assumption 7](#) (noting that (a) matches [Assumption 5\(a\)](#), while (d) is ensured by the overlap/weight-moment assumptions and local smoothness); part (f) records

the plug-in score domination needed by the standard cone argument. Choose $\lambda \asymp \sqrt{\log(p)/N}$.

Then:

Remark 6 (On the choice of N in λ). The scaling $\lambda \asymp \sqrt{\log(p)/N}$ is stated in terms of the total sample size N for notational simplicity. In applications where the nuisance loss for a given arm/site pair is effectively averaged over an arm-specific subsample (e.g., treated units in the source), one may equivalently write $\lambda \asymp \sqrt{\log(p)/N_{\text{eff}}}$ where N_{eff} is the corresponding effective sample size. Under the sampling-fraction regime and overlap/positivity conditions stated above, $N_{\text{eff}} \asymp N$, so the rate statements are unchanged.

1. **(General)**

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

2. **(Special)** If, for each fold k , the conditional foldwise score at $\bar{\beta}$ satisfies

$$\mathbb{E}\left[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}\right] = \mathbf{0},$$

then the plug-in score term disappears and

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right).$$

Moreover, we can convert the ℓ_2 -rate to an ℓ_1 -rate by an explicit cone argument. Let $\Delta := \hat{\beta} - \bar{\beta}$ and $\mathcal{A}_\beta := \text{supp}(\bar{\beta})$ with $|\mathcal{A}_\beta| = s_\beta$. From the proof below (see the KKT-based cone inequality leading to [Eq. \(15\)](#)), on a high-probability event we have

$$\|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 3\|\Delta_{\mathcal{A}_\beta}\|_1 + \frac{2}{\lambda}\|B\|_\infty\|\Delta\|_1,$$

where $\|B\|_\infty$ is the plug-in score term induced by cross-fitting. By condition (f), $(2/\lambda)\|B\|_\infty \leq 1/2$ w.p.a.1, and hence

$$\|\Delta\|_1 = \|\Delta_{\mathcal{A}_\beta}\|_1 + \|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 8\|\Delta_{\mathcal{A}_\beta}\|_1 \leq 8\sqrt{s_\beta}\|\Delta\|_2.$$

Combining this with the ℓ_2 bounds above yields the corresponding ℓ_1 -rate:

$$\|\hat{\beta} - \bar{\beta}\|_1 = O_p\left(s_\beta \sqrt{\frac{\log(p)}{N}}\right) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Instantiation for FACE-HD calibrated losses. In our application, β is instantiated as either α^1 or γ_1 . If [Assumption 7\(d\)](#) holds (as in the foldwise score-centering condition used by [Hou et al. \(2025\)](#)), then we may invoke the **(Special)** case above and obtain

$$\begin{aligned} \|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_q &= O_p\left(s_\alpha^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \\ \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_q &= O_p\left(s_\gamma^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \quad q \in \{1, 2\}. \end{aligned}$$

Proof. We give a detailed argument in our notation; it follows the structure of [Hou et al. \(2025\)](#) (proof of Lemma A21). Write the (random) fold-averaged empirical loss as

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}), \quad \hat{\beta} \in \arg \min_{\beta} \{L_N(\beta) + \lambda \|\beta\|_1\}.$$

Let $\Delta := \hat{\beta} - \bar{\beta}$, and let $\mathcal{A}_\beta := \text{supp}(\bar{\beta})$ with $|\mathcal{A}_\beta| = s_\beta$.

Step 1 (Basic inequality). By optimality,

$$L_N(\bar{\beta} + \Delta) + \lambda \|\bar{\beta} + \Delta\|_1 \leq L_N(\bar{\beta}) + \lambda \|\bar{\beta}\|_1,$$

so

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \leq \lambda(\|\bar{\beta}\|_1 - \|\bar{\beta} + \Delta\|_1) \leq \lambda(\|\Delta_{\mathcal{A}_\beta}\|_1 - \|\Delta_{\mathcal{A}_\beta^c}\|_1).$$

Step 2 (RSC expansion). By convexity,

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \geq \nabla L_N(\bar{\beta})^T \Delta + \mathcal{R}_N(\Delta),$$

where the remainder $\mathcal{R}_N(\Delta)$ satisfies the RSC lower bound ([Assumption 7\(a\)](#) as applied to this loss)

$$\mathcal{R}_N(\Delta) \geq \kappa_1 \|\Delta\|_2^2 - \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2$$

on the standard restricted cone. Combining with Step 1 yields

$$\kappa_1 \|\Delta\|_2^2 \leq \|\nabla L_N(\bar{\beta})\|_\infty \|\Delta\|_1 + \lambda(\|\Delta_{\mathcal{A}_\beta}\|_1 - \|\Delta_{\mathcal{A}_\beta^c}\|_1) + \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2. \quad (14)$$

Step 3 (Score decomposition and choice of λ). For each fold, write

$$S_k := \tilde{E}_k[\nabla_\beta \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)})], \quad m_k := \mathbb{E}[S_k \mid \mathcal{D}^{(-k)}].$$

Decompose the fold-averaged score as

$$\nabla L_N(\bar{\beta}) = \underbrace{\frac{1}{K_f} \sum_{k=1}^{K_f} (S_k - m_k)}_{\text{stochastic term } A} + \underbrace{\frac{1}{K_f} \sum_{k=1}^{K_f} m_k}_{\text{plug-in term } B}.$$

For each fixed k , conditional on $\mathcal{D}^{(-k)}$, cross-fitting makes S_k an average of independent sub-exponential random vectors with uniformly controlled coordinates. Bernstein/Hoeffding concentration and a union bound over the coordinates give $\|S_k - m_k\|_\infty = O_p\{\sqrt{\log(p)/N_k}\}$. Because K_f is fixed and $N_k \asymp N$, summing these foldwise bounds (without conditioning simultaneously on the overlapping training samples) yields

$$\|A\|_\infty = O_p\left(\sqrt{\frac{\log(p)}{N}}\right).$$

Choose $\lambda \asymp \sqrt{\log(p)/N}$ so that $\lambda \geq 2\|A\|_\infty$ w.p.a.1.

For the plug-in term, note that

$$B - \mathbb{E}[\nabla_\beta \ell(D; \bar{\beta}, \bar{\gamma})] = \frac{1}{K_f} \sum_{k=1}^{K_f} \mathbb{E}[\nabla_\beta \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) - \nabla_\beta \ell(D; \bar{\beta}, \bar{\gamma}) \mid \mathcal{D}^{(-k)}],$$

and the weight-regularity/Lipschitz condition implies

$$\|B - \mathbb{E}[\nabla_\beta \ell(D; \bar{\beta}, \bar{\gamma})]\|_2 \leq C \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2.$$

This L_2 perturbation bound controls the additive plug-in term in the final oracle inequality.

The separate ℓ_∞ domination required for the cone step is imposed in condition (f). In the special case, $\mathbb{E}[\nabla_\beta \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}] = 0$ for all k , hence $B = \mathbf{0}$.

Step 4 (Cone property and ℓ_2 rate). Plugging $\|\nabla L_N(\bar{\beta})\|_\infty \leq \|A\|_\infty + \|B\|_\infty$ into Eq. (14) and using $\lambda \geq 2\|A\|_\infty$ yields the cone inequality

$$\|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 3\|\Delta_{\mathcal{A}_\beta}\|_1 + \frac{2}{\lambda}\|B\|_\infty\|\Delta\|_1. \quad (15)$$

On the event $\{(2/\lambda)\|B\|_\infty \leq 1/2\}$, Eq. (15) implies $\|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 7\|\Delta_{\mathcal{A}_\beta}\|_1$ and hence $\|\Delta\|_1 \leq 8\|\Delta_{\mathcal{A}_\beta}\|_1 \leq 8\sqrt{s_\beta}\|\Delta\|_2$. Plugging this back into the RSC lower bound (Assumption 7(a) as applied to this loss), using $B^T\Delta \leq \|B\|_2\|\Delta\|_2$ for the plug-in score contribution, and absorbing the $\kappa_2(\log(p)/N)\|\Delta\|_1^2$ term yields

$$\|\Delta\|_2 = O_p(\sqrt{s_\beta}\lambda) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

This is the general-case ℓ_2 rate. In the special case, $B = \mathbf{0}$ and the second term drops.

Step 5 (ℓ_1 rate). From the cone property and $\|\Delta_{\mathcal{A}_\beta}\|_1 \leq \sqrt{s_\beta}\|\Delta\|_2$,

$$\|\Delta\|_1 = O_p(s_\beta\lambda) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Finally, for FACE-HD, the **(Special)** conclusion may be invoked for $\beta = \alpha^1$ and $\beta = \gamma_1$ only when the foldwise score-centering condition is assumed/verified as in Assumption 7(d). Otherwise, one should use the **(General)** bounds, which explicitly propagate the plug-in nuisance error. $\square$

Lemma 5 (Orthogonality from Calibrated Losses). *Under Assumption 7(c), let $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$ be the population targets of the calibrated losses in the two-level algorithm (see Section S1; Eq. (6) and Eq. (7)):*

$$\begin{aligned} L_\gamma(\gamma; \alpha^{1,*}) &= \mathbb{E}_t \left[\nabla_\alpha g(\phi(\mathbf{X}); \alpha^{1,*})^T \gamma \right] + \mathbb{E}_s \left[I(A=1) e^{-\phi(\mathbf{X})^T \gamma} g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*})) \right] \\ L_\alpha(\alpha; \gamma_1^*) &= \mathbb{E}_s \left[w_\tau(\mathbf{X}; \gamma_1^*) \left(-Y \cdot (\phi(\mathbf{X})^T \alpha) + \Psi(\phi(\mathbf{X})^T \alpha) \right) \right] \end{aligned}$$

(Ignoring regularization terms, which vanish in population). For canonical-link GLMs,

$$\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X}) g'(\phi(\mathbf{X})^T \alpha).$$

Remark 7 (Penalization and approximate orthogonality). The population moment conditions below are derived from the unpenalized population losses. In finite samples, the ℓ_1 penalty induces a KKT subgradient term of order λ in the estimating equations. Under the nuisance rates in [Lemma 4](#) and the sparsity scaling used in [Theorem 3](#), this penalty-induced perturbation contributes at most an $o_p(N^{-1/2})$ term to the final DR remainder and therefore does not affect the first-order limit theory.

Under asymptotically inactive truncation ([Assumption 6\(c\)](#)), the population first-order optimality conditions are therefore equivalent to the following moment conditions at ϑ^* :

$$\begin{aligned} (i) \quad & \mathbb{E}_t \left[\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] - \mathbb{E}_s \left[w(\mathbf{X}; \gamma_1^*) \nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] = \mathbf{0} \\ (ii) \quad & \mathbb{E}_s \left[w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \{Y - g(\phi(\mathbf{X}); \alpha^{1,*})\} \right] = \mathbf{0}. \end{aligned}$$

Proof. We derive the moment conditions by taking the gradients of the population loss functions L_{γ} and L_{α} with respect to the parameter being optimized and setting them to zero at the minimum ϑ^* .

Condition (i) from FOC of $L_{\gamma}(\gamma; \alpha^{1,*})$ w.r.t. γ : The gradient of $L_{\gamma}(\gamma; \alpha^{1,*})$ with respect to γ is:

$$\begin{aligned} \nabla_{\gamma} L_{\gamma}(\gamma; \alpha^{1,*}) &= \mathbb{E}_t \left[\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] \\ &\quad + \mathbb{E}_s \left[I(A=1) (-\phi(\mathbf{X})) e^{-\phi(\mathbf{X})^T \gamma} g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*})) \right] \end{aligned}$$

Setting this gradient to zero at $\gamma = \gamma_1^*$ gives:

$$\mathbb{E}_t \left[\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] = \mathbb{E}_s \left[\phi(\mathbf{X}) w(\mathbf{X}; \gamma_1^*) g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*})) \right].$$

This equation is constructed to enforce the moment condition (i). Note that for standard GLMs with canonical links,

$$\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X}) g'(\phi(\mathbf{X})^T \alpha).$$

Truncation Error Analysis: Define the event $\mathcal{E}_{\alpha, N} = \{|\phi(\mathbf{X})^T \alpha^{1,*}| \leq M_{\tau, N}\}$. By [Assumption 6](#)(c), $P(\mathcal{E}_{\alpha, N}^c \mid R = s) = o(N^{-1})$. On $\mathcal{E}_{\alpha, N}$, $\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}) = \phi(\mathbf{X})^T \alpha^{1,*}$, so $g'(\mathcal{T}(\cdot)) = g'(\cdot)$ exactly. Therefore, replacing $g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}))$ by $g'(\phi(\mathbf{X})^T \alpha^{1,*})$ in the population FOC changes the source-side moment by at most an $o(1)$ term, and condition (i) holds asymptotically.

Condition (ii) from FOC of $L_{\alpha}(\alpha; \gamma_1^*)$ w.r.t. α : The loss L_{α} is the population weighted GLM log-likelihood. Its gradient with respect to α is:

$$\nabla_{\alpha} L_{\alpha}(\alpha; \gamma_1^*) = \mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \gamma_1^*) \left(-Y \cdot \phi(\mathbf{X}) + \nabla_{\alpha} \Psi(\phi(\mathbf{X})^T \alpha) \right) \right]$$

Using the GLM identity $\nabla_{\alpha} \Psi(u) = g(u) \phi(\mathbf{X})$ where $u = \phi(\mathbf{X})^T \alpha$:

$$\nabla_{\alpha} L_{\alpha}(\alpha; \gamma_1^*) = \mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \left(g(\phi(\mathbf{X})^T \alpha) - Y \right) \right]$$

Setting this gradient to zero at $\alpha = \alpha^{1,*}$ gives:

$$\mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \left(Y - g(\phi(\mathbf{X})^T \alpha^{1,*}) \right) \right] = \mathbf{0}.$$

Finally, under [Assumption 6](#)(c), truncation is asymptotically inactive at γ_1^* , so $w_{\tau}(\mathbf{X}; \gamma_1^*) = w(\mathbf{X}; \gamma_1^*)$ w.p.a.1. This yields condition (ii). $\square$

Lemma 6 (Bias-Variance Bounds). *Let $\vartheta_n = (\alpha_n^1, \gamma_{1,n})$ be a sequence of nuisance parameters converging to $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$.*

1. **(Bias Bound - Double Robustness)** *If ϑ^* minimizes the population calibrated losses ([Lemma 5](#) holds), then:*

$$\begin{aligned} |\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n)] - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)]| &\leq C \cdot \left(\|\alpha_n^1 - \alpha^{1,*}\|_2^2 + \|\gamma_{1,n} - \gamma_1^*\|_2^2 \right. \\ &\quad \left. + \|\alpha_n^1 - \alpha^{1,*}\|_2 \cdot \|\gamma_{1,n} - \gamma_1^*\|_2 \right) \end{aligned}$$

2. (Variance Bound)

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)) \leq C \cdot (\|\alpha_n^1 - \alpha^{1,*}\|_2^2 + \|\gamma_{1,n} - \gamma_1^*\|_2^2)$$

Proof. We analyze the bias term $\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)]$. Let $\Delta_\alpha = \alpha_n^1 - \alpha^{1,*}$ and $\Delta_\gamma = \gamma_{1,n} - \gamma_1^*$. We decompose the bias using an intermediate step:

$$\text{Bias} = \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha_n^1, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n})]}_{\text{Bias from } \alpha} + \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]}_{\text{Bias from } \gamma}$$

Step 1 (Bias from α : mean-value expansion + orthogonality). Define the path $h_\alpha(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})]$ for $t \in [0, 1]$. By the fundamental theorem of calculus (and dominated convergence, justified by [Assumption 6\(b\)](#) and local boundedness of the derivatives), we have

$$h_\alpha(1) - h_\alpha(0) = \int_0^1 h'_\alpha(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})]^T \Delta_\alpha dt.$$

Hence

$$\begin{aligned} \text{Bias from } \alpha &= \left(\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right)^T \Delta_\alpha + \int_0^1 \left\{ \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] \right. \\ &\quad \left. - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right\}^T \Delta_\alpha dt. \end{aligned}$$

The first term is zero by [Lemma 5\(i\)](#). For the second term, we use Cauchy–Schwarz:

$$|\text{Bias from } \alpha| \leq \|\Delta_\alpha\|_2 \int_0^1 \left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right\|_2 dt.$$

The mapping $\vartheta \mapsto \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \vartheta)]$ is locally Lipschitz in ϑ by the explicit derivative formula below, the bounded-slope/Lipschitz condition on g' ([Assumption 6\(b\)](#)), and the bounded-weights/moment conditions ([Assumption 6\(a,b\)](#)). Therefore, for ϑ_n sufficiently close to ϑ^* ,

$$\left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right\|_2 \leq C (\|\Delta_\alpha\|_2 + \|\Delta_\gamma\|_2),$$

uniformly over $t \in [0, 1]$, and hence

$$|\text{Bias from } \alpha| \leq C (\|\Delta_\alpha\|_2^2 + \|\Delta_\alpha\|_2 \|\Delta_\gamma\|_2).$$

Step 2 (Bias from γ : mean-value expansion + orthogonality). Similarly define

$h_\gamma(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]$. Then

$$\text{Bias from } \gamma = \int_0^1 \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]^T \Delta_\gamma dt.$$

Adding and subtracting $\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)]$, the leading term is zero by [Lemma 5\(ii\)](#), and the remainder is bounded (again by local Lipschitz continuity and Cauchy–Schwarz) as

$$|\text{Bias from } \gamma| \leq C \|\Delta_\gamma\|_2^2.$$

Step 3 (Collecting). Combining the two displays yields the stated second-order bias bound.

Derivative identities used above. For example, the gradient w.r.t. $\boldsymbol{\alpha}$ at $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ is:

$$\mathbb{E}[\nabla_{\boldsymbol{\alpha}} \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] = \mathbb{E}_t[\nabla_{\boldsymbol{\alpha}} g(\cdot; \boldsymbol{\alpha}^{1,*})] - \mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \nabla_{\boldsymbol{\alpha}} g(\cdot; \boldsymbol{\alpha}^{1,*})]$$

where $w(\mathbf{X}; \boldsymbol{\gamma}) = I(A = 1)/e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}}$. This is exactly $\mathbf{0}$ by [Lemma 5\(i\)](#). Similarly, the gradient w.r.t. $\boldsymbol{\gamma}$ is:

$$\begin{aligned} \mathbb{E}[\nabla_{\boldsymbol{\gamma}} \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] &= \mathbb{E}_s[\nabla_{\boldsymbol{\gamma}} w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] \\ &= -\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X}) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] \end{aligned}$$

This is exactly $\mathbf{0}$ by [Lemma 5\(ii\)](#). This proves part 1.

Part 2 (Variance): explicit L_2 -Lipschitz bound. Write the difference as

$$\tilde{\varphi}_{t,s}(D; \boldsymbol{\vartheta}_n) - \tilde{\varphi}_{t,s}(D; \boldsymbol{\vartheta}^*) = T_t(D) + T_s(D),$$

where

$$T_t(D) := \frac{I(R=t)}{\rho_t} \{g(\phi(\mathbf{X}); \boldsymbol{\alpha}_n^1) - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\},$$

and

$$\begin{aligned} T_s(D) &:= \frac{I(R=s)}{\rho_s} \left(w(\mathbf{X}; \boldsymbol{\gamma}_{1,n}) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}_n^1)\} - w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\} \right) \\ &= \frac{I(R=s)}{\rho_s} \{T_{s,1}(D) + T_{s,2}(D)\}. \end{aligned}$$

where

$$T_{s,1}(D) := \{w(\mathbf{X}; \gamma_{1,n}) - w(\mathbf{X}; \gamma_1^*)\} \{Y - g(\phi(\mathbf{X}); \alpha^{1,*})\},$$

$$T_{s,2}(D) := w(\mathbf{X}; \gamma_{1,n}) \{g(\phi(\mathbf{X}); \alpha^{1,*}) - g(\phi(\mathbf{X}); \alpha_n^1)\}.$$

By $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$ and $\text{Var}(Z) \leq \mathbb{E}[Z^2]$,

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)) \leq 3 \mathbb{E}[T_t(D)^2] + 3 \mathbb{E}[T_{s,1}(D)^2] + 3 \mathbb{E}[T_{s,2}(D)^2].$$

For T_t and $T_{s,2}$, the bounded-slope assumption on g ([Assumption 6\(b\)](#)) implies $|g(u_1) - g(u_2)| \leq L_g |u_1 - u_2|$, hence

$$\mathbb{E}[T_t(D)^2] + \mathbb{E}[T_{s,2}(D)^2] \leq C \mathbb{E}[\|\phi(\mathbf{X})\|_2^2] \|\Delta_\alpha\|_2^2,$$

using the weight-moment condition ([Assumption 6\(a\)](#)) and finite second moments of $\phi(\mathbf{X})$ ([Assumption 5\(a\)](#)). For $T_{s,1}$, local L_2 -Lipschitz continuity of $w(\mathbf{X}; \gamma)$ in γ (implied by the exponential form of w , the overlap/weight-moment assumptions, and the moment condition in [Assumption 6\(b\)](#)) yields

$$\mathbb{E}[T_{s,1}(D)^2] \leq C \|\Delta_\gamma\|_2^2.$$

Combining the bounds gives the stated variance inequality. $\square$

Lemma 7 (Double Robustness of Bias Terms). *Let $\Delta_\alpha = \alpha_n^1 - \alpha^{1,*}$ and $\Delta_\gamma = \gamma_{1,n} - \gamma_1^*$.*

Define the two first-order bias components:

$$\mathcal{B}_\alpha(\alpha_n^1; \gamma_1^*) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha_n^1, \gamma_1^*) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]$$

$$\mathcal{B}_\gamma(\gamma_{1,n}; \alpha^{1,*}) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]$$

Under the conditions:

1. If [Lemma 5\(i\)](#) holds (from calibration w.r.t. γ), then $\mathcal{B}_\alpha(\alpha_n^1; \gamma_1^*) = O(\|\Delta_\alpha\|_2^2)$.
2. If [Lemma 5\(ii\)](#) holds (from calibration w.r.t. α), then $\mathcal{B}_\gamma(\gamma_{1,n}; \alpha^{1,*}) = O(\|\Delta_\gamma\|_2^2)$.

Note: For identification of the target parameter, [Theorem 3](#) still requires at least one nuisance model to be correctly specified so that $\mu^1(\vartheta^) = \mu^1$. This is formalized in [Lemma 1](#).*

Proof. 1. Analyzing $\mathcal{B}_\alpha$ using Lemma 5(i) (mean-value expansion). Let $h_1(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)]$ for $t \in [0, 1]$. Then $\mathcal{B}_\alpha = h_1(1) - h_1(0)$ and

$$\mathcal{B}_\alpha = \int_0^1 h_1'(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)]^T \Delta_\alpha dt.$$

Add and subtract $\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)]$ inside the integral to get

$$\begin{aligned} \mathcal{B}_\alpha &= \left(\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right)^T \Delta_\alpha + \int_0^1 \left\{ \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)] \right. \\ &\quad \left. - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\}^T \Delta_\alpha dt. \end{aligned}$$

The first term is zero by Lemma 5(i). For the second term, we bound the gradient difference. From the definition of $\tilde{\varphi}_{t,s}$ and the canonical-link GLM identity $\nabla_\alpha g(\phi(\mathbf{X}); \boldsymbol{\alpha}) = \phi(\mathbf{X})g'(\phi(\mathbf{X})^T \boldsymbol{\alpha})$,

$$\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}, \boldsymbol{\gamma}_1^*) = \frac{I(R=t)}{\rho_t} \phi(\mathbf{X})g'(\phi(\mathbf{X})^T \boldsymbol{\alpha}) - \frac{I(R=s)}{\rho_s} w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X})g'(\phi(\mathbf{X})^T \boldsymbol{\alpha}).$$

Using the Lipschitz property of g' (Assumption 6(b)) and the weight-moment condition (Assumption 6(a)), we have

$$\left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\|_2 \leq C \|\Delta_\alpha\|_2,$$

uniformly over $t \in [0, 1]$, where $C < \infty$ depends on ρ_t, ρ_s , the moment bound on $\phi(\mathbf{X})$ (Assumption 5(a)), and $L_{g'}$. Therefore, by Cauchy–Schwarz,

$$|\mathcal{B}_\alpha| \leq \int_0^1 C \|\Delta_\alpha\|_2 \cdot \|\Delta_\alpha\|_2 dt = O(\|\Delta_\alpha\|_2^2).$$

2. Analyzing $\mathcal{B}_\gamma$ using Lemma 5(ii) (mean-value expansion). Let $h_2(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]$. Then $\mathcal{B}_\gamma = h_2(1) - h_2(0)$ and

$$\mathcal{B}_\gamma = \int_0^1 \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]^T \Delta_\gamma dt.$$

Add and subtract the gradient at $\boldsymbol{\gamma}_1^*$:

$$\begin{aligned} \mathcal{B}_\gamma &= \left(\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right)^T \Delta_\gamma + \int_0^1 \left\{ \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)] \right. \\ &\quad \left. - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\}^T \Delta_\gamma dt. \end{aligned}$$

The first term is zero by [Lemma 5\(ii\)](#). For the remainder, recall $w(\mathbf{X}; \gamma) = I(A = 1) \exp(-\phi(\mathbf{X})^T \gamma)$ so that $\nabla_\gamma w(\mathbf{X}; \gamma) = -w(\mathbf{X}; \gamma) \phi(\mathbf{X})$. Thus

$$\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma) = \frac{I(R=s)}{\rho_s} \nabla_\gamma w(\mathbf{X}; \gamma) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}.$$

Under [Assumption 6\(a,b\)](#), the mapping $\gamma \mapsto \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma)]$ is locally Lipschitz in γ , hence

$$\left\| \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma_1^* + t\Delta_\gamma)] - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma_1^*)] \right\|_2 \leq C \|\Delta_\gamma\|_2$$

uniformly over $t \in [0, 1]$. Therefore $|\mathcal{B}_\gamma| \leq \int_0^1 C \|\Delta_\gamma\|_2^2 dt = O(\|\Delta_\gamma\|_2^2)$.

Clarification on Double Robustness: The key insight is that the calibrated loss functions are designed such that:

- [Lemma 5\(i\)](#) holds *by construction* of L_γ at any ϑ^* that minimizes L_γ , regardless of whether the weighting model is correct.
- [Lemma 5\(ii\)](#) holds *by construction* of L_α at any ϑ^* that minimizes L_α , regardless of whether the outcome model is correct.

Thus the first-order sensitivity to each nuisance can be removed by calibration; combined with model correctness of at least one nuisance for identification, this yields the double-robust conclusion in [Theorem 3](#). □

S2.5 Extension to the Control Mean and the TATE

The results above were stated for the treated potential outcome mean $\mu^1 = \mathbb{E}_t[Y(1)]$ for notational simplicity. The same proof strategy applies to the control mean μ^0 by replacing the treatment level 1 by 0 throughout (including the corresponding nuisance parameters $(\boldsymbol{\alpha}^0, \gamma_0)$ and weights). In particular, define the control-arm weight

$$w^0(\mathbf{X}; \gamma_0) := \frac{I(A=0)}{e^{\phi(\mathbf{X})^T \gamma_0}},$$

and the cross-fitted estimator

$$\hat{\mu}_{t,s}^{0,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k} \left[g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s}^{0,(-k)}) \right] + \tilde{E}_{s,k} \left[w^0(\mathbf{X}; \hat{\gamma}_{s,0}^{(-k)}) (Y - g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s}^{0,(-k)})) \right] \right\}.$$

Let $\vartheta^{0,*} = (\boldsymbol{\alpha}^{0,*}, \gamma_0^*)$ denote the population targets of the control-arm calibrated losses (the same construction as for μ^1 but with $I(A=0)$), and define the corresponding centered influence function $\varphi_{t,s}^0(D; \vartheta^{0,*})$ analogously.

Theorem 4 (Asymptotic Normality of $\hat{\mu}_{t,s}^{0,\text{CF}}$). *Under the analogues of [Assumptions 5 to 7](#) for the control arm, and assuming identification of $\mu^0(\vartheta^{0,*}) = \mu^0$ under outcome-model or weight-model correctness (control-arm analogue of [Lemma 1](#)), if the total sparsity satisfies $\frac{s_{\text{tot}}^2 \log^2(p)}{N} \rightarrow 0$, then*

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{0,\text{CF}} - \mu^0 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{0,*}), \quad V_{t,s}^{0,*} := \text{Var}(\varphi_{t,s}^0(D; \vartheta^{0,*})).$$

Proof. The proof is identical to that of [Theorem 3](#) after replacing the treatment level 1 by 0 in the definition of the score contribution, the calibrated moment conditions, and the corresponding nuisance estimators. $\square$

Corollary 1 (Asymptotic Normality of the TATE). *With $\tau = \mu^1 - \mu^0$ as defined above, define the cross-fitted TATE estimator*

$$\hat{\tau}_{t,s}^{\text{CF}} := \hat{\mu}_{t,s}^{1,\text{CF}} - \hat{\mu}_{t,s}^{0,\text{CF}}.$$

Under the conditions of [Theorem 3](#) and [Theorem 4](#), we have the asymptotic linear representation

$$\sqrt{N} \left(\hat{\tau}_{t,s}^{\text{CF}} - \tau \right) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \left(\varphi_{t,s}(D_i; \vartheta^*) - \varphi_{t,s}^0(D_i; \vartheta^{0,*}) \right) + o_p(1),$$

and hence

$$\sqrt{N} \left(\hat{\tau}_{t,s}^{\text{CF}} - \tau \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{\tau,*}), \quad V_{t,s}^{\tau,*} := \text{Var}(\varphi_{t,s}(D; \vartheta^*) - \varphi_{t,s}^0(D; \vartheta^{0,*})).$$

The corollary gives a sufficient route through separate arm identification. For aggregation, a weaker estimand-level route is possible. Let $\mu_{t,s}^{a,*} = \text{plim } \hat{\mu}_{t,s}^{a,\text{CF}}$. If $\mu_{t,s}^{1,*} - \mu_{t,s}^{0,*} = \tau$ and the centered arm contrast admits the same influence-function expansion with an $o_p(N^{-1/2})$ remainder, then the source is first-order valid for the TATE even when neither arm-specific pseudo-limit equals μ^a . The contrast expansion is imposed explicitly in [Assumption 8\(e\)](#); equality of the two pseudo-limit biases alone is not sufficient.

S3 Additional Simulation Results

The direct-TATE rerun uses $p = 100$ for C(1)–C(3). The archived figures below report the preceding treated-mean results at $p = 50$: C(2) omits the quadratic outcome terms, while C(3) omits the quadratic site/propensity terms. Because the skew-normal source-to-target density ratio is not exactly log-quadratic, C(1) is a rich working-model configuration rather than literal joint parametric correctness. These figures are retained for reproducibility and are not part of the new direct-TATE simulation array. They show that FACE-HD is markedly more stable than the non-adaptive aggregators as source heterogeneity grows, but they should not be interpreted as showing the lowest RMSE at every value of ρ or nominal coverage in every C(3) setting. Direct-TATE rerun figures at $p = 100$ are reported separately using the same seeds and data-generating settings.

[Table 4](#) records the bounded pilot used to lock the production Wald penalty-activation cutoff. The pilot used ten seeds (501–510) disjoint from production seeds 1–500 and did not use coverage in the selection rule. All candidates satisfied the no-deviation efficiency screen, and $c = 1$ minimized the worst RMSE at $\rho \in \{1, 2.5\}$. These ten-run values are a tuning audit rather than final performance estimates; the 500-replicate cutoff sensitivity is reported with the production analysis.

To isolate the C(3) centering mechanism before adaptive source aggregation, we also ran

Table 4: Coverage-blind cutoff-locking pilot for $C(1)$, $p = 100$, and $K = 2$. Each RMSE uses ten disjoint pilot replicates. “Eligible” means that the $\rho = 0$ RMSE is at most 1.05 times the smallest value in that column.

Cutoff c	Multiplier λ	RMSE ($\rho = 0$)	RMSE ($\rho = 1$)	RMSE ($\rho = 2.5$)	Eligible/selected
1.0	1.000	0.01957	0.03105	0.02938	Yes/Yes
1.5	0.667	0.01936	0.03623	0.03847	Yes/No
2.0	0.500	0.01941	0.04239	0.05061	Yes/No
2.5	0.400	0.01959	0.04677	0.06400	Yes/No
3.0	0.333	0.01970	0.04964	0.07414	Yes/No

the target-only TATE estimator for 500 replicates at $p = 100$, $K = 4$, and $\rho = 0$, replacing either nuisance prediction by its known DGP value on each held-out fold. [Table 5](#) reports the result. The fitted–fitted estimator has a material negative bias even though its standard error is slightly conservative relative to the empirical spread. Either oracle nuisance removes nearly all of that bias and restores nominal coverage, which is the finite-sample signature of a doubly robust product remainder. This focused experiment diagnoses the target anchor; it is not substituted for the full direct-TATE comparison array.

Because the four variants use the same replicate and folds, their paired inclusion–exclusion contrast isolates the interaction of the two nuisance errors. Specifically, fitted/fitted minus oracle-outcome minus oracle-propensity plus oracle/oracle has mean -0.0159 and Monte Carlo SE 0.0003 . This accounts for nearly all of the fitted/fitted centering bias and supports the product-remainder diagnosis rather than a variance-scale failure. Under the one-standard-error target rule, the same interaction remainder is -0.0231 (Monte Carlo SE 0.0003). Its paired difference from the primary rule is -0.00721 (Monte Carlo SE 0.00017), while the paired fitted/fitted estimate shift is -0.00723 (Monte Carlo SE 0.00024). The

Table 5: C(3) target-only TATE oracle-remainder diagnostic, based on 500 paired replicates at $p = 100$, $K = 4$, and $\rho = 0$. SD is the empirical Monte Carlo standard deviation and mean SE is the average influence-function standard error. The primary simulation analysis uses the minimum-validation-loss rule; the lower panel changes only the fitted target nuisance rule.

Nuisance predictions	Bias	SD	Mean SE	RMSE	Coverage
<i>Minimum-validation-loss rule (primary)</i>					
Fitted outcome, fitted propensity	−0.0178	0.0290	0.0302	0.0341	0.918
Oracle outcome, fitted propensity	−0.0021	0.0279	0.0290	0.0280	0.952
Fitted outcome, oracle propensity	−0.0018	0.0319	0.0324	0.0319	0.952
Oracle outcome, oracle propensity	−0.0019	0.0307	0.0311	0.0307	0.950
<i>One-standard-error rule (target-nuisance sensitivity)</i>					
Fitted outcome, fitted propensity	−0.0251	0.0292	0.0302	0.0384	0.872
Oracle outcome, fitted propensity	−0.0022	0.0275	0.0287	0.0275	0.954
Fitted outcome, oracle propensity	−0.0018	0.0324	0.0328	0.0324	0.958
Oracle outcome, oracle propensity	−0.0019	0.0307	0.0311	0.0307	0.950

double-oracle rows agree exactly replicate by replicate, so the deterioration is not a change in the generated data, folds, or truth. Stronger target regularization therefore increases rather than removes the finite-sample product remainder; the primary simulation minimum-loss rule is retained without coverage-based retuning.

The $\rho = 0$ endpoint fixes the conditional treatment log-odds shift at one in every site, but it does not force the raw site-specific marginal risk differences to be identical. [Table 6](#) evaluates those true marginal effects directly under each site’s covariate law. This separates the small covariate-shift/non-collapsibility discrepancy from nuisance estimation and adaptive

Table 6: True site-marginal TATEs at $\rho = 0$, $p = 100$, and $K = 4$.

Site	Skewness ν_r	Marginal TATE	Difference from target
Target	0.00	0.2063	0.0000
s_1	0.05	0.2037	-0.0026
s_2	0.10	0.2014	-0.0048
s_3	0.15	0.1993	-0.0070
s_4	0.20	0.1970	-0.0093

weighting, and helps explain why SS or IVW can exchange a small centering error for a larger variance reduction at this endpoint. The four source-law treated means exceed the target $\mu^1 = 0.6927$ by 0.0079, 0.0153, 0.0218, and 0.0292, while their control-mean differences are 0.0105, 0.0202, 0.0288, and 0.0384; subtracting the arms partially cancels these shifts and yields the smaller TATE differences in the table. The calculation uses 500,000 reference draws per site, with every integration Monte Carlo SE for a TATE contrast below 9×10^{-5} .

Legacy lower-dimensional setting ($p = 50$). The archived treated-mean study used the lower dimension $p = 50$ (with $n = 1000$ observations per site, so $N = 1000(K + 1)$ in total). [Figure 6](#) shows C(1), which uses quadratic working bases for both nuisances. At $\rho = 0$, all sites share the conditional outcome mechanism and treatment log-odds shift. A non-adaptive estimator may attain smaller finite-sample RMSE when its variance reduction outweighs the modest covariate-shift-induced differences in site-specific marginal effects, and no adaptation is needed. As ρ increases, FACE-HD gives a substantially more stable bias-coverage tradeoff, whereas SS, IVW, Federated-DR, and Pooled-DR become biased and lose coverage. The C(2)–C(3) figures at $p = 50$ display the same qualitative robustness pattern, with some finite-sample undercoverage under C(3).

One-round versus two-round. [Figure 7](#) compares the archived arm-wise TATE implementations of one-round and two-round FACE-HD in a continuous-outcome setting where the sources remain mean-compatible but their conditional outcome models differ in shape from the target’s—a source-specific effect modification of strength s , centered so that each source’s target-population mean treatment effect is unchanged. Across the full range of s and at both $p = 10$ and $p = 50$, the one-round and two-round curves track one another closely in bias, RMSE, coverage, and interval length, and both are more efficient than the target-only anchor. This is descriptive evidence rather than a formal equivalence test: the archived cells contain 80 completed replicates at $p = 10$ and 52–60 at $p = 50$. It supports adopting the communication-light one-round version as the default initialization; the two-round variant remains available as an alternative that relaxes the target-initialization alignment condition. This diagnostic predates the common-weight direct-TATE aggregation and is retained without relabeling it as the new estimator.

S4 Aggregated Estimator: FACE-HD

This section develops the aggregation theory for the direct-TATE estimator in the main manuscript. Arm-specific nuisances are fitted first, but their influence-function components are contrasted before weights, Wald statistics, or variances are calculated. The main result is [Theorem 5](#).

The proof proceeds in five steps. First, [Lemma 8](#) gives the quadratic TATE variance $L_\tau^*(\eta)$. Second, [Lemma 9](#) establishes uniform convergence of its empirical analogue. Third, [Lemma 11](#) proves TATE-level source selection. Fourth, [Lemma 12](#) makes estimated and oracle weights first-order equivalent. Finally, the multisite CLT and Slutsky’s theorem give valid TATE inference.

S4.1 Setup and Notation for Aggregation

Following Eq. (9), aggregate the source-assisted TATE estimators $\{\hat{\tau}_{t,s_j}\}_{j=1}^K$ with the target-only TATE $\hat{\tau}_{ot}$ using one common weight vector. Define the global sample size across all participating sites as

$$N_{\text{all}} := N_t + \sum_{j=1}^K N_{s_j},$$

and assume $N_t/N_{\text{all}} \rightarrow \rho_t^{\text{all}} \in (0, 1)$ and $N_{s_j}/N_{\text{all}} \rightarrow \rho_{s_j}^{\text{all}} \in (0, 1)$ for each fixed j .

Notation correspondence with the main manuscript:

- $\hat{\tau}_{ot} = \hat{\mu}_{ot}^1 - \hat{\mu}_{ot}^0$: target-only TATE;
- $\hat{\tau}_{t,s_j} = \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$: source-assisted TATE from s_j ;
- $\varphi_{ot,i}^\tau = \varphi_{ot,i}^1 - \varphi_{ot,i}^0$, $\zeta_{j,i}^\tau = \zeta_{j,i}^1 - \zeta_{j,i}^0$, and $\xi_{j,i}^\tau = \xi_{j,i}^1 - \xi_{j,i}^0$: treatment-contrast influence components.

The direct-TATE estimator is

$$\hat{\tau}_{\text{agg}}(\eta) = \hat{\tau}_{ot} + \sum_{j=1}^K \eta_j (\hat{\tau}_{t,s_j} - \hat{\tau}_{ot}).$$

Lemma 8 (Variance-Covariance Structure for Aggregation). *For each source site s_j , let $\varphi_{t,s_j,i}^\tau$ denote the population influence contribution of $\hat{\tau}_{t,s_j}$ under common $\sqrt{N_{\text{all}}}$ scaling, and let $\varphi_{ot,i}^\tau$ denote the target-only TATE contribution. Write*

$$\varphi_{t,s_j,i}^\tau = \zeta_{j,i}^\tau + \xi_{j,i}^\tau.$$

Then, for $j \neq k$,

$$\text{Cov}(\xi_j^\tau, \xi_k^\tau) = 0, \quad \text{Cov}(\zeta_j^\tau, \xi_k^\tau) = 0, \quad \text{Cov}(\varphi_{ot}^\tau, \xi_k^\tau) = 0,$$

because these terms are based on disjoint source samples or on target-versus-source samples that are independent across sites. Consequently, all cross-site covariance terms arise only

through the shared target component:

$$\text{Cov}(\varphi_{t,s_j}^\tau, \varphi_{t,s_k}^\tau) = \text{Cov}(\zeta_j^\tau, \zeta_k^\tau), \quad \text{Cov}(\varphi_{ot}^\tau, \varphi_{t,s_j}^\tau) = \text{Cov}(\varphi_{ot}^\tau, \zeta_j^\tau).$$

In particular, the population variance of the aggregated estimator can be written as a quadratic form

$$L_\tau^*(\eta) = \eta^T H_\tau \eta + \mathbf{d}_\tau^T \eta + c_\tau,$$

where the off-diagonal entries of H are induced entirely by the shared target sample.

Proof. The target sample is independent of every source sample, and source samples are mutually independent. Taking an arm contrast changes neither this independence nor the site on which a component is evaluated. Hence source–source and target–source covariances vanish as displayed. Expanding the variance of

$$\varphi_{\text{agg}}^\tau(\eta) = \varphi_{ot}^\tau + \sum_{j=1}^K \eta_j (\varphi_{t,s_j}^\tau - \varphi_{ot}^\tau)$$

then gives the stated quadratic representation, with off-diagonal terms coming solely from the shared target contribution. $\square$

Remark 8 (Relation to FACE as a special case). [Lemma 8](#) is compatible with the covariance decomposition underlying [Han et al. \(2025\)](#). In particular, if the source-assisted estimators are specialized to the FACE construction and the nuisance structure is simplified to the corresponding low-dimensional setting, then the same argument shows that cross-source covariance is induced only by the shared target sample. Thus the FACE variance formula can be viewed as a special case of the present decomposition, whereas the current statement is written to accommodate the more general high-dimensional calibrated nuisance estimators used in FACE-HD.

S4.2 Penalized Variance Estimation

Define the empirical direct-TATE loss:

$$\widehat{L}_\tau(\eta) = N_{\text{all}} \widehat{\text{Var}}\{\widehat{\tau}_{\text{agg}}(\eta)\}$$

which is a quadratic function of η (equivalently, N_{all} times the variance expression in [Eq. \(10\)](#) in the main manuscript):

$$\widehat{L}_\tau(\eta) = \eta^T \widehat{H}_\tau \eta + \widehat{\mathbf{d}}_\tau^T \eta + \widehat{c}_\tau$$

where every component is calculated after taking the arm contrast.

For the implemented two-level cross-fitting, let $\widehat{L}_{\tau,-k}$ denote this objective calculated only from the outer training sample $\mathcal{T}_k$, and let $\widehat{\eta}^{(-k)}$ be its penalized minimizer. In the supporting arguments below, $\widehat{L}_\tau$ and $\widehat{\eta}$ are shorthand for an arbitrary fixed outer fold's $\widehat{L}_{\tau,-k}$ and $\widehat{\eta}^{(-k)}$. Since K_f is fixed, every $o_p(1)$ statement holds uniformly over $k \leq K_f$. The final held-out pseudo-value variance $\widehat{V}_\tau$ is a distinct finite-sample quantity; both it and the foldwise training objectives evaluated at their minimizers converge to $L_\tau^*(\eta_\tau^*)$.

Lemma 9 (Consistency of Variance/Covariance Components). *Assume [Assumptions 5 to 7](#) for both arms, and let K be fixed. Let $H_\tau, \mathbf{d}_\tau, c_\tau$ denote the population analogues. Define*

$$a_N := \max_{a \in \{0,1\}} \max_{j \leq K} \max_{k \leq K_f} \left\{ \|\widehat{\boldsymbol{\alpha}}_{t,s_j}^{a,(-k)} - \boldsymbol{\alpha}_{t,s_j}^{a,*}\|_2 + \|\widehat{\boldsymbol{\gamma}}_{s_j,a}^{(-k)} - \boldsymbol{\gamma}_{s_j,a}^*\|_2 \right\} = o_p(1).$$

Then:

$$\|\widehat{H}_\tau - H_\tau\|_{\text{op}} + \|\widehat{\mathbf{d}}_\tau - \mathbf{d}_\tau\|_2 + |\widehat{c}_\tau - c_\tau| = o_p(1),$$

and consequently, for any fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$\sup_{\eta \in \mathcal{B}} |\widehat{L}_\tau(\eta) - L_\tau^*(\eta)| = o_p(1).$$

If the products of the relevant influence-function components have uniformly bounded second moments, the empirical-averaging part is $O_p(N_{\text{all}}^{-1/2})$, giving the sharper bound $O_p(N_{\text{all}}^{-1/2}) + O_p(a_N)$. This sharper rate is not needed for the FACE-HD oracle limit.

Proof. For readability in this proof only, suppress the subscript τ on $\widehat{H}_\tau, H_\tau, \widehat{\mathbf{d}}_\tau, \mathbf{d}_\tau, \widehat{c}_\tau, c_\tau, \widehat{L}_\tau$, and L_τ^* . We prove entrywise consistency and then convert to the stated matrix/vector norms using that K is fixed.

Step 1 (Generic moment form). Each element of $\widehat{H}$, $\widehat{\mathbf{d}}$ and $\widehat{c}$ is a (finite) linear combination of empirical averages of the form

$$\widehat{M}_r = \frac{1}{N_r} \sum_{i \in \mathcal{D}_r} f(D_i; \widehat{\vartheta}^{(-\text{fold}(i))}), \quad r \in \{t, s_1, \dots, s_K\},$$

where f is a product of two empirical influence-function components (including the cross-site covariance terms computed on the *same* target sample), and $\widehat{\vartheta}^{(-\text{fold}(i))}$ denotes the cross-fitted nuisance estimate used to evaluate the fold containing i . Let M_r be the population analogue obtained by replacing the sample average with $\mathbb{E}_r[\cdot]$ and $\widehat{\vartheta}^{(-\text{fold}(i))}$ with the population target ϑ^* .

Step 2 (Conditional LLN and plug-in stability). Partition the site- r average by evaluation fold:

$$\widehat{M}_r = \sum_{k=1}^{K_f} \frac{N_{r,k}}{N_r} \widehat{M}_{r,k}, \quad \widehat{M}_{r,k} = \frac{1}{N_{r,k}} \sum_{i \in \mathcal{D}_{r,k}} f(D_i; \widehat{\vartheta}^{(-k)}).$$

For each fixed k , conditional on $\mathcal{D}^{(-k)}$, the nuisance estimate is fixed and the evaluation-fold summands in $\widehat{M}_{r,k}$ are independent. The conditional weak law of large numbers therefore gives

$$\widehat{M}_{r,k} - \mathbb{E}_r[f(D; \widehat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] = o_p(1).$$

Because K_f is fixed, summing these foldwise statements gives

$$\widehat{M}_r - \sum_{k=1}^{K_f} \frac{N_{r,k}}{N_r} \mathbb{E}_r[f(D; \widehat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] = o_p(1).$$

This conditions on one fold's training sample at a time, rather than jointly on the overlapping collection $\{\mathcal{D}^{(-k)}\}_{k=1}^{K_f}$. If the products have uniformly bounded second moments,

each foldwise empirical term is $O_p(N_{r,k}^{-1/2})$, and the fixed weighted sum is $O_p(N_r^{-1/2})$. **Assumption 6(b)** gives the required uniform integrability in a neighborhood of ϑ^* . Next, by **Assumption 7(d)**, $\widehat{\vartheta}^{(-k)} \rightarrow_p \vartheta^*$, and by the same local L_2 stability argument used in **Lemma 6** (Part 2), the mapping $\vartheta \mapsto f(D; \vartheta)$ is locally L_2 -Lipschitz. Therefore,

$$\max_{1 \leq k \leq K_f} \left| \mathbb{E}_r[f(D; \widehat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] - \mathbb{E}_r[f(D; \vartheta^*)] \right| = O_p(a_N),$$

and combining the two displays yields $\widehat{M}_r - M_r = o_p(1) + O_p(a_N) = o_p(1)$. Under the additional second-moment condition above, the empirical-averaging part is $O_p(N_r^{-1/2})$, hence $O_p(N_{\text{all}}^{-1/2})$ under the site-ratio condition $N_r/N_{\text{all}} \rightarrow \rho_r > 0$.

Step 3 (From entrywise to matrix/vector norms). Since $\widehat{H}$ has fixed dimension $K \times K$ and $\widehat{\mathbf{d}}$ has fixed dimension K , the preceding bound applies to each entry, hence

$$\|\widehat{H} - H\|_{\text{op}} \leq C_K \max_{j,k} |\widehat{H}_{jk} - H_{jk}| = o_p(1), \quad \|\widehat{\mathbf{d}} - \mathbf{d}\|_2 \leq \sqrt{K} \max_j |\widehat{d}_j - d_j| = o_p(1),$$

Finally, $\widehat{c}$ is a finite linear combination of the same sample averages $\{\widehat{M}_r\}$ (coming from the constant terms in the variance expansion), so the same entrywise bound implies $|\widehat{c} - c| = o_p(1)$.

Step 4 (Uniform convergence of the quadratic form). For η in a fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$|\widehat{L}(\eta) - L^*(\eta)| \leq \|\widehat{H} - H\|_{\text{op}} \|\eta\|_2^2 + \|\widehat{\mathbf{d}} - \mathbf{d}\|_2 \|\eta\|_2 + |\widehat{c} - c|,$$

and the right-hand side is $o_p(1)$ uniformly over $\eta \in \mathcal{B}$ because $\sup_{\eta \in \mathcal{B}} \|\eta\|_2 < \infty$. $\square$

The penalized objective uses the TATE-level truncated-Wald penalty. Recall $\widehat{L}_\tau(\eta) = N_{\text{all}} \widehat{\text{Var}}\{\widehat{\tau}_{\text{agg}}(\eta)\}$:

$$\widehat{\eta} = \arg \min_{\eta \in \mathbb{R}^K} \left\{ \widehat{L}_\tau(\eta) + \sum_{j=1}^K (\lambda \widehat{t}_j^\tau - 1)_+ |\eta_j| \right\}, \quad \widehat{t}_j^\tau := \frac{|\widehat{\tau}_{ot} - \widehat{\tau}_{t,s_j}|}{\widehat{\text{SE}}(\text{disc}_j^\tau)}, \quad (16)$$

where $\text{disc}_j^\tau = \widehat{\tau}_{ot} - \widehat{\tau}_{t,s_j}$ and

$$\widehat{\text{SE}}(\text{disc}_j^\tau)^2 = \frac{\widehat{V}_{ot}^\tau}{N_t} + \frac{\widehat{V}_{t,s_j}^\tau}{N_t} + \frac{\widehat{V}_{s_j}^\tau}{N_{s_j}} - \frac{2\widehat{C}_{ot,s_j}^\tau}{N_t} = \widehat{\text{Var}}(\widehat{\tau}_{t,s_j} - \widehat{\tau}_{ot}).$$

Equivalently, the implementation minimizes the unscaled variance plus $N_{\text{all}}^{-1} \sum_j (\lambda \hat{t}_j^\tau - 1)_+ |\eta_j|$. This exact rescaling is tested in the codebase. The penalty is inactive when $\hat{t}_j^\tau \leq 1/\lambda$. The oracle-selection regime is $1/\lambda \rightarrow \infty$ while $1/\lambda = o(\sqrt{N_{\text{all}}} b_{0,N}^\tau)$.

Remark 9 (FACE-style validation for λ). The main manuscript fixes the aggregation penalty-activation threshold at the locked pilot value $1/\lambda = 1$ and does not tune it on the analyzed data. The FACE sample-splitting validation rule described here is an optional alternative for those who instead wish to tune λ over a grid: for each outer evaluation fold, candidate weights are learned on inner-training summaries by solving the penalized aggregation problem, and these fixed weights are then evaluated on held-out inner-validation summaries through the pure validation-variance criterion. The penalty is used to construct each training-fold candidate but is not added again to its held-out validation score. This optional validation step is a finite-sample rule for choosing a stable point along the penalized aggregation path; it is not the same object as the nuisance-model cross-validation used for α and γ . For the asymptotic arguments below, $\lambda = \lambda_N$ determines the Wald-significance penalty-activation threshold of [Eq. \(16\)](#): it may be deterministic or data-adaptive, provided it is measurable with respect to the outer training data and satisfies [Assumption 8\(b\)–\(c\)](#) (i.e. $1/\lambda_N \rightarrow \infty$ while $1/\lambda_N = o(\sqrt{N_{\text{all}}} b_{0,N})$) with probability tending to one; it is *not* a penalty parameter driven to infinity. Under fixed separation this condition is the simpler $\lambda_N \rightarrow 0$ and $\lambda_N \sqrt{N_{\text{all}}} \rightarrow \infty$. The inner validation construction preserves the cross-fitting independence needed for the influence-function expansion because the training summaries used to compute $\hat{\eta}^{(-k,-m)}(\lambda)$, the validation summaries used to choose $\hat{\lambda}^{(-k)}$, and the final outer evaluation fold $\mathcal{D}_k$ are separated. Consequently neither $\hat{\lambda}^{(-k)}$ nor $\hat{\eta}^{(-k)}$ uses the outer evaluation fold $\mathcal{D}_k$. In the production implementation, the locked threshold $1/\lambda = 1$ activates the penalty when $\hat{t}_j^\tau > 1$. Crossing this threshold initiates soft shrinkage and is not interpreted as a stand-alone hypothesis test or as guaranteed source exclusion.

Remark 10 (Soft-thresholding interpretation). The penalty in Eq. (16) is adaptive ℓ_1 . For a diagonal Hessian of $\widehat{L}_\tau$, each coordinate minimizes

$$\widehat{H}_{\tau,jj}\eta_j^2 + \widehat{b}_{\tau,j}\eta_j + (\lambda\widehat{t}_j^\tau - 1)_+|\eta_j|, \quad \widehat{b}_{\tau,j} := \left. \frac{\partial \widehat{L}_\tau}{\partial \eta_j} \right|_{\eta=0} = N_{\text{all}} \frac{2(\widehat{C}_{ot,s_j}^\tau - \widehat{V}_{ot}^\tau)}{N_t},$$

where $\widehat{b}_{\tau,j} = O_p(1)$ is the asymptotic-variance gradient at $\eta = 0$. Completing the square yields the explicit solution

$$\widehat{\eta}_j = \text{SoftThreshold}\left(\eta_j^{\text{OLS}}, \frac{(\lambda\widehat{t}_j^\tau - 1)_+}{2\widehat{H}_{\tau,jj}}\right), \quad \eta_j^{\text{OLS}} := -\frac{\widehat{b}_{\tau,j}}{2\widehat{H}_{\tau,jj}},$$

where η_j^{OLS} is the unpenalized variance-minimizing coordinate. Equivalently, the KKT selection rule is

$$\widehat{\eta}_j = 0 \iff |\widehat{b}_{\tau,j}| \leq (\lambda\widehat{t}_j^\tau - 1)_+,$$

with the analogous coupled KKT conditions for a non-diagonal Hessian.

S4.3 Informative Source Indices

For each source, write $\mu_{t,s_j}^{a,*} = \text{plim } \widehat{\mu}_{t,s_j}^a$. Define the TATE-compatible source set

$$\mathcal{J}_\tau^* = \{j : \mu_{t,s_j}^{1,*} - \mu_{t,s_j}^{0,*} = \tau\} = \{j : \text{plim } \widehat{\tau}_{t,s_j} = \tau\}.$$

Define the oracle-restricted space:

$$\mathbb{R}_{\mathcal{J}_\tau^*} := \{\eta \in \mathbb{R}^K : \eta_j = 0 \text{ for all } j \notin \mathcal{J}_\tau^*\}.$$

Let η_τ^* be the oracle TATE weights:

$$\eta_\tau^* = \arg \min_{\eta \in \mathbb{R}_{\mathcal{J}_\tau^*}} L_\tau^*(\eta), \quad L_\tau^*(\eta) = \text{Var}\{\sqrt{N_{\text{all}}}(\widehat{\tau}_{\text{agg}}(\eta) - \tau)\}.$$

Assumption 8 (Aggregation Regularity). Assume:

- (a) (*Compatible sources and global curvature*) The set $\mathcal{J}_\tau^*$ is non-empty. The quadratic objective L_τ^* has a positive-definite Hessian on $\mathbb{R}^K$, with minimum eigenvalue bounded

away from zero. Hence L_τ^* is coercive and has a finite unique minimizer on every coordinate subspace, including $\mathbb{R}_{\mathcal{J}_\tau^*}$.

(b) (*Separation / “ β -min” for biased sites*) Define

$$b_{0,N}^\tau := \min_{j \notin \mathcal{J}_\tau^*} |\text{plim}(\hat{\tau}_{t,s_j}) - \tau|.$$

Assume $b_{0,N}^\tau \sqrt{N_{\text{all}}} \rightarrow \infty$ and that the asymptotic variance of each candidate TATE discrepancy $\sqrt{N_{\text{all}}} \{\hat{\tau}_{t,s_j} - \hat{\tau}_{ot}\}$ is bounded away from zero.

(c) (*Threshold rate*) The penalty-activation threshold satisfies

$$\lambda \rightarrow 0, \quad \lambda \sqrt{N_{\text{all}}} b_{0,N}^\tau \rightarrow \infty,$$

equivalently $1/\lambda = o(\sqrt{N_{\text{all}}} b_{0,N}^\tau)$ while $1/\lambda \rightarrow \infty$.

(d) (*Fixed source set*) The number of source sites K is fixed.

(e) (*Compatible-source contrast expansion*) For every $j \in \mathcal{J}_\tau^*$, the centered contrast satisfies

$$\sqrt{N_{t,s_j}} \left[\{\hat{\mu}_{t,s_j}^1 - \mu_{t,s_j}^{1,*}\} - \{\hat{\mu}_{t,s_j}^0 - \mu_{t,s_j}^{0,*}\} \right] = \frac{1}{\sqrt{N_{t,s_j}}} \sum_i \varphi_{t,s_j,i}^\tau + o_p(1),$$

with the corresponding multisite scaling, a centered finite-variance influence function, and the uniform foldwise version used by the cross-fitted estimator. Separate application of [Theorems 3](#) and [4](#) is sufficient. If compatibility instead arises only because the two arm-specific pseudo-limit biases cancel, this contrast expansion is an additional assumption.

If λ is selected over a pre-specified sensitivity grid, the separation and threshold-window conditions are interpreted as high-probability conditions on the selected sequence. The primary implementation instead fixes $1/\lambda = 1$ before examining the outer evaluation fold. The oracle support recovery and single normal limit stated below therefore concern the rate-indexed sequence in [Assumption 8\(c\)](#); they are not asserted for the fixed-cutoff implementation without an additional fixed-threshold argument.

Remark 11 (When separation fails / local alternatives). [Assumption 8\(b\)](#) is used only for support recovery. Under local TATE alternatives $b_{0,N}^\tau = O(N_{\text{all}}^{-1/2})$, exact selection can fail because $\hat{t}_j^\tau$ remains $O_p(1)$. The added first-order bias is controlled by

$$\left| \sum_{j \notin \mathcal{J}_\tau^*} \eta_j \{ \text{plim}(\hat{\tau}_{t,s_j}) - \tau \} \right|,$$

which is negligible when the included TATE deviations are $o(N_{\text{all}}^{-1/2})$.

S4.4 Connection to the Original FACE Squared-Discrepancy Objective

The main FACE-HD estimator uses the truncated-Wald objective in [Eq. \(16\)](#). This subsection records the direct analogue of the original FACE adaptive-aggregation objective ([Han et al. 2025](#), Equation (11)), which uses the raw squared discrepancy rather than the studentized Wald statistic. With the notation of [Section S4.3](#), define

$$\hat{\eta}_{\text{sq}} = \arg \min_{\eta \in \mathbb{R}^K} \left\{ \hat{L}_\tau(\eta) + \lambda_{\text{sq},N} \sum_{j=1}^K |\eta_j| (\text{disc}_j^\tau)^2 \right\}, \quad \text{disc}_j^\tau = \hat{\tau}_{ot} - \hat{\tau}_{t,s_j}, \quad (17)$$

where $\hat{L}_\tau(\eta)$ is the same rescaled TATE-variance objective used in [Eq. \(16\)](#).

The oracle argument is the same as the proof of [Lemma 11](#), but the rate window is different. Suppose [Assumptions 5](#) to [7](#) hold and that the global-curvature and fixed- K parts of [Assumption 8\(a\),\(d\)](#) hold. If the raw-discrepancy tuning parameter satisfies

$$\lambda_{\text{sq},N} \rightarrow \infty, \quad \lambda_{\text{sq},N} = o(N_{\text{all}}^{1/2}), \quad \lambda_{\text{sq},N} (b_{0,N}^\tau)^2 \rightarrow \infty, \quad (18)$$

then the squared-discrepancy objective has the same oracle-selection behavior:

$$P(\hat{\eta}_{\text{sq},j} = 0 \text{ for all } j \notin \mathcal{J}_\tau^*) \rightarrow 1, \quad \|\hat{\eta}_{\text{sq},\mathcal{J}_\tau^*} - \eta_{\tau,\mathcal{J}_\tau^*}^*\|_2 = o_p(1).$$

For each $j \in \mathcal{J}_\tau^*$, the two arm expansions imply $\text{disc}_j^\tau = O_p(N_{\text{all}}^{-1/2})$, hence

$$\lambda_{\text{sq},N} (\text{disc}_j^\tau)^2 = O_p(\lambda_{\text{sq},N}/N_{\text{all}}) = o_p(N_{\text{all}}^{-1/2}),$$

so the penalty is negligible on the oracle subspace. For each $j \notin \mathcal{J}_\tau^*$, $|\text{disc}_j^\tau| \geq b_{0,N}^\tau/2$ w.p.a.1; therefore

$$\lambda_{\text{sq},N}(\text{disc}_j^\tau)^2 \geq \frac{1}{4}\lambda_{\text{sq},N}(b_{0,N}^\tau)^2 \rightarrow \infty \quad \text{w.p.a.1.}$$

As in [Lemma 11](#), coercivity first gives $\|\hat{\eta}_{\text{sq}}\|_2 = O_p(1)$, and the gradient of the quadratic variance part is therefore $O_p(1)$ along the segment joining the optimizer to its coordinate-zero competitor. The diverging adaptive penalty makes that competitor strictly favorable unless $\hat{\eta}_{\text{sq},j} = 0$, forcing zero weights for biased sources.

Under fixed TATE separation, this reduces to the original FACE scaling $\lambda_{\text{sq},N} \rightarrow \infty$ and $\lambda_{\text{sq},N} = o(N_{\text{all}}^{1/2})$. The truncated-Wald objective needs only the weaker studentized TATE separation in [Assumption 8](#).

S4.5 Aggregation Main Results

For later use in the aggregation proofs, we restate the pairwise input result that is repeatedly invoked below.

Theorem (restated). *Suppose [Assumptions 5 to 7](#) hold. Assume ϑ^* is the unique population target pair induced by the calibrated losses under the alignment condition in [Assumption 7\(c\)](#), implying [Lemma 5](#) holds. If further, at least one nuisance model (outcome or weighting) is correctly specified in the sense of [Lemma 1](#) (so $\mu^1(\vartheta^*) = \mu^1$), the calibrated nuisance sparsities satisfy*

$$\frac{(s_\alpha + s_\gamma)^2 \log^2(p)}{N} \rightarrow 0,$$

and $V_{t,s}^ > 0$, then the cross-fitted estimator is asymptotically normal. Here s_α and s_γ denote the calibrated sparsities from [Definition 1](#):*

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

If the initial nuisance estimators enter the two-level algorithm separately from the calibrated

update, their stochastic effect is absorbed through [Assumption 7\(c\)–\(d\)](#). The finite-sample KKT perturbation from the ℓ_1 penalties is likewise absorbed in these nuisance-rate conditions and is $o_p(N^{-1/2})$ under the stated sparsity scaling.

Theorem 5 (Asymptotic Normality of Direct-TATE FACE-HD). *Suppose the pairwise assumptions hold and [Assumption 8](#) holds. For $j \in \mathcal{J}_\tau^*$, let $N_{t,s_j} = N_t + N_{s_j}$ and let $s_{\text{tot},j}$ bound the combined arm-specific nuisance sparsity. If*

$$\max_{j \in \mathcal{J}_\tau^*} \frac{s_{\text{tot},j}^2 \log^2(p)}{N_{t,s_j}} \rightarrow 0,$$

and $L_\tau^*(\eta_\tau^*) > 0$, then

$$\sqrt{N_{\text{all}}}(\hat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}\{0, L_\tau^*(\eta_\tau^*)\}.$$

Moreover, the within-site-centered pseudo-value estimator in [Eq. \(11\)](#) is consistent:

$$\hat{V}_\tau \xrightarrow{p} L_\tau^*(\eta_\tau^*),$$

while, uniformly over the fixed outer folds, $\hat{L}_{\tau,-k}\{\hat{\eta}^{(-k)}\} \xrightarrow{p} L_\tau^*(\eta_\tau^*)$; these two estimators need not be numerically identical in finite samples. Furthermore,

$$\max_{k \leq K_f} \|\hat{\eta}^{(-k)} - \eta_\tau^*\|_2 = o_p(1), \quad P\left(\hat{\eta}_j^{(-k)} = 0 \text{ for every } j \notin \mathcal{J}_\tau^*, k \leq K_f\right) \rightarrow 1,$$

and

$$\sqrt{\frac{N_{\text{all}}}{\hat{V}_\tau}}(\hat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}(0, 1).$$

Proposition 1 (TATE Efficiency Gain). *Under the conditions of [Theorem 5](#), direct-TATE FACE-HD is asymptotically at least as efficient as target-only TATE:*

$$L_\tau^*(\eta_\tau^*) \leq L_\tau^*(0) = V_{\tau, \text{target}},$$

with strict inequality whenever there exists $j \in \mathcal{J}_\tau^*$ such that

$$\text{Var}(\varphi_{t,s_j}^\tau - \varphi_{ot}^\tau) > 0 \quad \text{and} \quad \text{Cov}(\varphi_{ot}^\tau, \varphi_{t,s_j}^\tau - \varphi_{ot}^\tau) \neq 0.$$

S4.6 Aggregation Supporting Lemmas

To keep the remaining proofs readable, use the generic shorthand

$$\theta = \tau, \quad \hat{\theta}_{ot} = \hat{\tau}_{ot}, \quad \hat{\theta}_{t,s_j} = \hat{\tau}_{t,s_j}, \quad \hat{\theta}_{\text{agg}} = \hat{\tau}_{\text{agg}}, \quad \mathcal{J}^* = \mathcal{J}_\tau^*, \quad \eta^* = \eta_\tau^*, \quad L^* = L_\tau^*,$$

and suppress the superscript τ on the treatment-contrast influence components. Thus every object in the supporting lemmas below is TATE-level.

Lemma 10 (Influence Function Representation). *Under [Assumptions 5](#) to [7](#) and [Assumption 8\(a\),\(e\)](#), the aggregated estimator admits the asymptotic linear representation:*

$$\sqrt{N_{\text{all}}} \{ \hat{\theta}_{\text{agg}}(\eta^*) - \theta \} = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1)$$

where $\varphi_{\text{agg},i}(\eta^*)$ is the aggregated influence function:

$$\varphi_{\text{agg},i}(\eta^*) = \left(1 - \sum_{j \in \mathcal{J}^*} \eta_j^* \right) \varphi_{ot,i} + \sum_{j \in \mathcal{J}^*} \eta_j^* \varphi_{t,s_j,i},$$

with $\varphi_{t,s_j,i}$ denoting the (rescaled) IF contribution of source-assisted estimator j under $\sqrt{N_{\text{all}}}$ scaling. **Convention:** Each $\varphi_{ot,i}$ (resp. $\varphi_{t,s_j,i}$) is understood as zero for observations not in the target data $\mathcal{D}_t$ (resp. not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$), and includes the constant rescaling needed to convert its natural $\sqrt{N_t}$ (resp. $\sqrt{N_{t,s_j}}$) linear representation into the common $\sqrt{N_{\text{all}}}$ scaling.

Proof. By the compatible-source contrast expansion in [Assumption 8\(e\)](#)—which follows by applying the pairwise expansion to both arms when they are separately identified—for every $j \in \mathcal{J}^*$,

$$\sqrt{N_{t,s_j}} \{ \hat{\theta}_{t,s_j} - \text{plim}(\hat{\theta}_{t,s_j}) \} = \frac{1}{\sqrt{N_{t,s_j}}} \sum_{i=1}^{N_{\text{all}}} \bar{\varphi}_{t,s_j,i} + o_p(1),$$

where $\bar{\varphi}_{t,s_j,i}$ is understood as zero for observations not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$. Since $N_{t,s_j}/N_{\text{all}} \rightarrow c_j \in (0, 1)$, the above is equivalent to

$$\sqrt{N_{\text{all}}} \{ \hat{\theta}_{t,s_j} - \text{plim}(\hat{\theta}_{t,s_j}) \} = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{t,s_j,i} + o_p(1)$$

for a rescaled IF $\varphi_{t,s_j,i}$. The target-only estimator has the analogous form:

$$\sqrt{N_{\text{all}}}(\hat{\theta}_{ot} - \theta) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{ot,i} + o_p(1).$$

For $j \in \mathcal{J}^*$, $\text{plim}(\hat{\theta}_{t,s_j}) = \theta$.

By linearity, the aggregated estimator with fixed $\eta = \eta^*$ satisfies:

$$\sqrt{N_{\text{all}}}(\hat{\theta}_{\text{agg}}(\eta^*) - \theta) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1).$$

The remainder remains $o_p(1)$ because K is fixed. $\square$

Lemma 11 (Selection Consistency). *Under [Assumptions 5 to 7](#) and [Assumption 8](#), the penalized estimator $\hat{\eta}$ from [Eq. \(16\)](#) satisfies:*

1. (**Oracle-subspace consistency**) $\|\hat{\eta} - \eta^*\|_2 = o_p(1)$
2. (**Selection**) $\hat{\eta}_j = 0$ for all $j \notin \mathcal{J}^*$ with probability approaching 1.

Proof. We give a self-contained argument; it parallels the FACE proof in [Han et al. \(2025\)](#) (Lemma S5) but the objective here is an explicit quadratic form. Recall $\hat{L}(\eta) = \eta^T \hat{H} \eta + \hat{\mathbf{d}}^T \eta + \hat{c}$ and its population analogue $L^*(\eta) = \eta^T H \eta + \mathbf{d}^T \eta + c$.

Part 1 (Consistency on the oracle subspace). Define the oracle-restricted estimator

$$\tilde{\eta} = \arg \min_{\eta \in \mathbb{R}_{\mathcal{J}^*}} \left\{ \hat{L}(\eta) + \sum_{j=1}^K (\lambda \hat{t}_j^\tau - 1)_+ |\eta_j| \right\}.$$

For $j \in \mathcal{J}^*$, $\text{disc}_j^\tau = \hat{\theta}_{ot} - \hat{\theta}_{t,s_j} = O_p(N_{\text{all}}^{-1/2})$ and $\hat{t}_j^\tau = O_p(1)$. Under $1/\lambda \rightarrow \infty$, the penalty is zero on every oracle coordinate w.p.a.1, so $\tilde{\eta}$ coincides with the unpenalized TATE-variance minimizer on $\mathbb{R}_{\mathcal{J}^*}$.

Next, by [Lemma 9](#), the quadratic coefficients satisfy $\|\hat{H} - H\|_{\text{op}} = o_p(1)$ and $\|\hat{\mathbf{d}} - \mathbf{d}\|_2 = o_p(1)$.

Let $H_{\mathcal{J}^*}$ and $\hat{H}_{\mathcal{J}^*}$ denote the principal submatrices corresponding to $\mathcal{J}^*$, and similarly let $\mathbf{d}_{\mathcal{J}^*}$ and $\hat{\mathbf{d}}_{\mathcal{J}^*}$ denote the corresponding subvectors. Because K is fixed, $H_{\mathcal{J}^*}$ is a fixed-dimensional positive definite matrix by the global-curvature condition, hence $\|H_{\mathcal{J}^*}^{-1}\|_{\text{op}} < \infty$

and $\|\widehat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}} = O_p(1)$. More explicitly, by Weyl's inequality and [Lemma 9](#), $\lambda_{\min}(\widehat{H}_{\mathcal{J}^*}) \geq \lambda_{\min}(H_{\mathcal{J}^*}) - \|\widehat{H}_{\mathcal{J}^*} - H_{\mathcal{J}^*}\|_{\text{op}}$, which is bounded away from 0 w.p.a.1, so $\widehat{H}_{\mathcal{J}^*}$ is invertible and $\|\widehat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}} = O_p(1)$. Because the penalty is zero on the oracle subspace w.p.a.1 and both quadratic objectives are coercive and strictly convex there, their unconstrained minimizers satisfy the first-order conditions $2\widehat{H}_{\mathcal{J}^*}\eta + \widehat{\mathbf{d}}_{\mathcal{J}^*} = 0$ and $2H_{\mathcal{J}^*}\eta + \mathbf{d}_{\mathcal{J}^*} = 0$, so

$$\eta_{\mathcal{J}^*}^* = -\frac{1}{2}H_{\mathcal{J}^*}^{-1}\mathbf{d}_{\mathcal{J}^*}, \quad \tilde{\eta}_{\mathcal{J}^*} = -\frac{1}{2}\widehat{H}_{\mathcal{J}^*}^{-1}\widehat{\mathbf{d}}_{\mathcal{J}^*} + o_p(1).$$

To make the perturbation step explicit, write

$$\tilde{\eta}_{\mathcal{J}^*} - \eta_{\mathcal{J}^*}^* = -\frac{1}{2}\left(\widehat{H}_{\mathcal{J}^*}^{-1}\widehat{\mathbf{d}}_{\mathcal{J}^*} - H_{\mathcal{J}^*}^{-1}\mathbf{d}_{\mathcal{J}^*}\right) + o_p(1).$$

Then

$$\widehat{H}_{\mathcal{J}^*}^{-1}\widehat{\mathbf{d}}_{\mathcal{J}^*} - H_{\mathcal{J}^*}^{-1}\mathbf{d}_{\mathcal{J}^*} = \widehat{H}_{\mathcal{J}^*}^{-1}(\widehat{\mathbf{d}}_{\mathcal{J}^*} - \mathbf{d}_{\mathcal{J}^*}) + (\widehat{H}_{\mathcal{J}^*}^{-1} - H_{\mathcal{J}^*}^{-1})\mathbf{d}_{\mathcal{J}^*}.$$

The first term is $o_p(1)$ since $\|\widehat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}} = O_p(1)$ and $\|\widehat{\mathbf{d}} - \mathbf{d}\|_2 = o_p(1)$. For the second term, use $\widehat{H}_{\mathcal{J}^*}^{-1} - H_{\mathcal{J}^*}^{-1} = \widehat{H}_{\mathcal{J}^*}^{-1}(H_{\mathcal{J}^*} - \widehat{H}_{\mathcal{J}^*})H_{\mathcal{J}^*}^{-1}$, so

$$\|\widehat{H}_{\mathcal{J}^*}^{-1} - H_{\mathcal{J}^*}^{-1}\|_{\text{op}} \leq \|\widehat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}}\|\widehat{H} - H\|_{\text{op}}\|H_{\mathcal{J}^*}^{-1}\|_{\text{op}} = o_p(1),$$

and hence the second term is also $o_p(1)$. Therefore $\|\tilde{\eta} - \eta^*\|_2 = o_p(1)$. Finally, since $\hat{\eta}$ agrees with $\tilde{\eta}$ w.p.a.1 by Part 2 below, the same consistency holds for $\hat{\eta}$.

Part 2 (Selection of biased sites). First, the unconstrained penalized minimizer is stochastically bounded. By [Assumption 8\(a\)](#), coefficient consistency, and Weyl's inequality, there is a constant $h_0 > 0$ such that $\lambda_{\min}(\widehat{H}) \geq h_0$ w.p.a.1. Since the penalty is nonnegative and the objective at $\hat{\eta}$ is no larger than its value at zero,

$$h_0\|\hat{\eta}\|_2^2 - \|\widehat{\mathbf{d}}\|_2\|\hat{\eta}\|_2 \leq 0.$$

Because $\|\widehat{\mathbf{d}}\|_2 = O_p(1)$, this implies $\|\hat{\eta}\|_2 = O_p(1)$. Hence the quadratic gradient is $O_p(1)$ uniformly along any line segment joining $\hat{\eta}$ to a vector formed by setting one of its coordinates

to zero:

$$\sup_{\bar{\eta} \text{ on such a segment}} \left| \frac{\partial \hat{L}}{\partial \eta_j}(\bar{\eta}) \right| \leq 2 \|\widehat{H}\|_{\text{op}} \|\hat{\eta}\|_2 + \|\widehat{\mathbf{d}}\|_2 = O_p(1).$$

For $j \notin \mathcal{J}^*$, let $b_{j,N}^\tau = |\text{plim}(\hat{\theta}_{t,s_j}) - \theta|$. Then

$$\hat{t}_j^\tau = \frac{|\text{disc}_j^\tau|}{\widehat{\text{SE}}(\text{disc}_j^\tau)} = \Theta_p(\sqrt{N_{\text{all}} b_{j,N}^\tau}) \rightarrow \infty,$$

and $(\lambda \hat{t}_j^\tau - 1)_+ \rightarrow \infty$ by [Assumption 8](#)(b)–(c).

Let $\hat{\eta}^{(j=0)}$ equal $\hat{\eta}$ with coordinate j set to zero. This is a feasible point in $\mathbb{R}^K$. The mean-value theorem and the preceding gradient bound give

$$\hat{L}\{\hat{\eta}^{(j=0)}\} - \hat{L}(\hat{\eta}) \leq \sup_{\bar{\eta} \text{ on the segment}} \left| \frac{\partial \hat{L}}{\partial \eta_j}(\bar{\eta}) \right| |\hat{\eta}_j| = O_p(1) |\hat{\eta}_j|.$$

The penalty term decreases by $(\lambda \hat{t}_j^\tau - 1)_+ |\hat{\eta}_j|$ when $\hat{\eta}_j$ is set to zero. Since this coefficient diverges while the quadratic-gradient bound is $O_p(1)$, every incompatible coordinate is zero w.p.a.1. Therefore $\hat{\eta} \in \mathbb{R}_{\mathcal{J}^*}$ w.p.a.1, hence $\hat{\eta} = \tilde{\eta}$ w.p.a.1. $\square$

Lemma 12 (Asymptotic Equivalence). *Under [Assumptions 5 to 7](#) and [Assumption 8](#):*

$$\sqrt{N_{\text{all}}} \{\hat{\theta}_{\text{agg}}(\hat{\eta}) - \hat{\theta}_{\text{agg}}(\eta^*)\} = o_p(1)$$

Proof. The difference is:

$$\sqrt{N_{\text{all}}} \{\hat{\theta}_{\text{agg}}(\hat{\eta}) - \hat{\theta}_{\text{agg}}(\eta^*)\} = \sum_{j=1}^K (\hat{\eta}_j - \eta_j^*) \sqrt{N_{\text{all}}} (\hat{\theta}_{t,s_j} - \hat{\theta}_{ot})$$

For $j \in \mathcal{J}^*$, weight consistency and the two arm-specific pairwise expansions give $|\hat{\eta}_j - \eta_j^*| = o_p(1)$ and $\sqrt{N_{\text{all}}} (\hat{\theta}_{t,s_j} - \hat{\theta}_{ot}) = O_p(1)$.

For $j \notin \mathcal{J}^*$: By [Lemma 11](#)(2), $\hat{\eta}_j = 0$ w.p.a.1, and $\eta_j^* = 0$ by definition (oracle excludes biased sites). Thus the difference is zero w.p.a.1. Because K is fixed, summing over j yields $o_p(1)$. $\square$

Remark 12 (Growing number of sources K). The fixed- K argument extends only with additional concentration. A sufficient condition for the plug-in term is

$$\|\hat{\eta} - \eta^*\|_1 \max_{j \leq K} \left| \sqrt{N_{\text{all}}} (\hat{\theta}_{t,s_j} - \hat{\theta}_{ot}) \right| = o_p(1),$$

which typically requires both a sparsity assumption on η^* and concentration bounds scaling with $\log K$.

S4.7 Proofs of Aggregation Results

Proof of Theorem 5. Combining Lemma 10, Lemma 11, and Lemma 12:

$$\sqrt{N_{\text{all}}}(\hat{\theta}_{\text{agg}} - \theta) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1).$$

As in Lemma 3, the observations are independent with a multi-site structure and site proportions converging to constants. The aggregated influence function contributions $\{\varphi_{\text{agg},i}(\eta^*)\}_{i=1}^{N_{\text{all}}}$ are therefore independent with mean zero and finite variance $L^*(\eta^*)$. Under Assumption 6(b), these contributions have a uniform $(2 + \delta)$ moment, so the same Lyapunov argument as in Lemma 3 applies and yields:

$$\frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*)).$$

For variance estimation consistency, first fix an outer fold. By Lemma 11(1), $\|\hat{\eta}^{(-k)} - \eta^*\|_2 = o_p(1)$. By the uniform convergence of $\hat{L}_{\tau,-k}$ to L^* and local Lipschitz continuity of the quadratic map $L^*(\eta) = \eta^T H \eta + \mathbf{d}^T \eta + c$ in a neighborhood of η^* (e.g., on $\{\eta : \|\eta - \eta^*\|_2 \leq R\}$, $|L^*(\eta_1) - L^*(\eta_2)| \leq (2\|H\|_{\text{op}}R + \|\mathbf{d}\|_2)\|\eta_1 - \eta_2\|_2$):

$$\left| \hat{L}_{\tau,-k}\{\hat{\eta}^{(-k)}\} - L^*(\eta^*) \right| = o_p(1).$$

This holds uniformly over $k \leq K_f$ because the number of folds is fixed. For the distinct held-out variance estimator, evaluate each observation's pseudo-value with its fold-specific weight.

Weight consistency, the local L_2 stability of the cross-fitted influence components, and the within-site conditional law of large numbers imply that its within-site first and second moments differ by $o_p(1)$ from those of the fixed oracle-weight pseudo-values. Therefore the WSS construction in [Remark 5](#) gives $\widehat{V}_\tau \xrightarrow{p} L^*(\eta^*)$.

Finally, write

$$\sqrt{\frac{N_{\text{all}}}{\widehat{V}_\tau}}(\widehat{\theta}_{\text{agg}} - \theta) = \left\{ \sqrt{\frac{L^*(\eta^*)}{\widehat{V}_\tau}} \right\} \left\{ \frac{\sqrt{N_{\text{all}}}(\widehat{\theta}_{\text{agg}} - \theta)}{\sqrt{L^*(\eta^*)}} \right\}.$$

The second factor converges to $\mathcal{N}(0, 1)$ and $\widehat{V}_\tau \xrightarrow{p} L^*(\eta^*)$, so the prefactor converges to one.

Therefore the product converges in distribution to $\mathcal{N}(0, 1)$ by Slutsky's theorem. $\square$

Proof of [Proposition 1](#). Since η^* minimizes the TATE variance and $\eta = 0$ is feasible, $L^*(\eta^*) \leq L^*(0) = V_{\tau, \text{target}}$.

For strict improvement, consider one informative source $j \in \mathcal{J}^*$. The optimal projection is:

$$\eta_j^* = -\frac{\text{Cov}(\varphi_{ot}^\tau, \varphi_{t,s_j}^\tau - \varphi_{ot}^\tau)}{\text{Var}(\varphi_{t,s_j}^\tau - \varphi_{ot}^\tau)}.$$

If the covariance is non-zero and the denominator is positive, the minimizing weight is non-zero and strictly decreases variance relative to $\eta = 0$, so $L^*(\eta^*) < L^*(0)$. $\square$

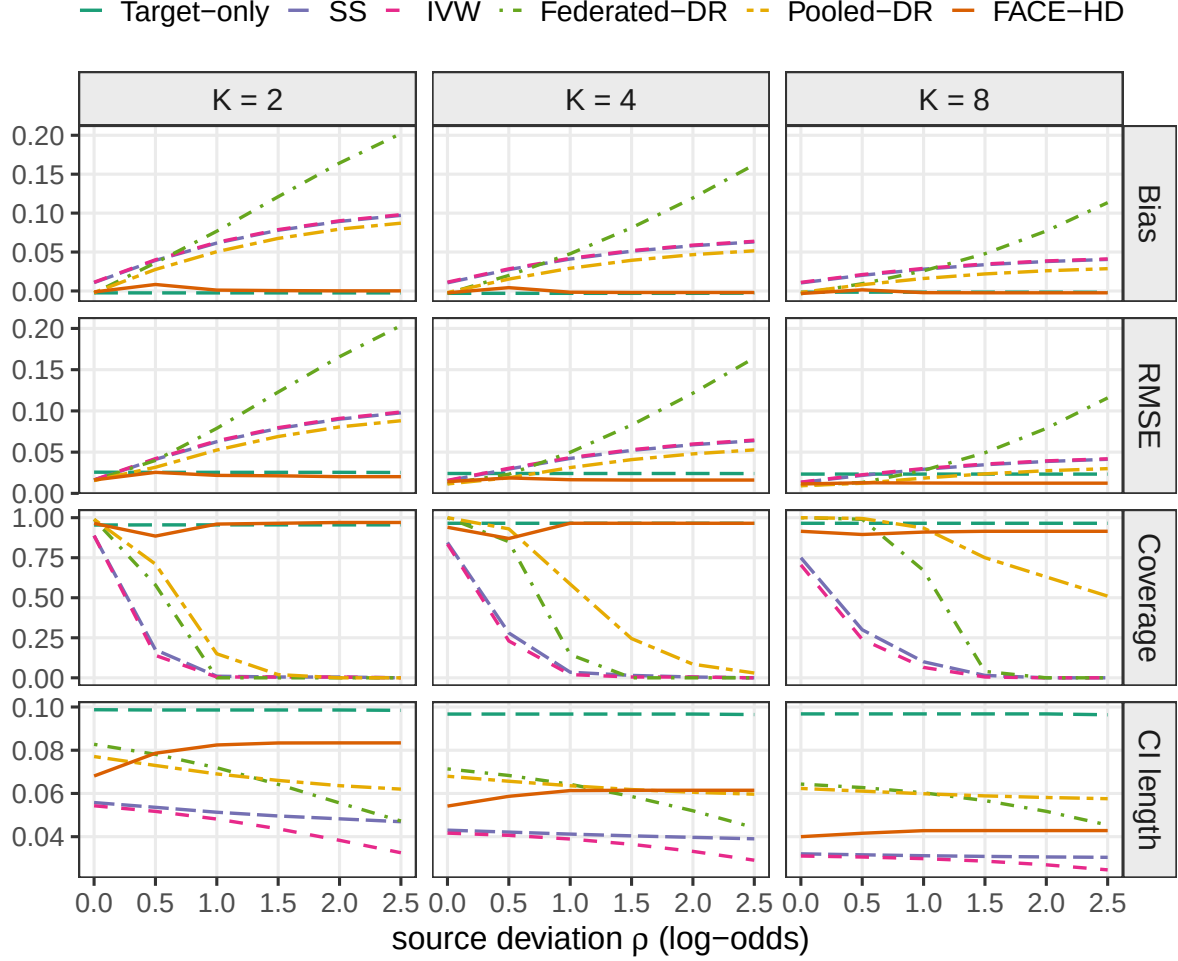


Figure 4: Legacy treated-mean negative-transfer simulation, $C(2)$, $p = 50$. Panels show bias, RMSE, 95% coverage, and CI length for μ^1 as the source deviation ρ increases, with columns for $K = 2, 4, 8$ source sites. FACE-HD remains stable while the naive aggregators lose coverage.

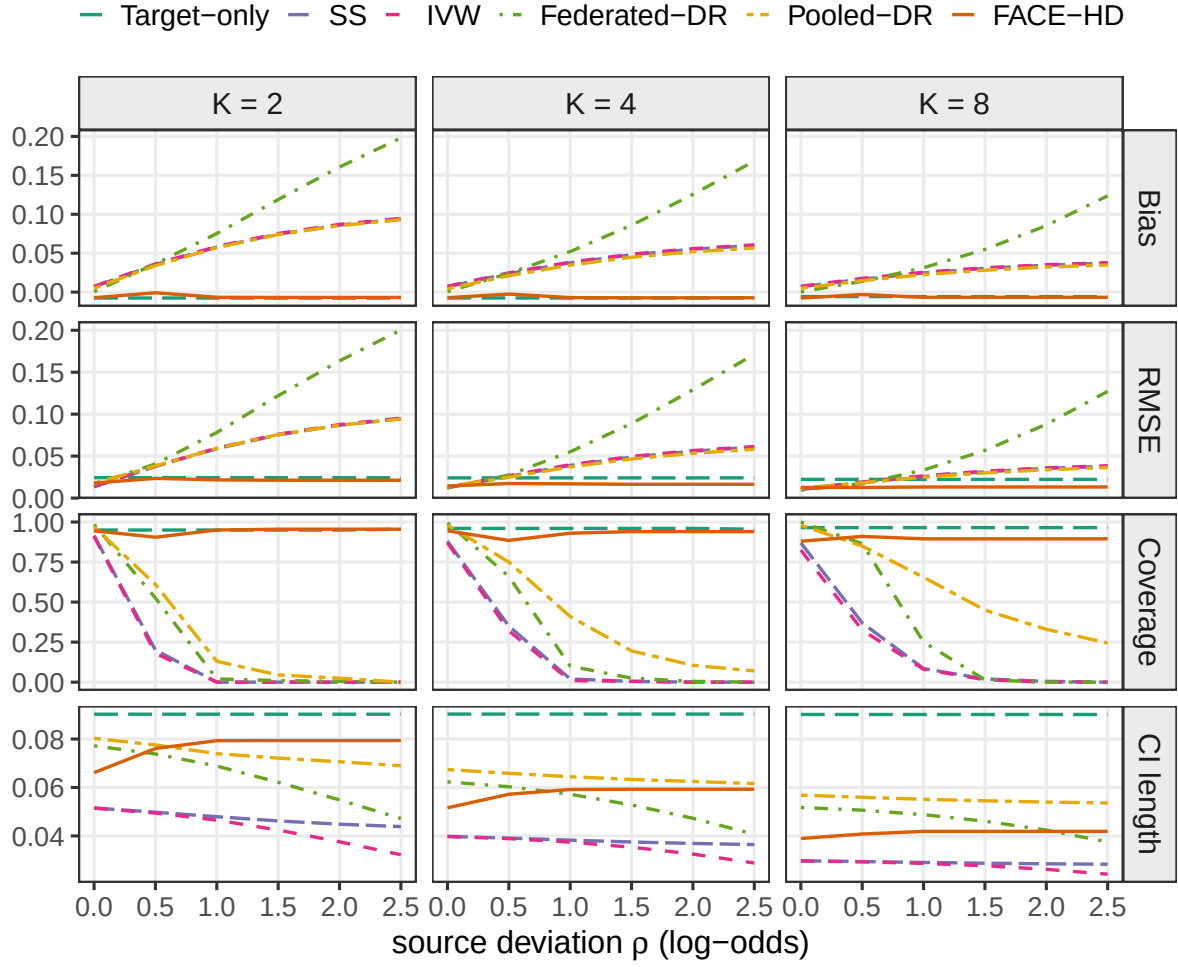


Figure 5: Legacy treated-mean negative-transfer simulation, $C(3)$, $p = 50$. Same layout as [figure 4](#); the site/propensity model is misspecified.

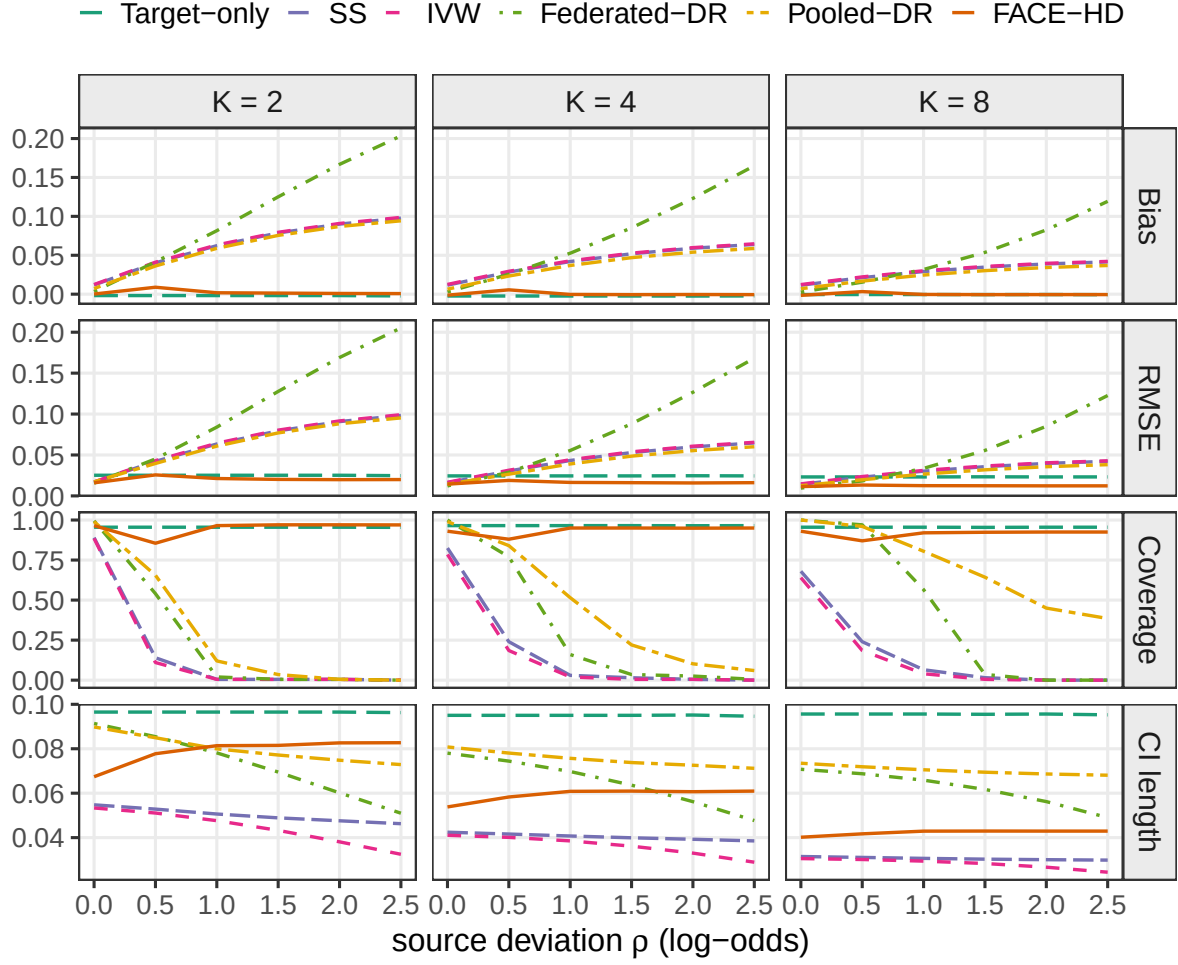


Figure 6: Legacy treated-mean negative-transfer simulation, $C(1)$, $p = 50$. Same layout as [figure 4](#); the conclusions match the $p = 100$ case.

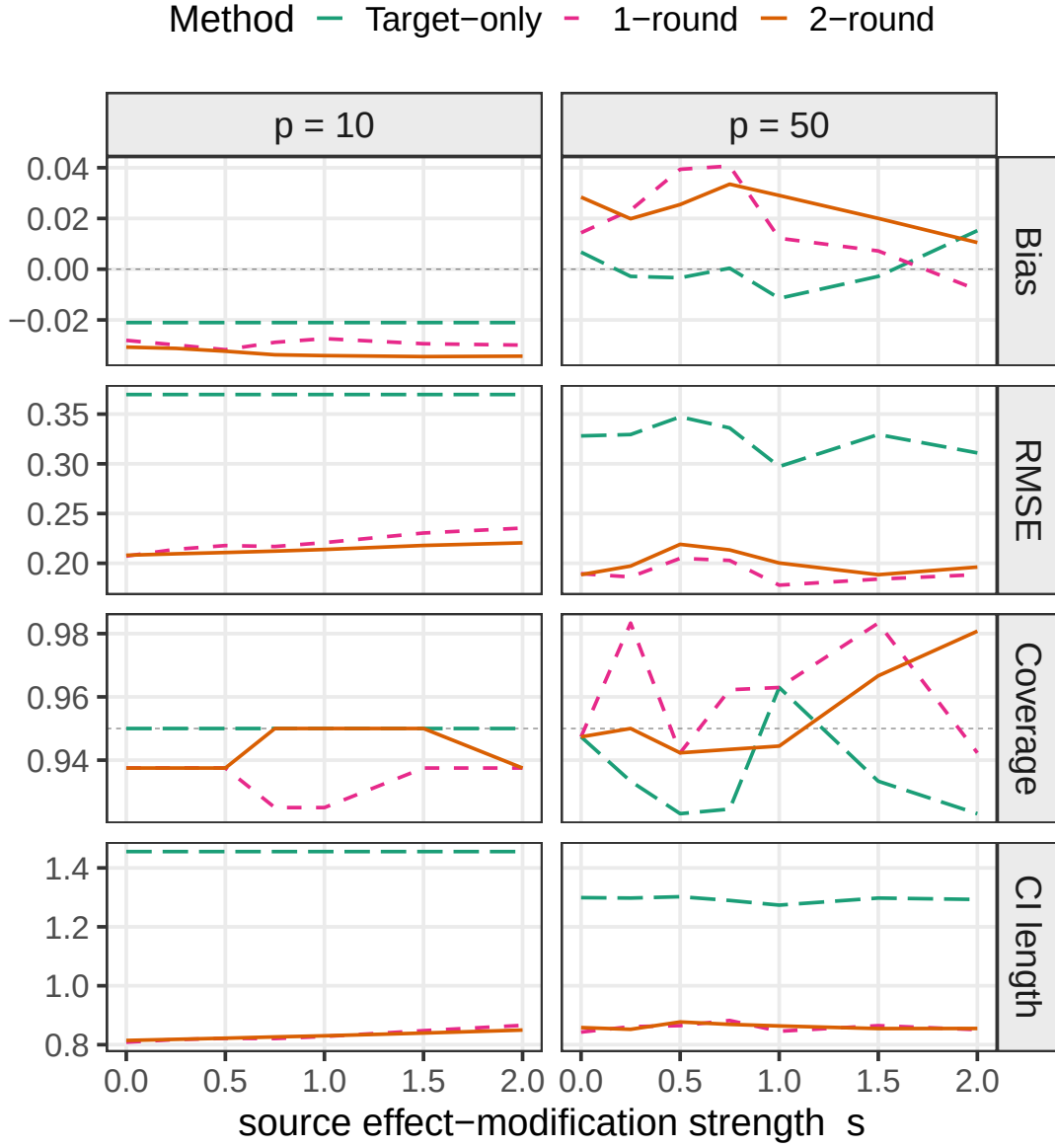


Figure 7: Archived one-round versus two-round arm-wise TATE FACE-HD for the continuous-outcome experiment. Panels show TATE bias, RMSE, 95% coverage, and CI length as the effect-modification strength s increases, for $p = 10$ and $p = 50$. Cells contain 80 completed replicates at $p = 10$ and 52–60 at $p = 50$; the comparison is descriptive, and the two variants track closely.